



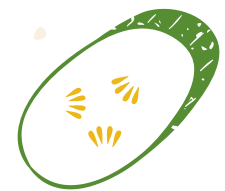
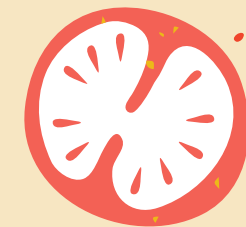
TERAPI MEDIK GIZI PADA PENYAKIT SISTEM GEH

A. YASMIN SYAUKI
DEPARTEMEN ILMU GIZI
FK UNHAS
2024

SKDI 2012

Outline :

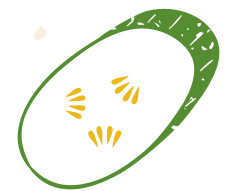
- Penyakit – penyakit sisitem GEH dalam SKDI 2012
- Review fisiologi saluran cerna
- Macam-macam konsistensi makanan
- Faktor-faktor risiko terkait gizi pada penyakit sistem GEH
- Terapi medik gizi pada penyakit terkait mulut dan esofagus
- Terapi medik gizi pada penyakit terkait gaster dan duodenum
- Terapi medik gizi pada penyakit terkait kolon, rectum dan anus
- Terapi medik gizi pada penyakit terkait hepar dan kandung empedu



SKDI 2012

Level 4 :

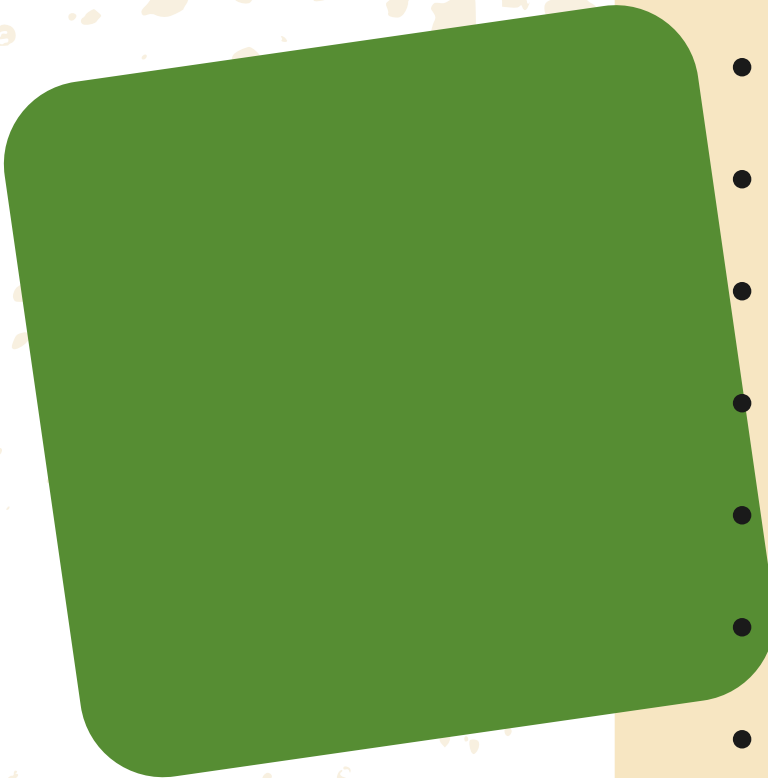
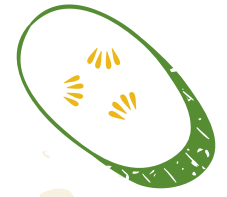
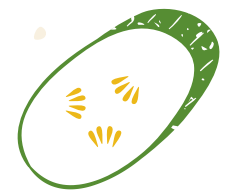
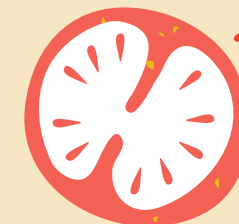
- Kandidiasis mulut
- Ulkus mulut (aptosa, herpes)
- Gastritis
- GERD
- Hepatitis A
- Rujuk balik Hepatitis B
- Rujuk balik Hepatitis C
- Gastroenteritis (kolera, giardiasis)
- Disentri basiler, amuba
- Hemoroid grade 1-2
- Intoleransi makanan
- Alergi makanan



SKDI 2012

Level 3A :

- Perlemakan hepar
- Hepatitis B
- Hepatitis C
- Esofagitis refluks
- Ulkus (gaster, duodenum)
- Kolitis
- Irritable Bowel Syndrome
- Malabsorpsi
- Abses hepar amoeba
- Hemoroid grade 3-4
- Fisura anus
- Prolaps rectum, anus



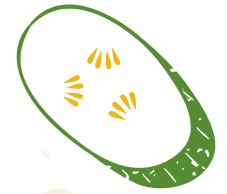
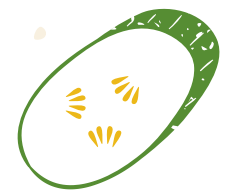
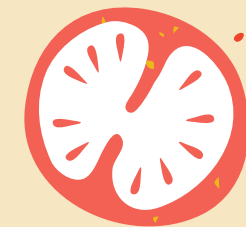
SKDI 2012

Level 3A :

- Abses hepar amoeba
- Proktitis
- Glositis
- Angina Ludwig
- Karies gigi
- Infeksi gusi

Level 3B :

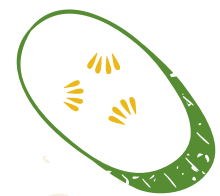
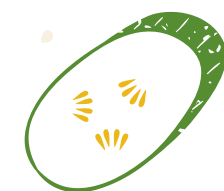
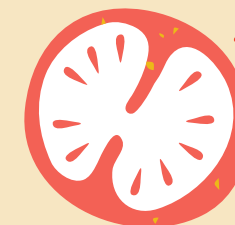
- Perdarahan gastrointestinal
- Kolesistitis
- Pankreatitis
- Peritonitis
- Ileus obstruksi



SKDI 2012

Level 3B :

- Perforasi usus
- Apendisitis akut
- Abses apendiks
- Lesi korosif pada esofagus
- Keracunan makanan
- Keracunan insektisida
- Botulisme



Saluran Pencernaan dan Organ Pendukung

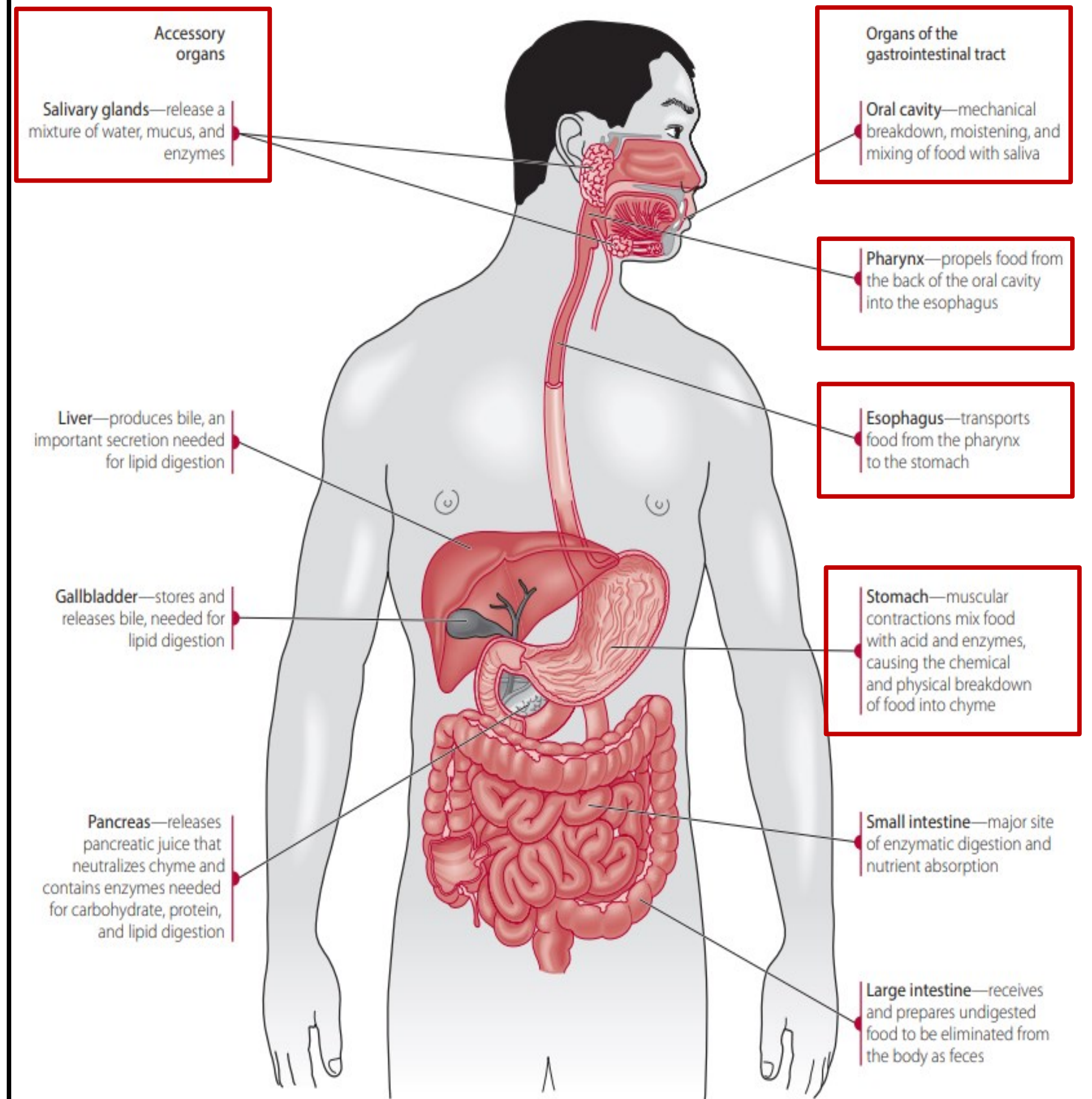


Figure 2.1 The digestive tract and its accessory organs.

Source: Beerman/McGuire, *Nutritional Sciences*, 1/e. © Cengage Learning.

Mechanical and Chemical Digestion

Mechanical Digestion: Gastrointestinal Motility

- Muscles and nerves work together to produce steady motility.

Muscles. provide two general types of movement:

- (1) muscle tone or tonic contraction, which ensures the continuous passage of the food mass and valve control along the way;
- (2) periodic muscle contraction and relaxation, which are rhythmic waves that mix the food mass and move it forward.

Nerves. *intramural nerve plexus* extends from the esophagus to the anus.

- 1) control muscle tone in the wall;
- 2) regulate the rate and intensity of the alternating muscle contractions;
- 3) coordinate all of the various movements.

Mechanical and Chemical Digestion

Chemical Digestion: Gastrointestinal Secretions

Hydrochloric Acid and Buffer Ions

- produce the correct pH that is necessary for enzymatic activity

Enzymes.

- proteins of a specific kind and quantity for breaking down nutrients.

Mucus

- lubricate and protect the mucosal tissues that line the GI tract, and they help to mix the food mass

Water and Electrolytes

- products of digestion are carried and circulated through the GI tract and into the tissues by water and electrolytes

Bile

- divides fat into smaller pieces to expose more surface area for the actions of fat-splitting enzymes

Table 2.1 Digestive Enzymes and Their Actions

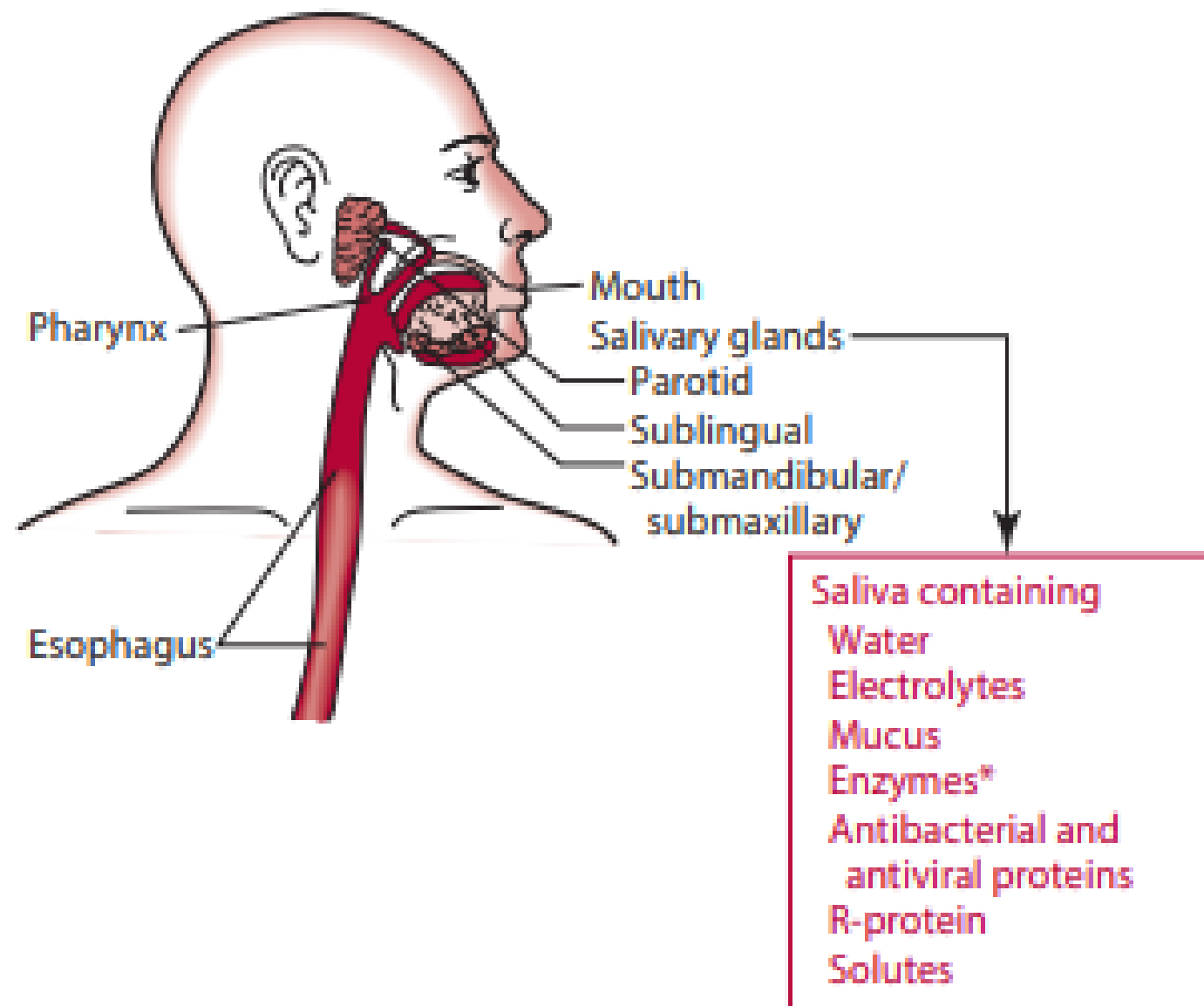
Enzyme or Zymogen/Enzyme	Site of Secretion	Preferred Substrate(s)	Primary Site of Action
Salivary α -amylase	Mouth	α (1-4) bonds in starch, dextrins	Mouth, stomach
Lingual lipase	Mouth	Triacylglycerol	Mouth, stomach
Pepsinogen/pepsin	Stomach	Carboxyl end of phe, tyr, trp, met, leu, glu, asp [®]	Stomach
Gastric lipase	Stomach	Triacylglycerol (mostly medium chain)	Stomach

Secretory cells in the intestinal tract and the accessory organs produce each of the preceding substances for specific jobs in chemical digestion.

The secretory action of these cells or glands is stimulated by the following:

- (1) the presence of food;
- (2) nerve impulses;
- (3) hormonal stimuli.

The mouth and pharynx



*Main enzyme in saliva is salivary amylase, which hydrolyzes α (1-4) bonds in starch.

Figure 2.3 Secretions of the oral cavity.

<u>Mechanical</u>	Teeth and jaw muscles to chew food for ready swallowing
<u>Secretion</u>	Salivary Secretion by Parotid, sub mandibular and sub lingual Glands
<u>Chemical</u>	- salivary α-amylase (ptyalin) to hydrolyze internal bonds (1-4) in starch. - lingual lipase: hydrolyzes dietary triacylglycerols (triglycerides) especially after food is swallowed and in the stomach

Salivary Function

- Saliva lubricates and coats and protects the oral mucosa.
- Antibacterial and antiviral proteins in saliva including IgA antibodies and lysozyme enzymes lyse the cell walls of some bacteria.
- An R-protein in saliva functions in the stomach to enhance the absorption of vitamin B12
- Bicarbonates in saliva help in neutralizing the acids in the food consumed and the acids produced by bacteria inhabiting the oral cavity. (saliva pH 7).

Kandidiasis mulut/Ulkus mulut (aptosa, herpes)

- . Candida albicans is part of the healthy flora in the oral cavity.
- . It can also cause opportunistic infection, which can be deleterious.
- . The most typical type of chronic oral candidiasis is denture stomatitis, and C. albicans is identified as the most crucial organism in this situation.
- . Due to the development of the resistant form of candida, using conventional drugs can sometimes be ineffective.
- . Herbs and naturally imitative bioactive compounds could become a new source for antimycotic therapy.
- . Medical nutrition therapy : Diet liquid, blenderized or soft diet

Disorders of the Esophagus

Heartburn

- Midchest
- Usually occurs after eating and may last for several hours
- Repeated episodes : GERD



Damage Cell

- Edema
- Erosion and ulceration
- Fibrotic tissue -> stricture



Chronic Cough



Sour taste in the Mouth



Reflux oesophagitis

Management

- reduce abdominal fatness
- posture
- reduce gastric fullness (small meals)
- avoid foods which delay gastric emptying
 - fat
- avoid coffee, alcohol

The Stomach

Once the bolus of food has passed through the gastroesophageal sphincter, it enters the stomach

1. Cardia
 2. Fundus
 3. Body
 4. Antrum
- Empty : 50 mL (~2 oz)
 - Filled : 1 L - 1.5 L (~37–52 oz).

These glands include:

1. The cardiac glands, found in a narrow rim at the juncture of the esophagus and the stomach
2. The oxyntic glands, found in the fundus and body
3. The pyloric glands, located primarily in the antrum.

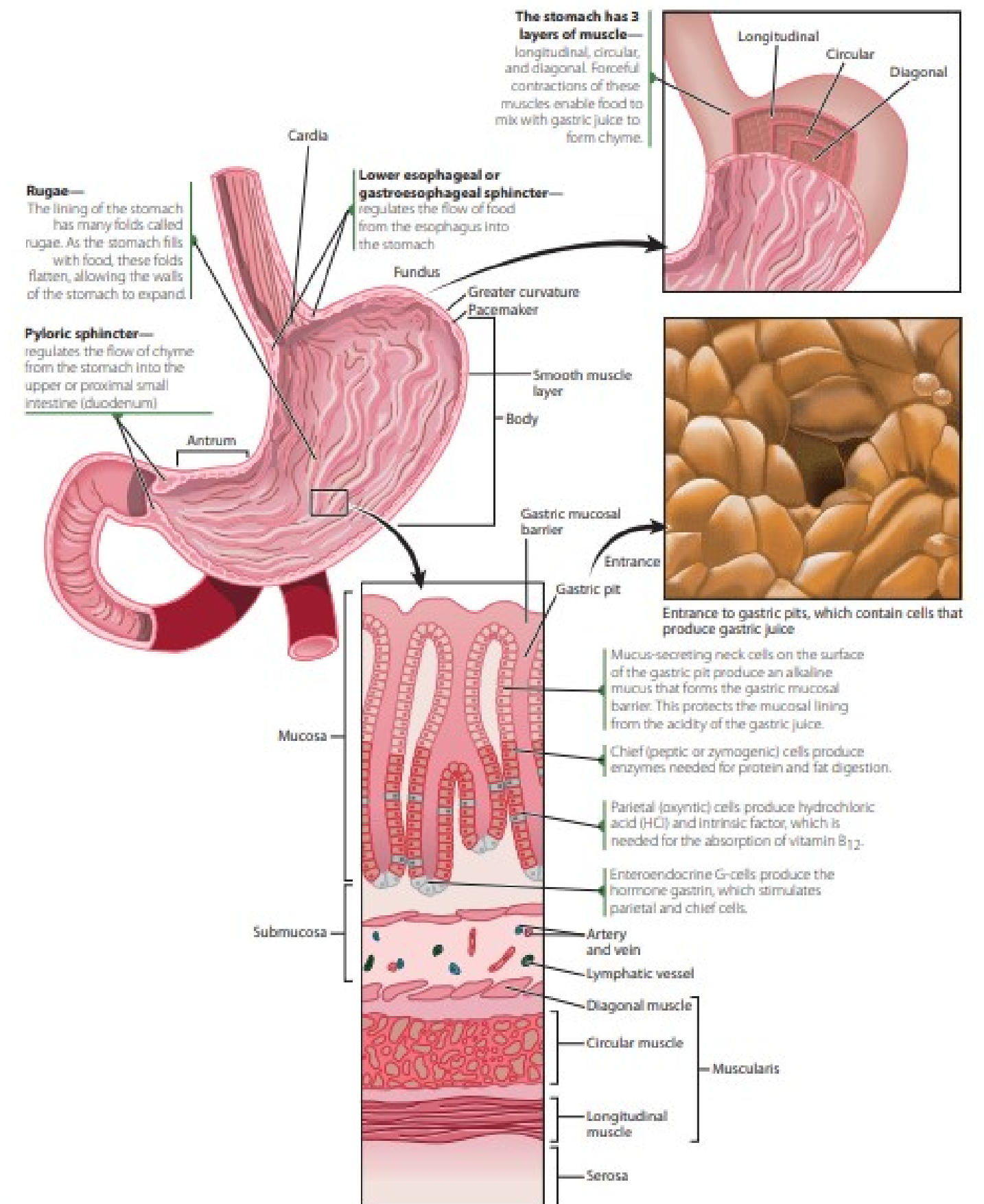
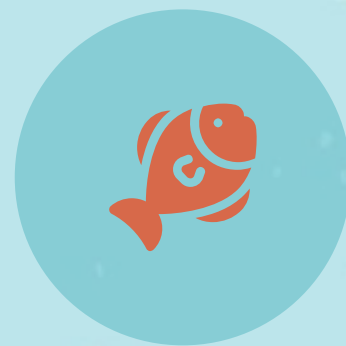


Figure 2.4 Structure of the stomach including a gastric gland and its secretions.
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Gland Secretion

Neck cells

Mucus



Parietal (Oxyntic)
Cells

Hydrochloric acid and intrinsic
factor

Chief (Peptic)
Cells

Pepsinogen dan
Lipase gaster

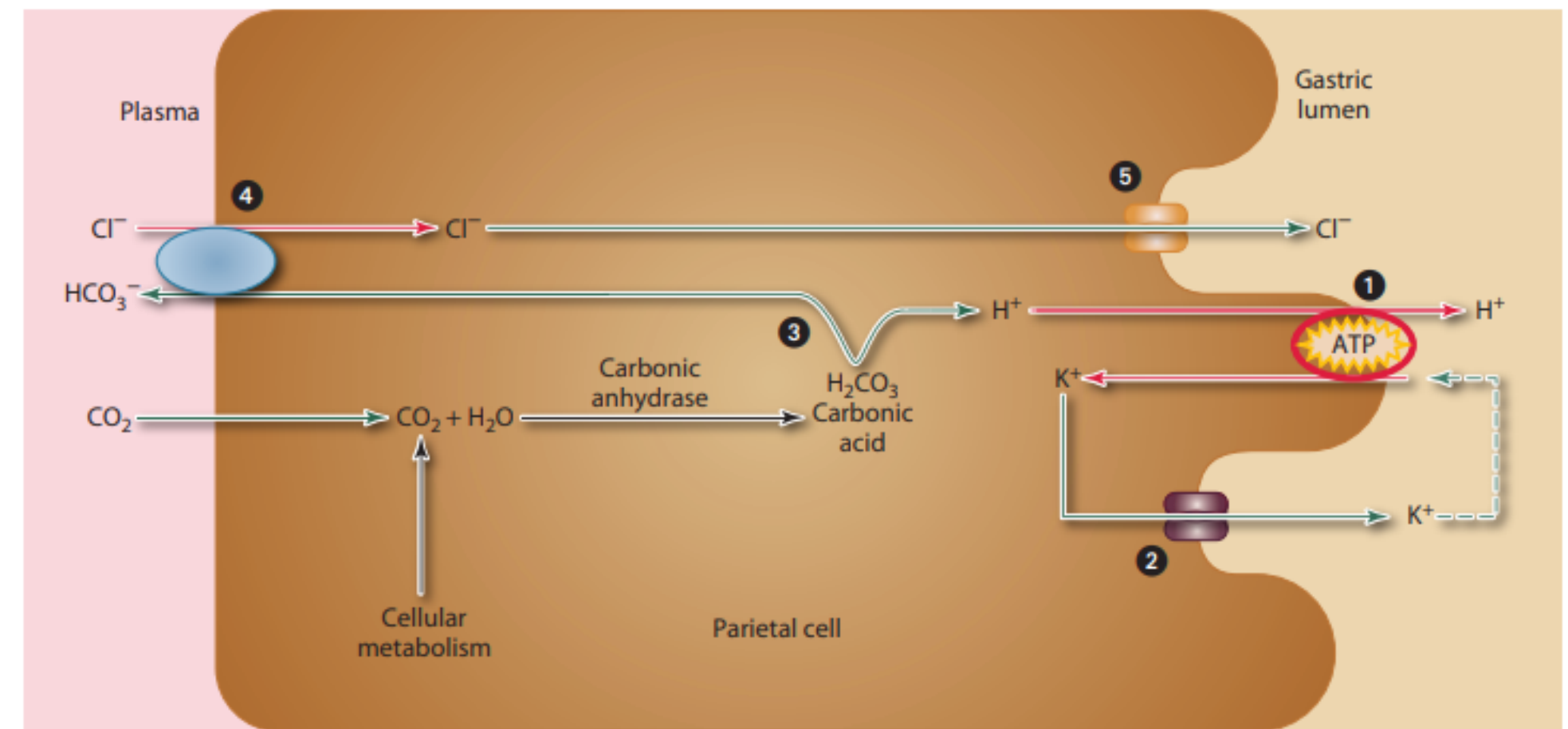


Enteroendokrin
Cells

Hormones



Mechanism of HCl Secretion



- 1 Parietal cells actively secrete hydrogen (H^+) and chloride (Cl^-) by two different transport systems. A hydrogen (proton) potassium ATPase exchange system (H^+ , K^+ -ATPase), also referred to as a proton pump, secretes hydrogens (protons) into the lumen in exchange for potassium ions (K^+) with each ATP molecule hydrolyzed.
- 2 Following the active exchange, the potassium ions typically diffuse out of the parietal cells and back into the lumen.
- 3 The hydrogen arises, along with bicarbonate, from the dissociation of carbonic acid (H_2CO_3). The carbonic acid is generated within the parietal cell from carbonic anhydrase, an enzyme found in high concentrations within parietal cells, using water and carbon dioxide. The water and carbon dioxide are produced within the cell from normal metabolism; the carbon dioxide also may arise in the cell following diffusion from the plasma.
- 4 The chloride ions needed to form hydrochloric acid arise initially from the plasma from which they are transported by a secondary active transport system in exchange for bicarbonate into the parietal cells. This antiporter carries simultaneously the bicarbonate down its concentration gradient into the plasma and the chloride against its concentration gradient into the parietal cell.
- 5 From the parietal cells, the chloride ions then diffuse out via a chloride channel into the gastric lumen joining the hydrogen ions to generate hydrochloric acid.

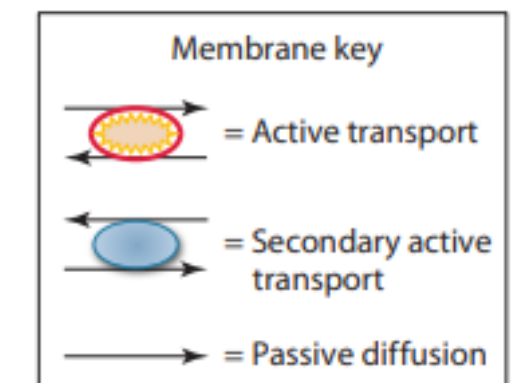


Figure 2.5 Mechanism of HCl secretion.

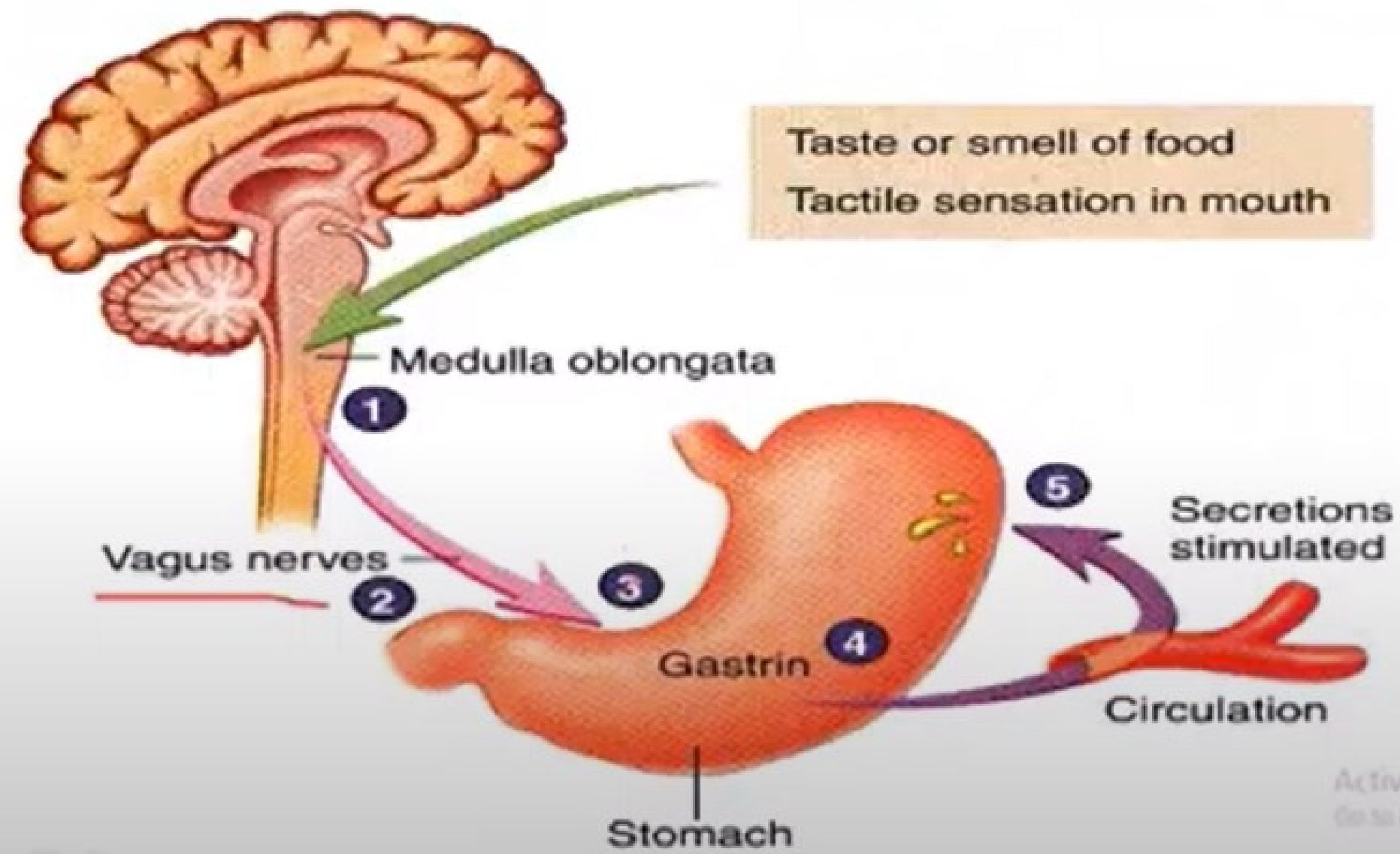
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Functions hydrochloric acid in Gastric Juice

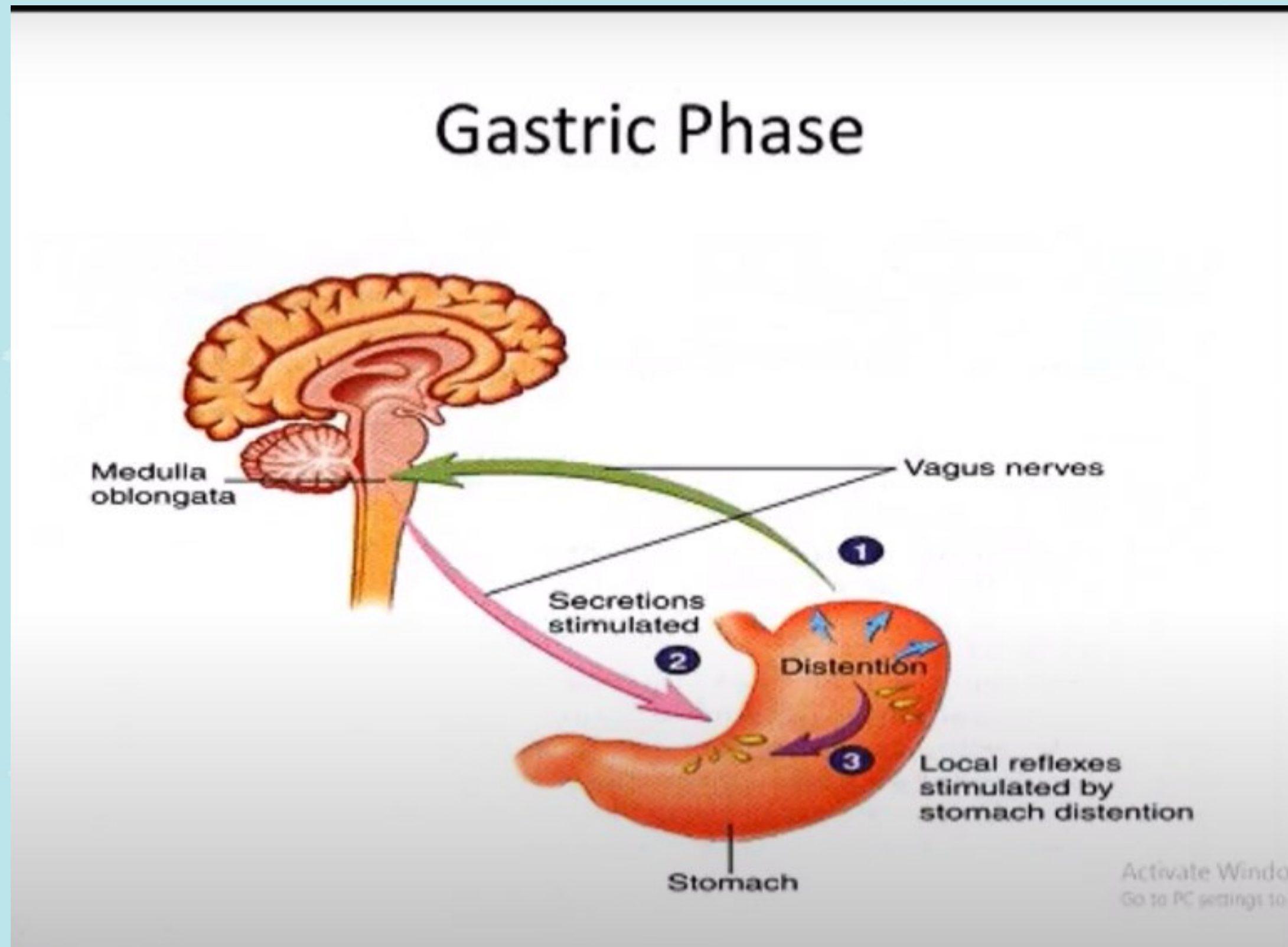


- ✓ Converting or activating the zymogen pepsinogen to form pepsin (needed for protein digestion)
- ✓ Denaturing proteins
- ✓ Releasing various nutrients such as minerals from organic complexes so absorption can occur
- ✓ Acting as a bactericide agent (needed to kill bacteria ingested along with food).

Cephalic Phase

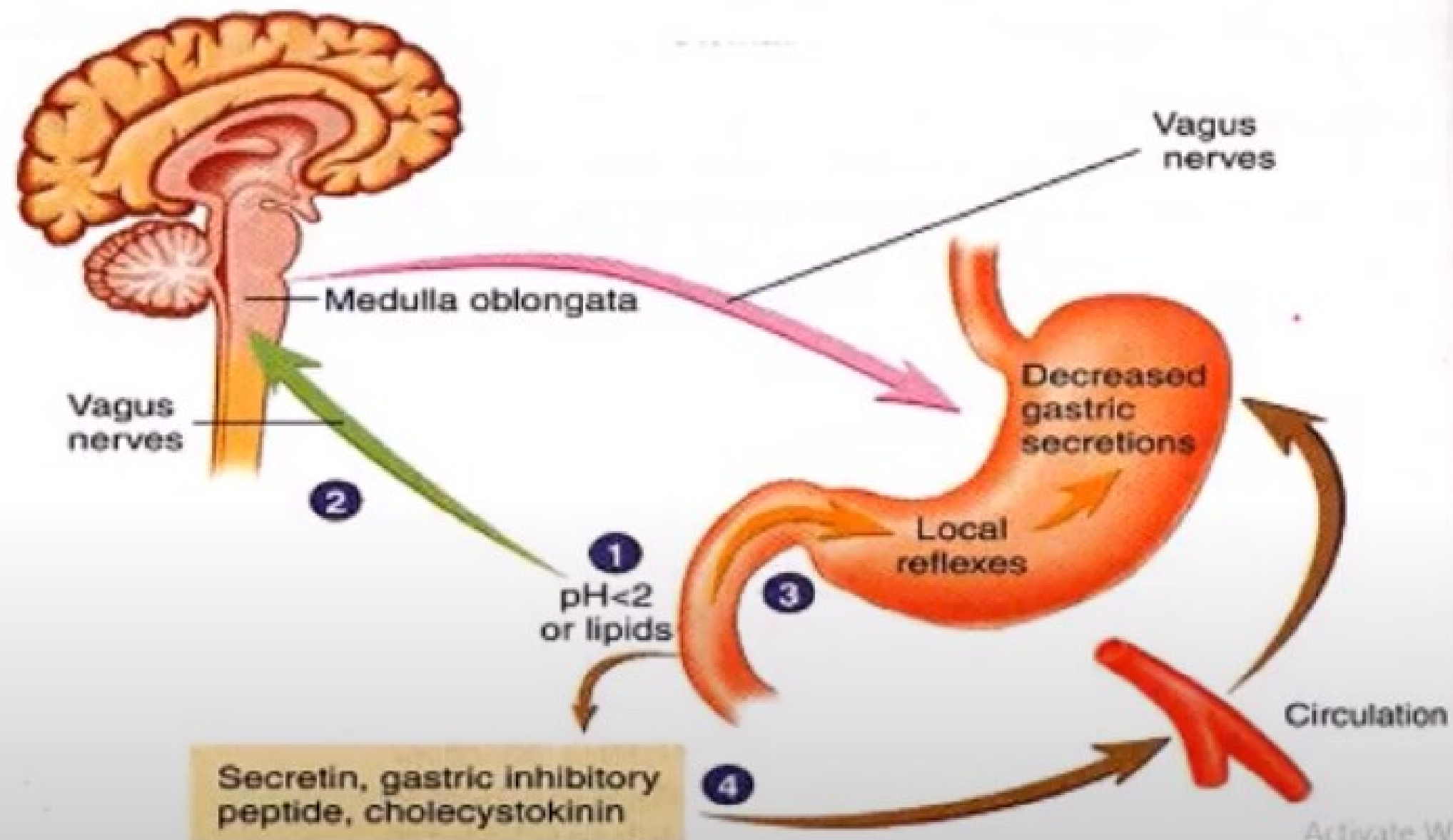


REGULASI SEKRESI GASTRIK



REGULASI SEKRESI GASTRIK

Intestinal Phase



Overview Of Protein Digestion

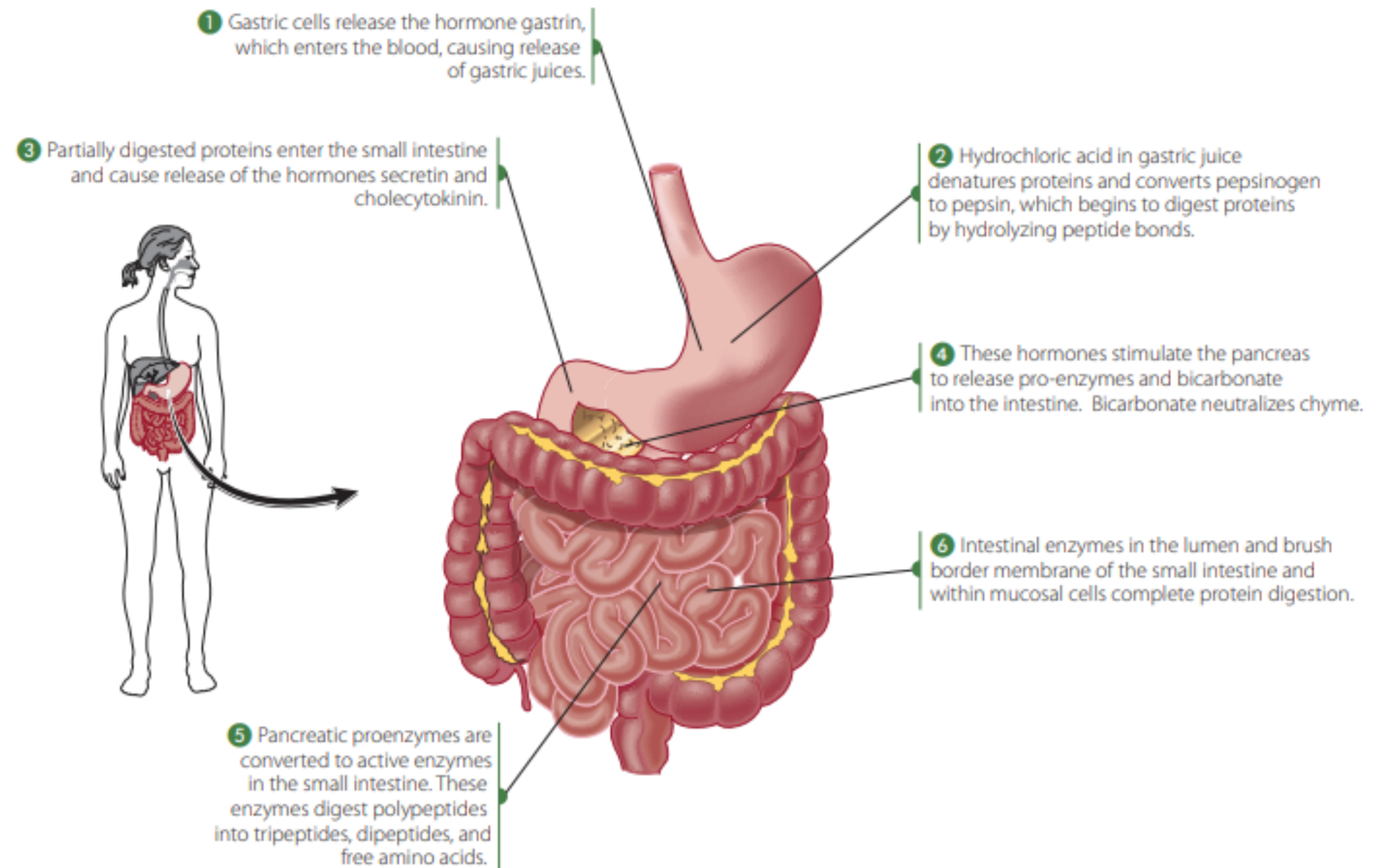


Figure 6.1 An overview of protein digestion.

Source: Derived from Beerman/McGuire, *Nutritional Sciences*, 1/e. © Cengage Learning.

Table 6.3 Major Enzymes Responsible for Protein Digestion

Zymogen	Enzyme or Activator	Enzyme	Site of Activity	Substrate (peptide bonds adjacent to)	End Product(s)
Pepsinogen	HCl or pepsin →	Pepsin	Stomach	Most amino acids, including aromatic, dicarboxylic, leu, met	Peptides
Trypsinogen	Enteropeptidase or Trypsin →	Trypsin	Intestine	Basic amino acids	Smaller peptides, some free amino acids
Chymotrypsinogen	Trypsin →	Chymotrypsin	Intestine	Aromatic amino acids, met, asn, his	Smaller peptides, some free amino acids
Procarboxypeptidases	Trypsin →	Carboxypeptidase	Intestine		
		A		C-terminal neutral amino acids	Free amino acids
		B		C-terminal basic amino acids	Free amino acids
		Aminopeptidases	Intestine	N-terminal amino acids	Free amino acids

MEDICAL NUTRITIONAL THERAPY GASTRITIS DAN GERD

1. To minimize reductions in sphincter pressure, high fat foods as well as chocolate, nicotine, alcohol, and carminatives (volatile oil extracts of plants, most often oils of spearmint and peppermint) should be avoided.
2. Increase gastric acid production (such as alcohol, excessive calcium, and decaffeinated and caffeinated coffee and tea) should also be avoided.
3. Eating smaller and drinking fluids between meals
4. Losing weight (if overweight or obese) and avoiding tight-fitting clothes
5. Avoiding lying down, lifting, or bending for at least 2 hours after eating

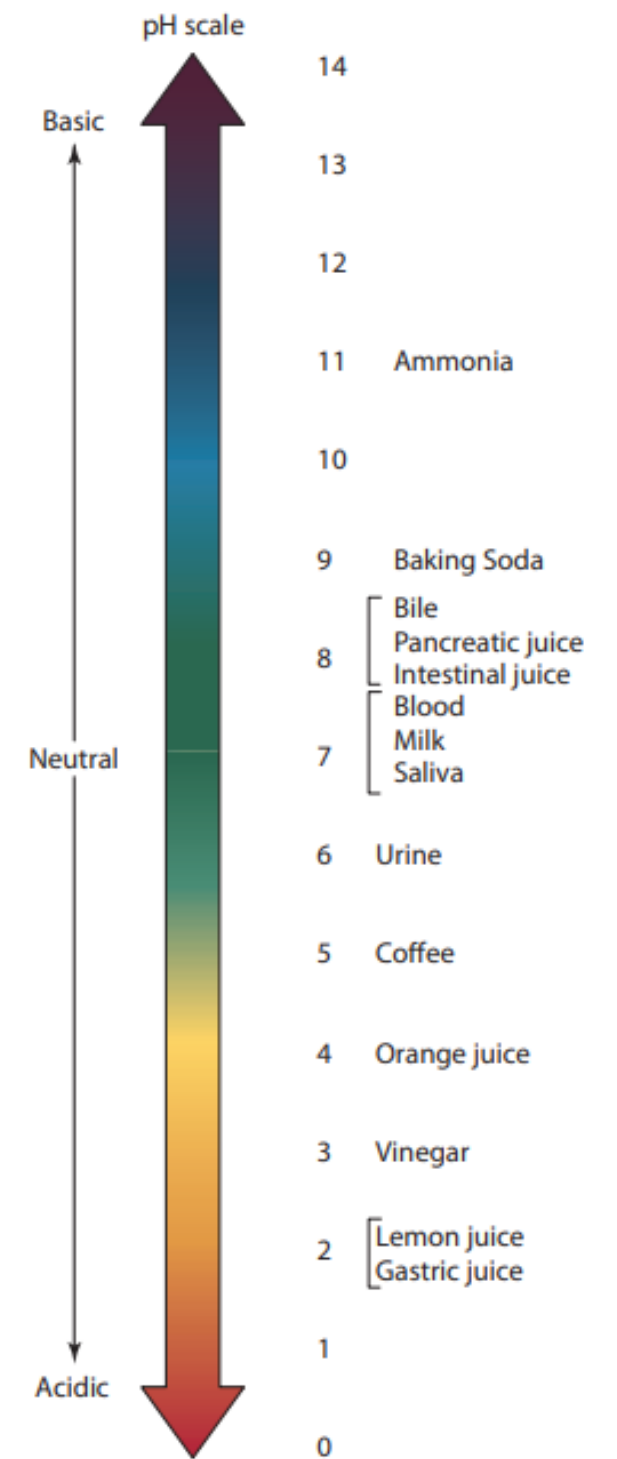


Figure 2.6 Approximate pHs of selected body fluids, compounds, and beverages.

MEDICAL NUTRITIONAL THERAPY GASTRIC ULCER AND GASTRIC BLEEDING

- Diet should be liquid or blenderized -> gastric ulcer
Diet should be stopped -> gastric bleeding

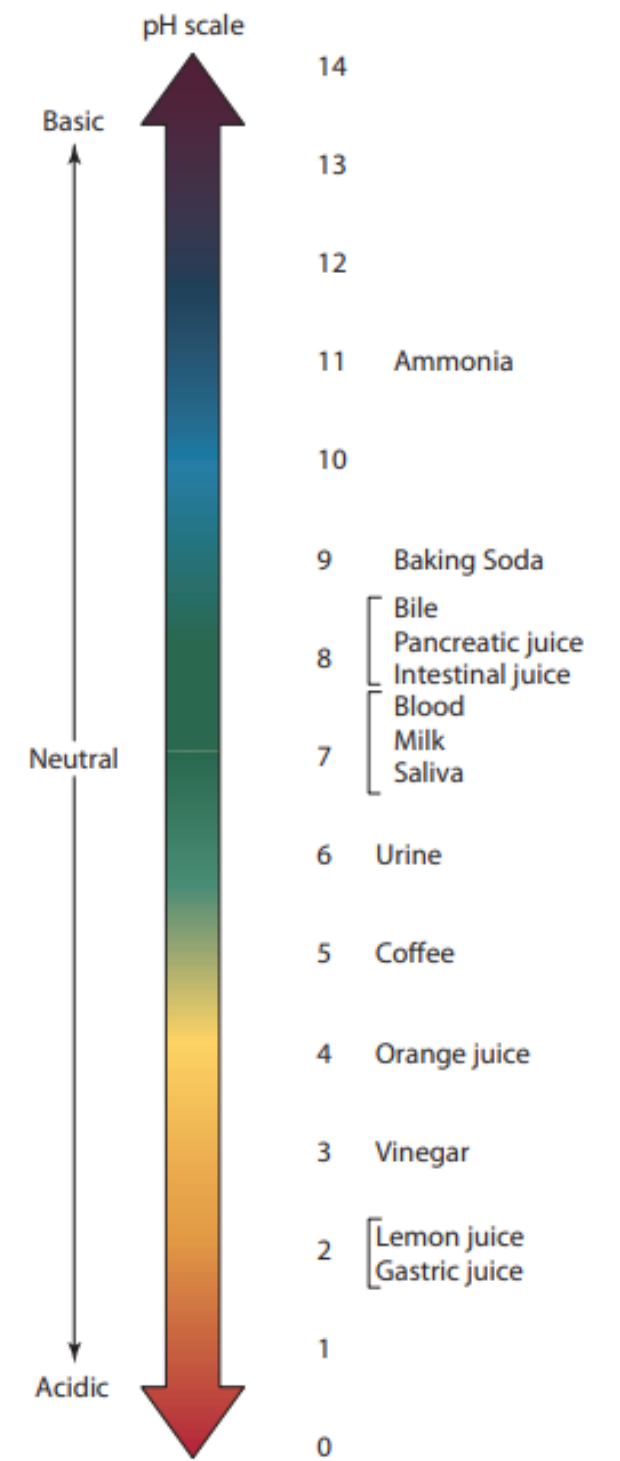


Figure 2.6 Approximate pHs of selected body fluids, compounds, and beverages.

Nutrition Prescription for GERD

- . Initiate weight-reduction program if overweight
- . Initiate smoking cessation (lowers LES pressure)
- . Improve clearing of materials from esophagus
- . Remain upright after eating
- . Avoid eating within 3 hours of bedtime
- . Wear loose-fitting clothing
- . Raise the head of bed for sleeping

Nutrition Prescription for GERD

Reduce gastric acidity by eliminating the following:

- . Black and red pepper
- . Coffee (caffeinated and decaffeinated)
- . Alcohol

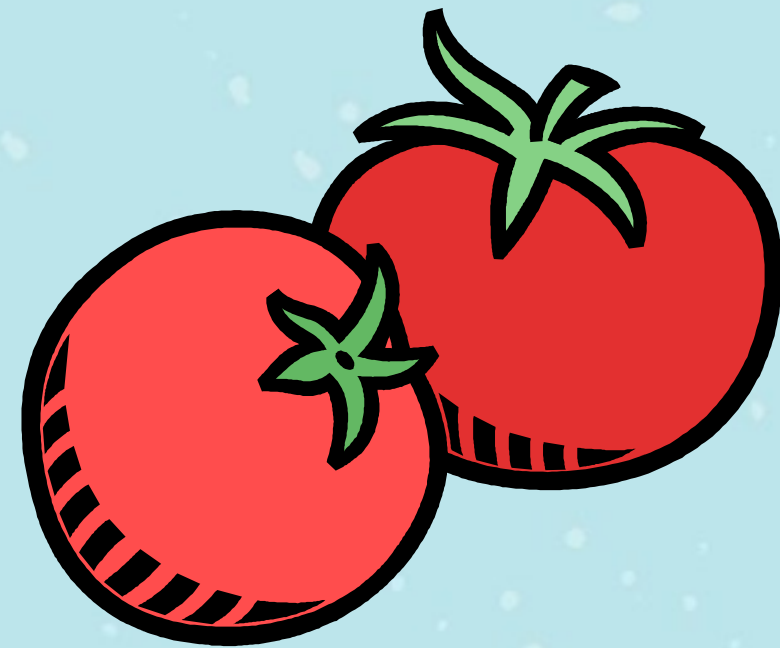
Substitute smaller more frequent meals

Restrict foods that lessen lower esophageal sphincter pressure by eliminating the following:

- . Chocolate
- . Mint
- . Foods with a high fat content.

Nutrition Prescription for GERD

- . Spicy, acidic foods may be irritating if esophagitis is present
- . Limitation of these foods should be based on individual tolerance



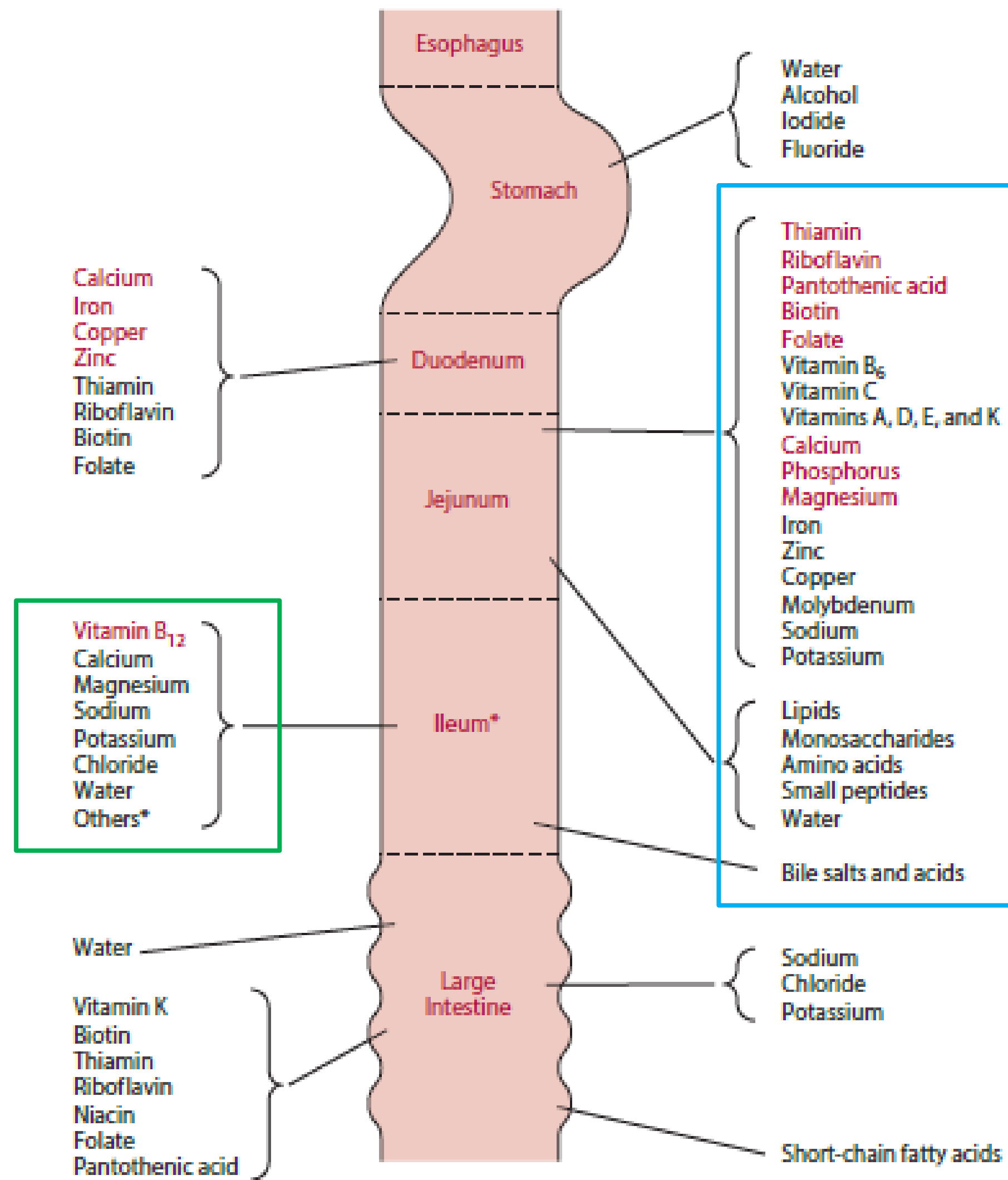
The Absorptive Process

”

- Following all the actions of the various secretions and enzymes from the mouth, stomach, pancreas, and small intestine and with the help of bile from the liver and gallbladder, the now digested nutrients are ready to be absorbed, that is, to enter the cells of the gastrointestinal tract

The absorption of most nutrients begins in the duodenum and continues throughout the jejunum and ileum

Generally, most absorption occurs in the proximal (upper) portion of the small intestine, but some nutrients are absorbed primarily in the distal ileum





The digestion and absorption of nutrients within the small intestine **are rapid**, with most of the carbohydrate, protein, and fat being absorbed within a few hours after chyme has reached the small intestine.

- Nutrients are absorbed across the brush border membrane into enterocytes primarily by diffusion, facilitated diffusion, active transport, and/or, occasionally, pinocytosis or endocytosis
- A few nutrients, however, gain entry into the body via a paracellular (between cells) route

The mechanism of absorption for a nutrient depends on several factors :

- solubility (fat versus water) of the nutrient
- the concentration or electrical gradient
- the size of the molecule to be absorbed.



Usus Halus mempunyai 2 fungsi :

01

PENCERNAAN

Proses Pencernaan disempurnakan sejumlah enzim dalam getah usus (sulus Enterikus)

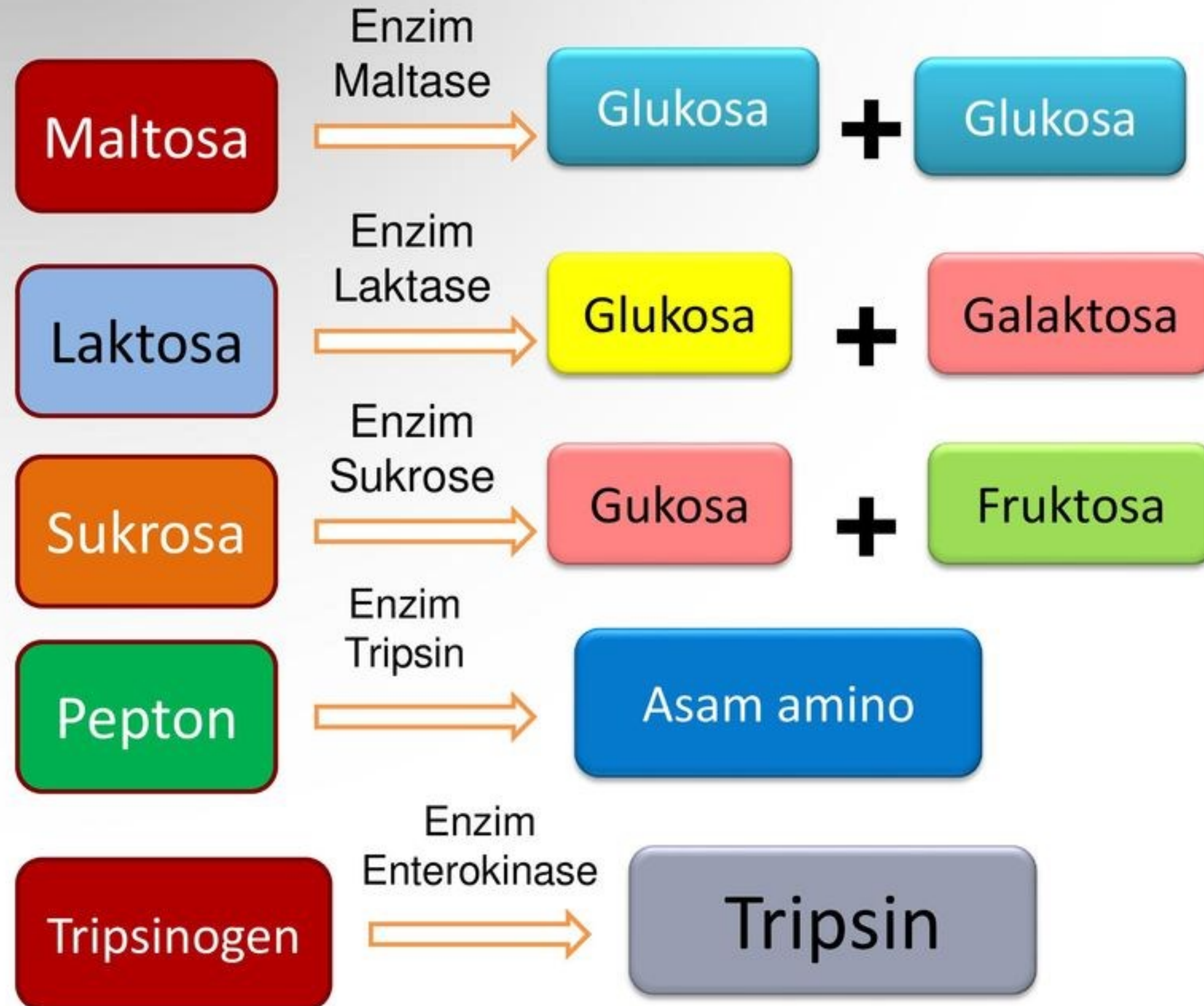
02

ABSORBSI

nutrisi dan air



Enzim – enzim dari usus halus



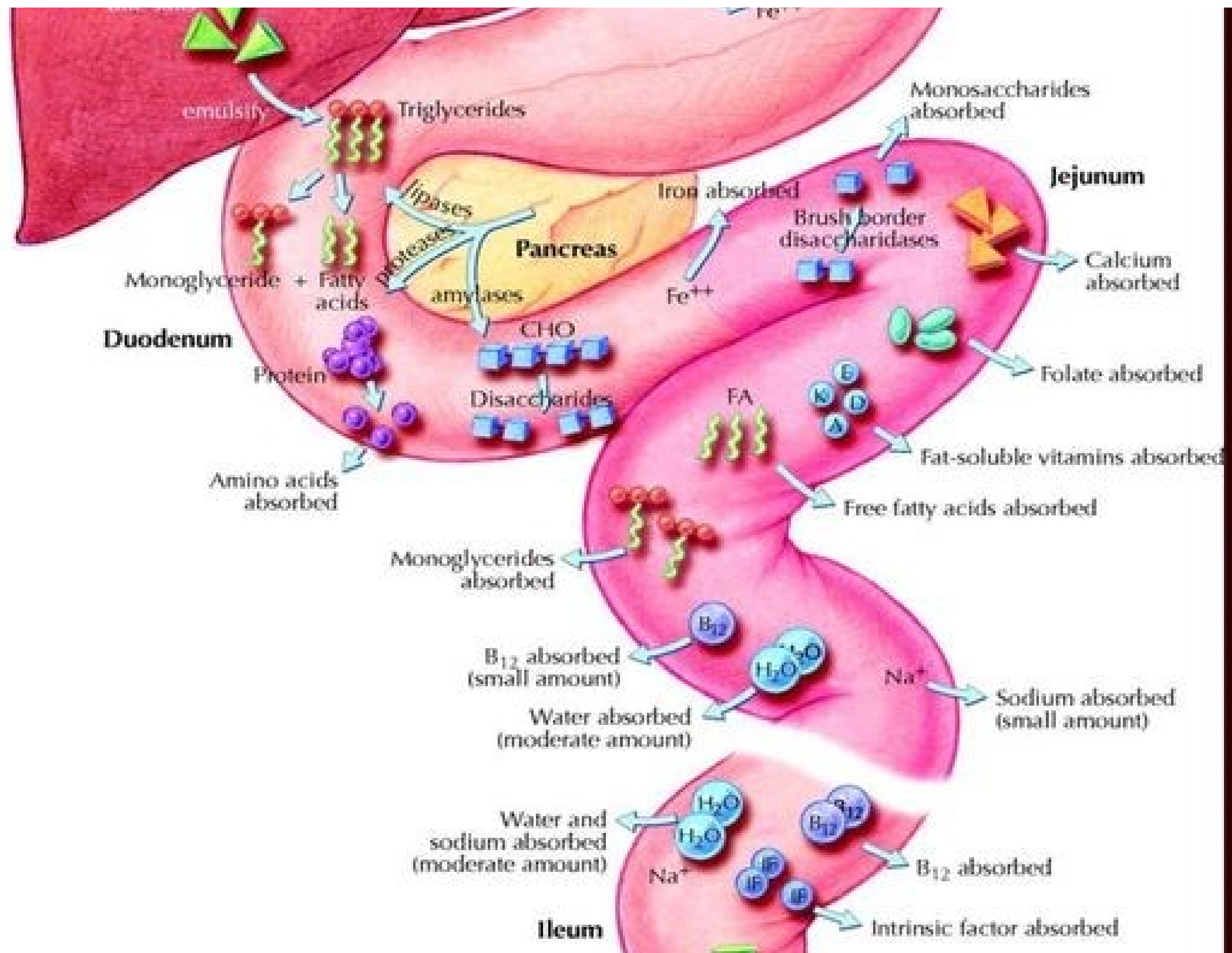
Usus Halus Tidak mensintesis enzim Pencernaan

Pencernaan di dalam lumen usus halus oleh---Enzim Pankreas

Pencernaan Lemak selesai ----- dalam Lumen usus Halus

Pencernaan Karbohidrat dan protein selesai ----- Brush Border

semua produk pencernaan akan diserap oleh usus halus kecuali penyerapan kalsium dan besi
yang disesuaikan kebutuhan Tubuh



<http://www.6/10/1297>

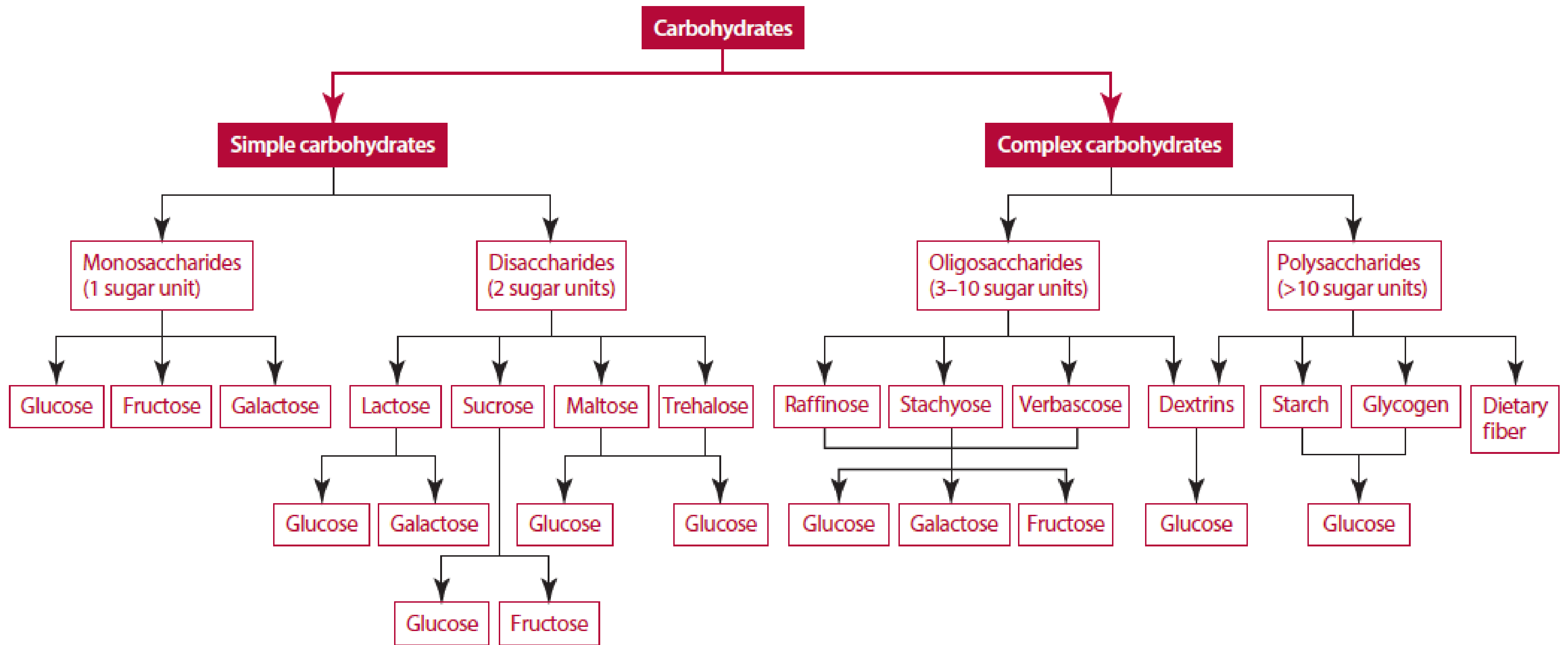
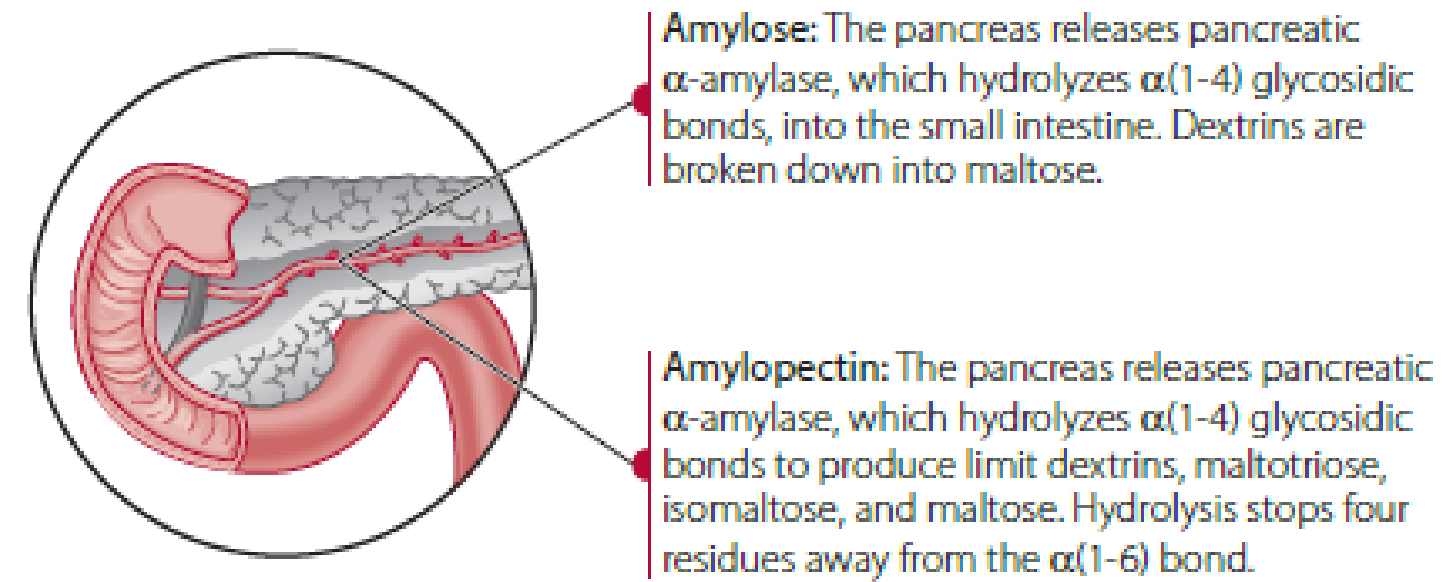
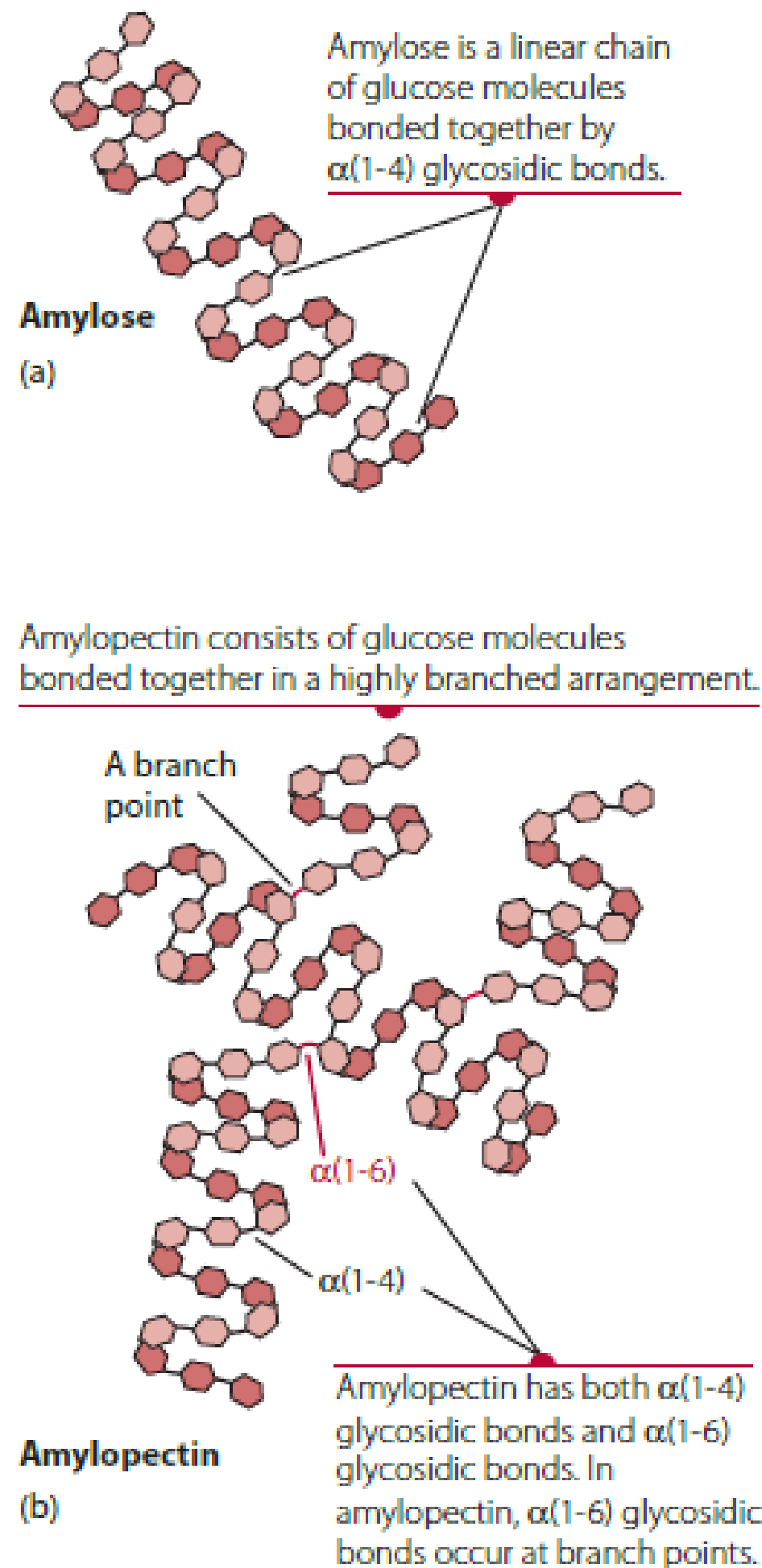


Figure 3.1 Classification of carbohydrates.

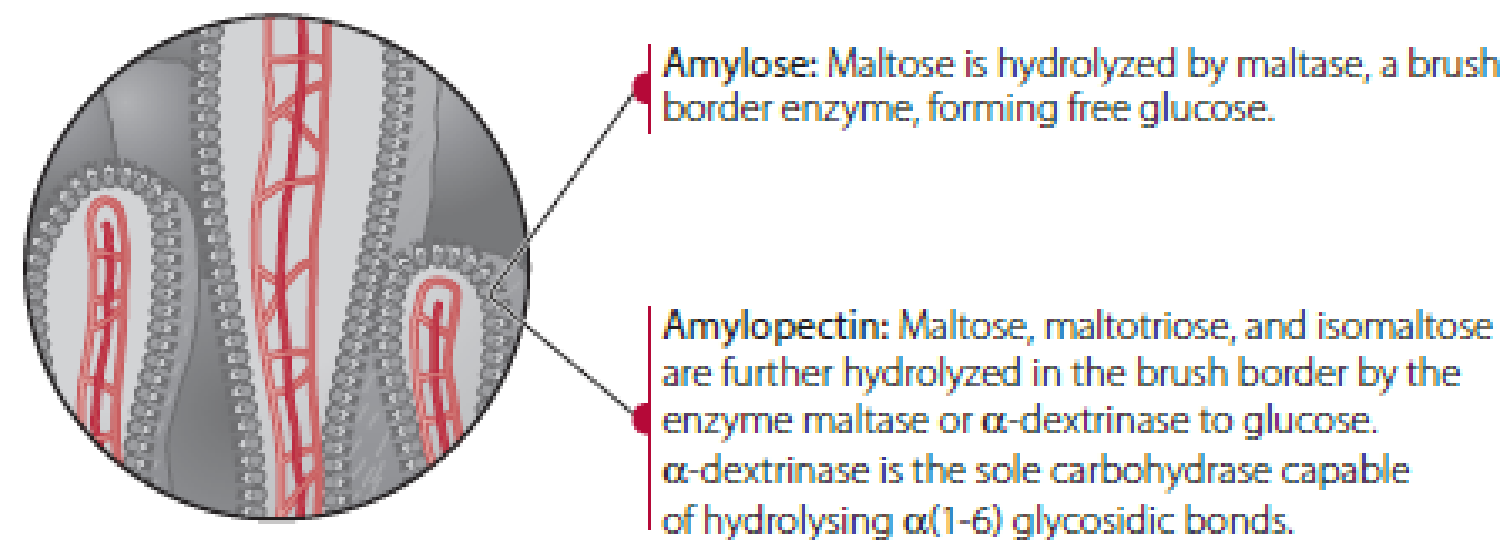
Source: Beerman/McGuire, *Nutritional Sciences*, 1/e. © Cengage Learning.

Poly-, tri-, and disaccharides must be hydrolyzed (enzymes involved are collectively called **glycosidases** or, alternatively, **carbohydases**)

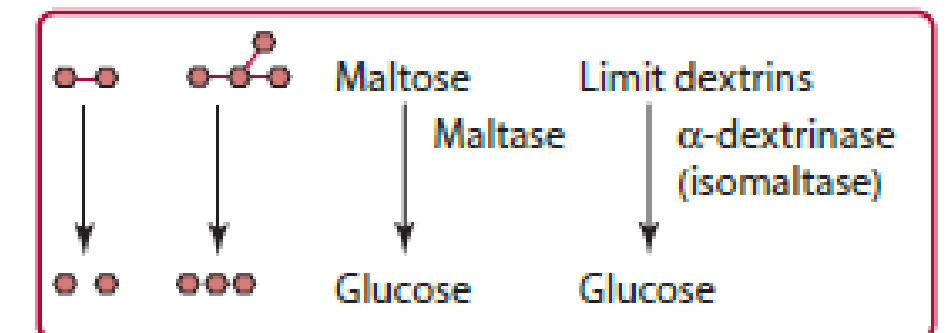
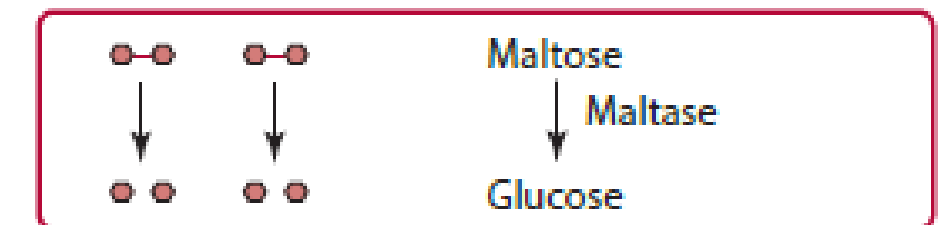
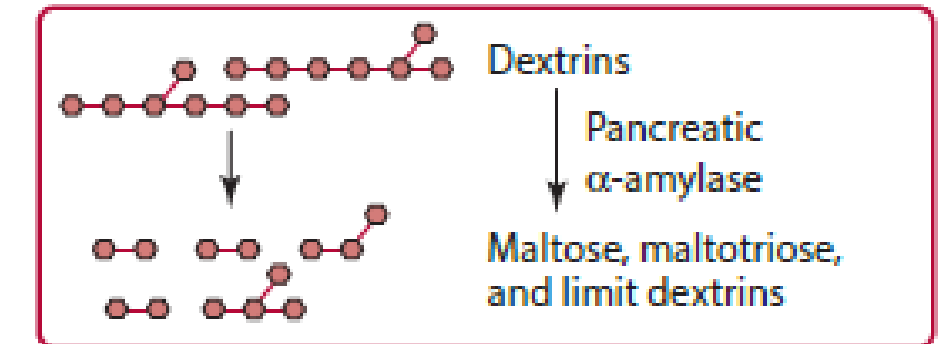
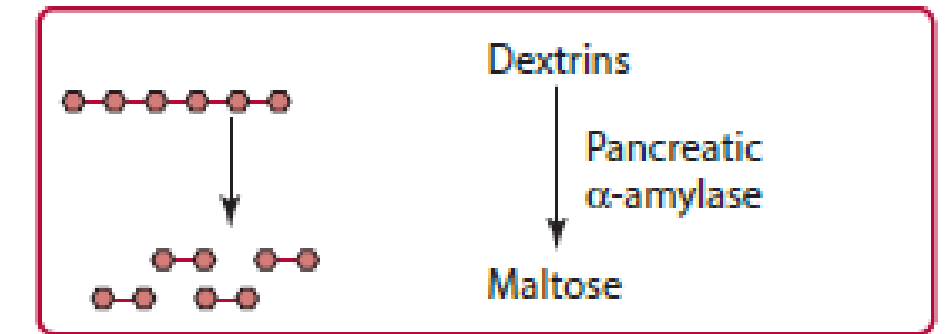
Polysaccharides



C. Digestion of amylose and amylopectin in the small intestine



D. Digestion of amylose and amylopectin on the brush border of the small intestine



Maltotriosa $\rightarrow$ Maltosa + glucose

Disaccharides

Digestion of disaccharides takes place almost entirely within the microvilli (the brush border) of the upper small intestine via **disaccharidase** activity $\rightarrow$ resulting monosaccharides $\rightarrow$ enter the enterocytes with the facilitation of specific transporters

From intestinal lumen → enterocytes

After carbohydrate digestion, glucose and galactose are absorbed into the enterocyte by the same mechanisms involving both active transport and facilitated transport

ACTIVE TRANSPORT
glucose and galactose
absorption via:

1. specific transporter protein, designated *sodium-glucose transporter 1 (SGLT1)* - requires energy as ATP

Passive FACILITATED
diffusion transport:
fructose, via:

1. **GLUT 5:**
 - does not require energy

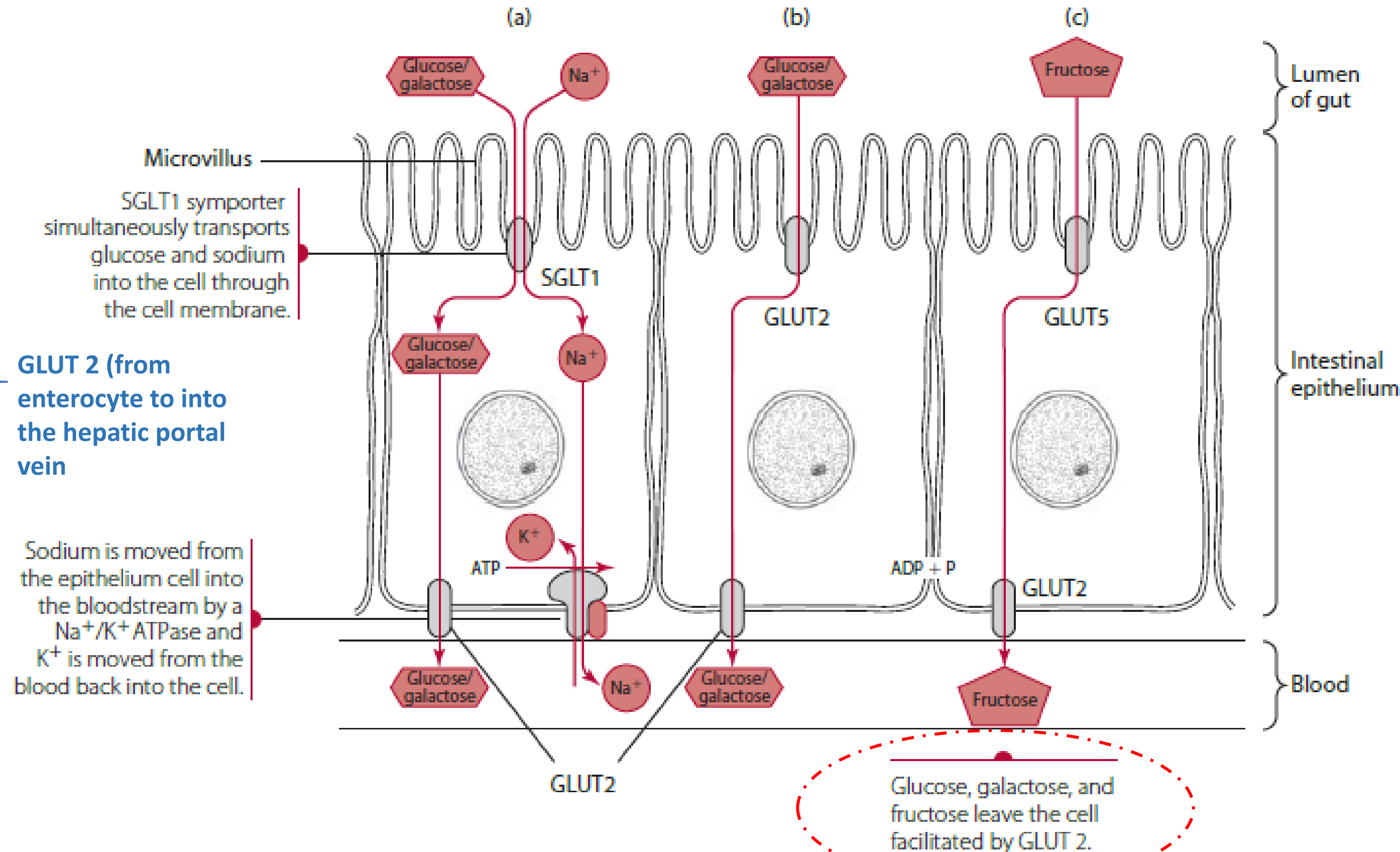
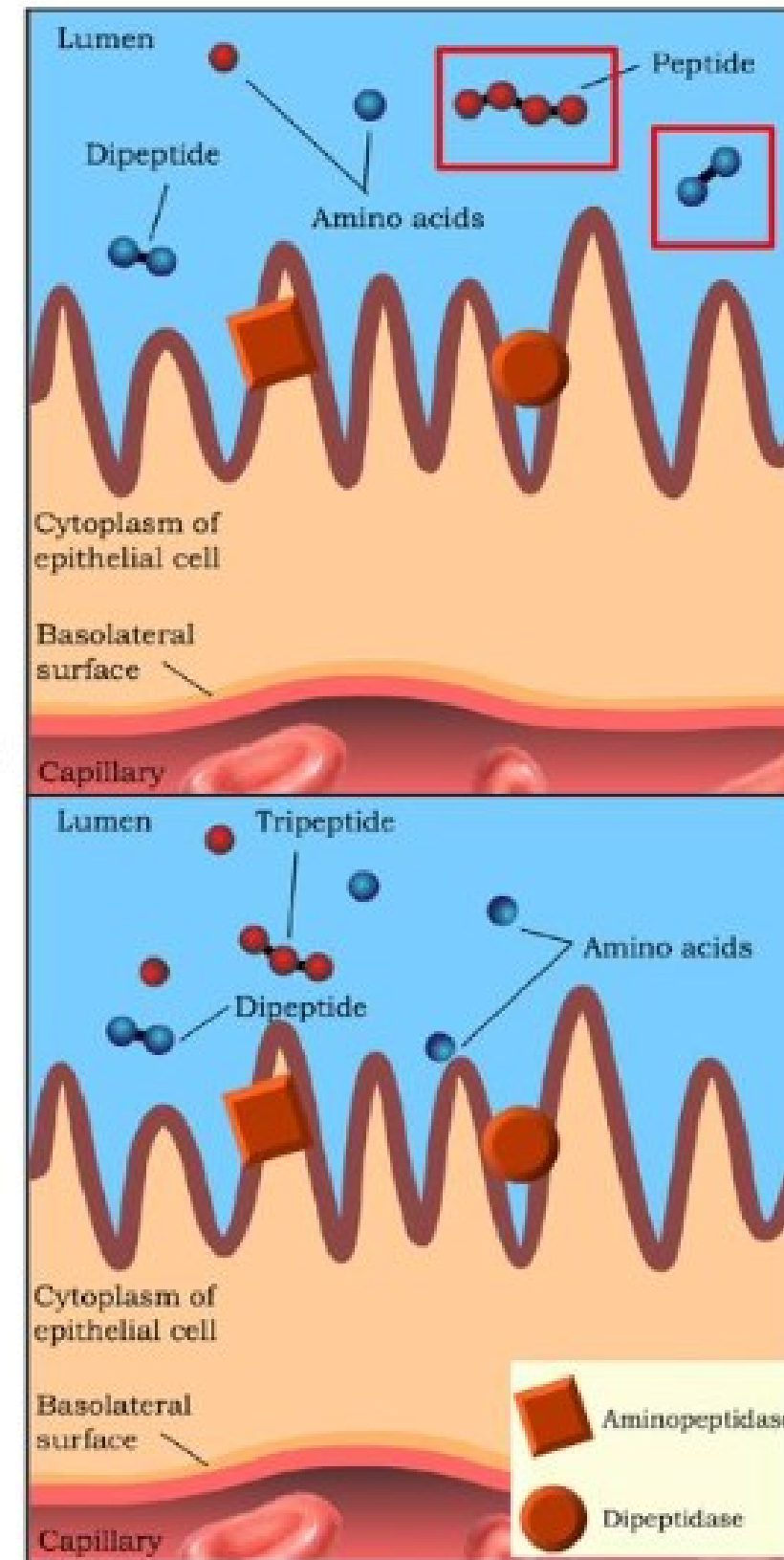
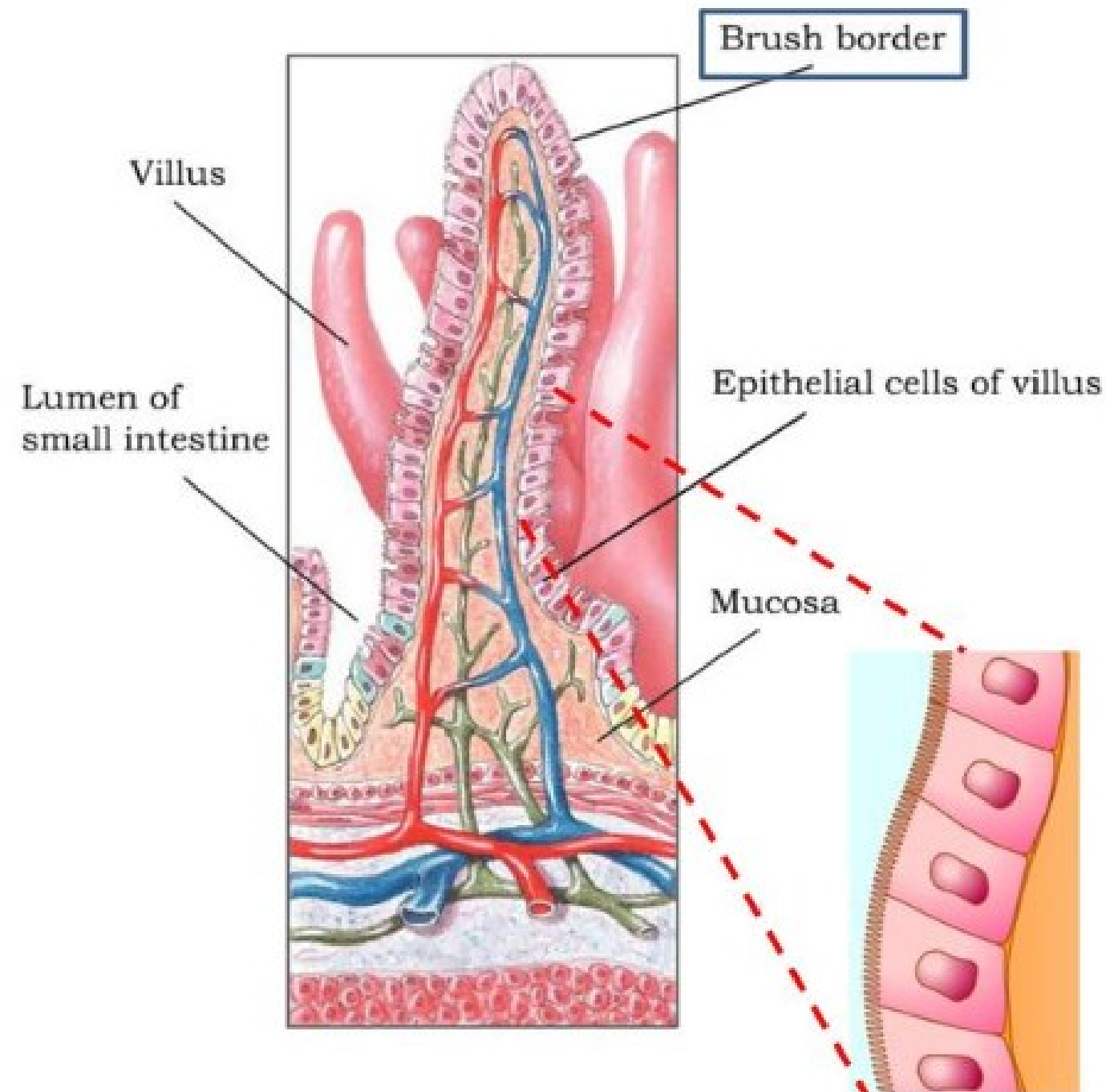


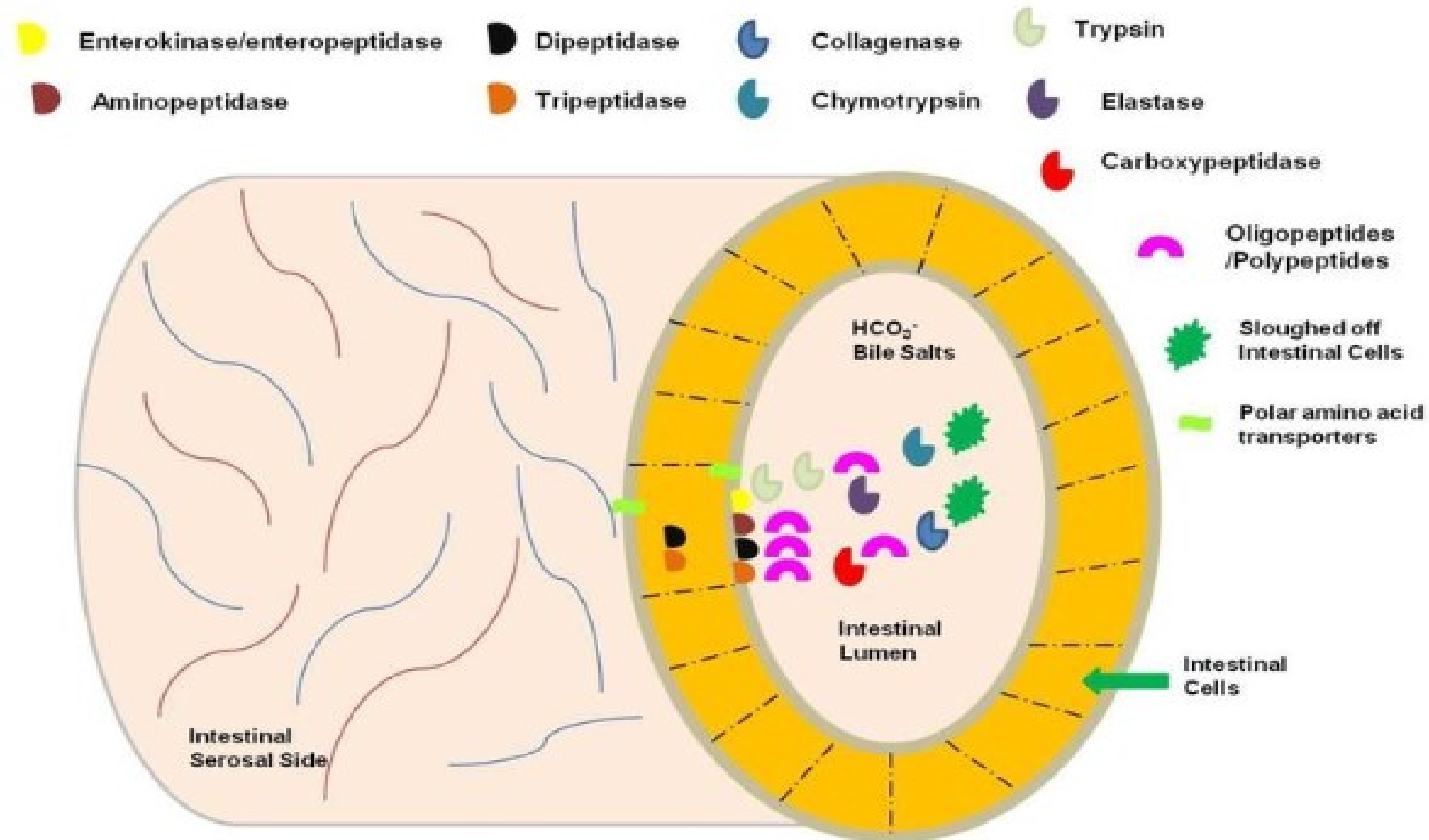
Table 3.2 Glucose Transporters (GLUT)

Transporter Protein	Substrates	Major Sites of Expression
GLUT1	Glucose, galactose, mannose, glucosamine	Erythrocytes, central nervous system, blood–brain barrier, placenta, fetal tissues in general
GLUT2	Glucose, galactose, fructose, mannose, glucosamine	Liver, β -cells of pancreas, kidney, small intestine
GLUT3	Glucose, galactose, mannose, xylose, dehydroascorbic acid	Brain (neurons), spermatozoa, placenta, preimplantation embryos
GLUT4 (insulin dependent)	Glucose, glucosamine, dehydroascorbic acid	Muscle, heart, brown and white adipocytes
GLUT5	Fructose, but not glucose	Intestine, kidney, brain, skeletal muscle, adipose tissue

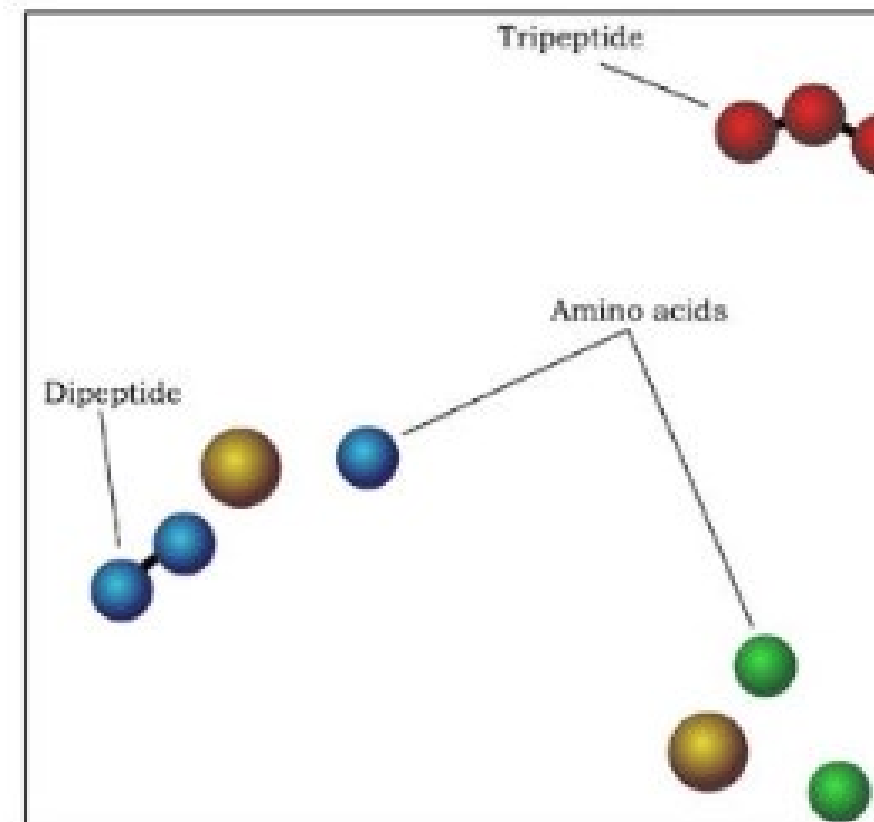
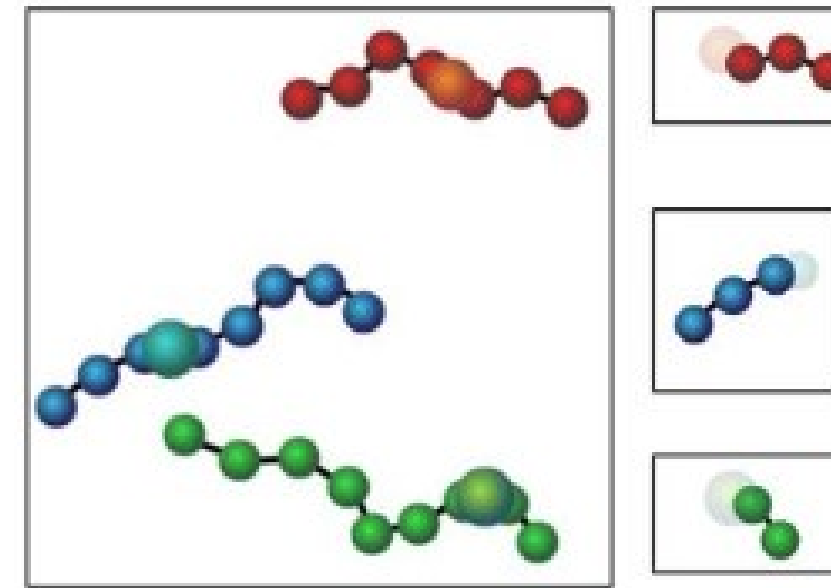
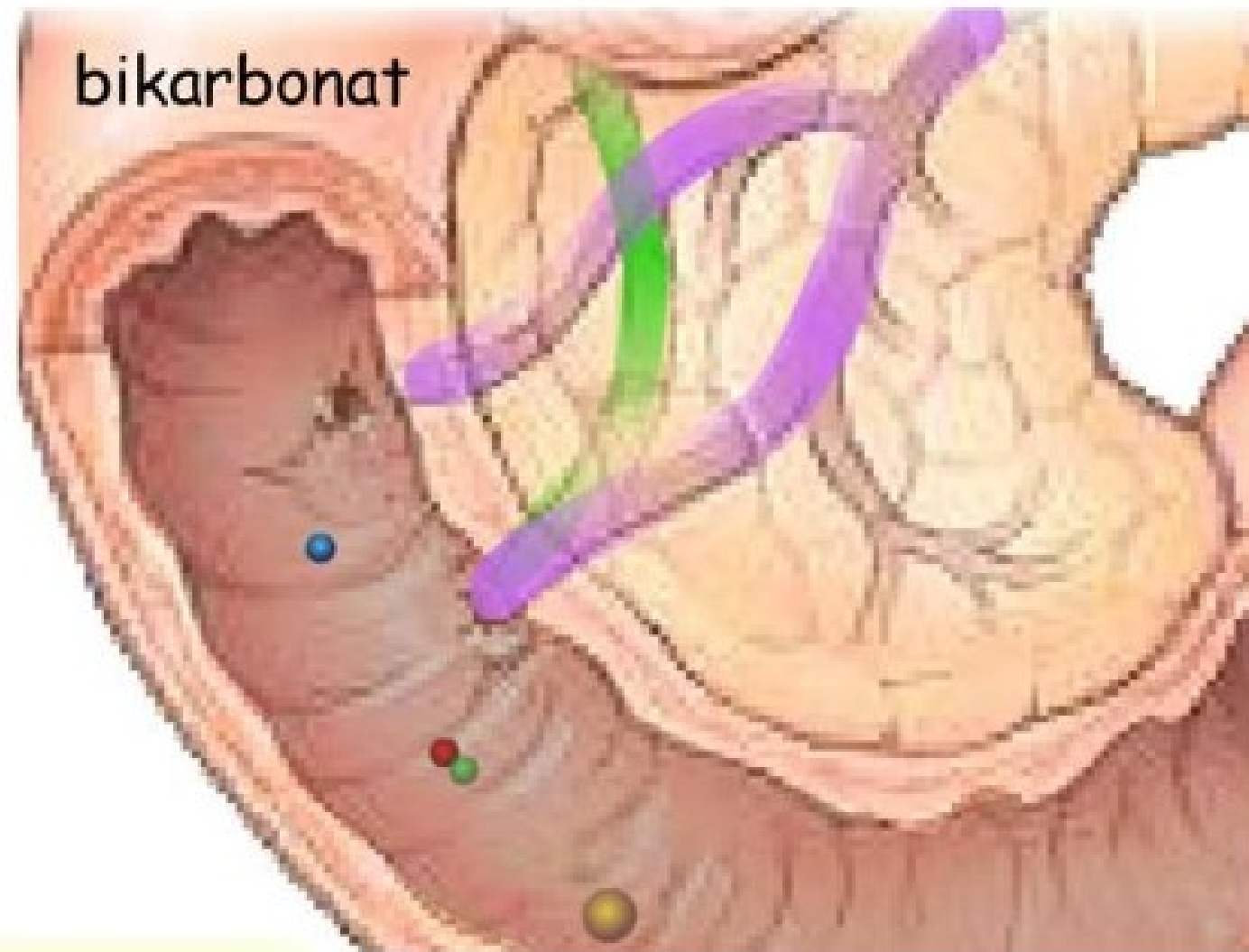
Pencernaan Protein di Usus Halus



Enzim permukaan sel usus halus



Pencernaan Protein oleh Enzim Pankreas



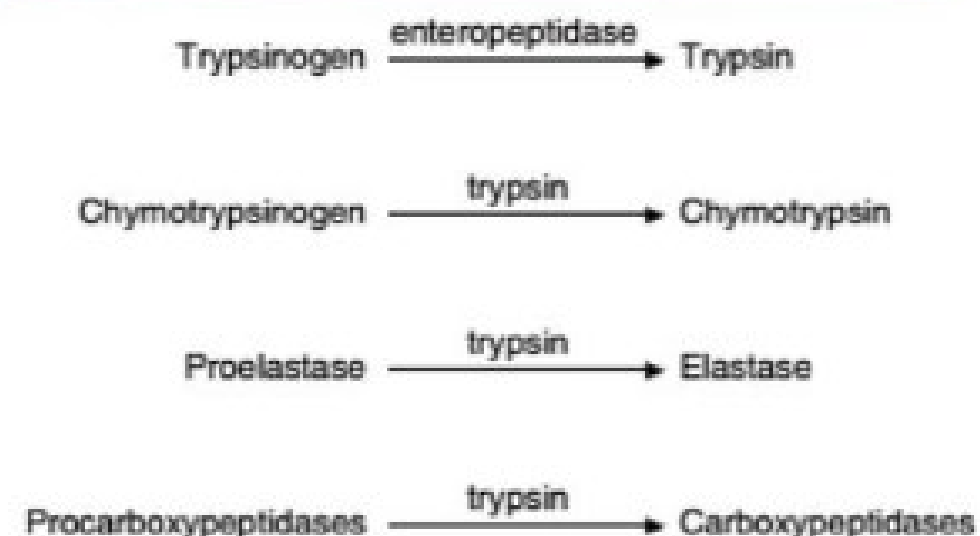
- Chyme → duodenum → stimulates the release of secretin and cholecystokinin hormones → stimulated to secrete zymogens → responsible for protein and polypeptide digestion

Enzyme proteolitik aktif

Protein digestion yields two main end products:

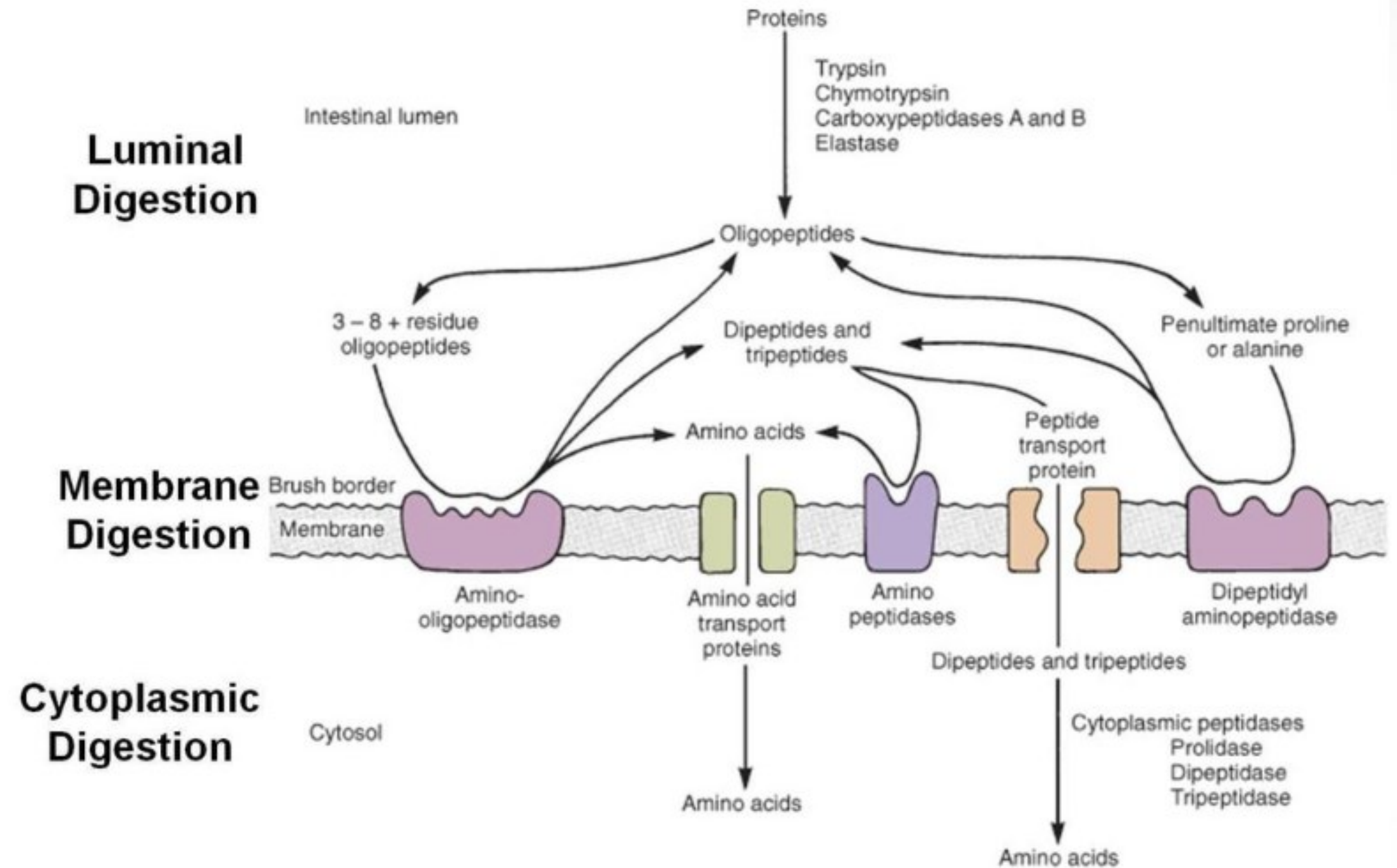
→ peptides (principally dipeptides and tripeptides) and free amino acids
→ must be absorbed.

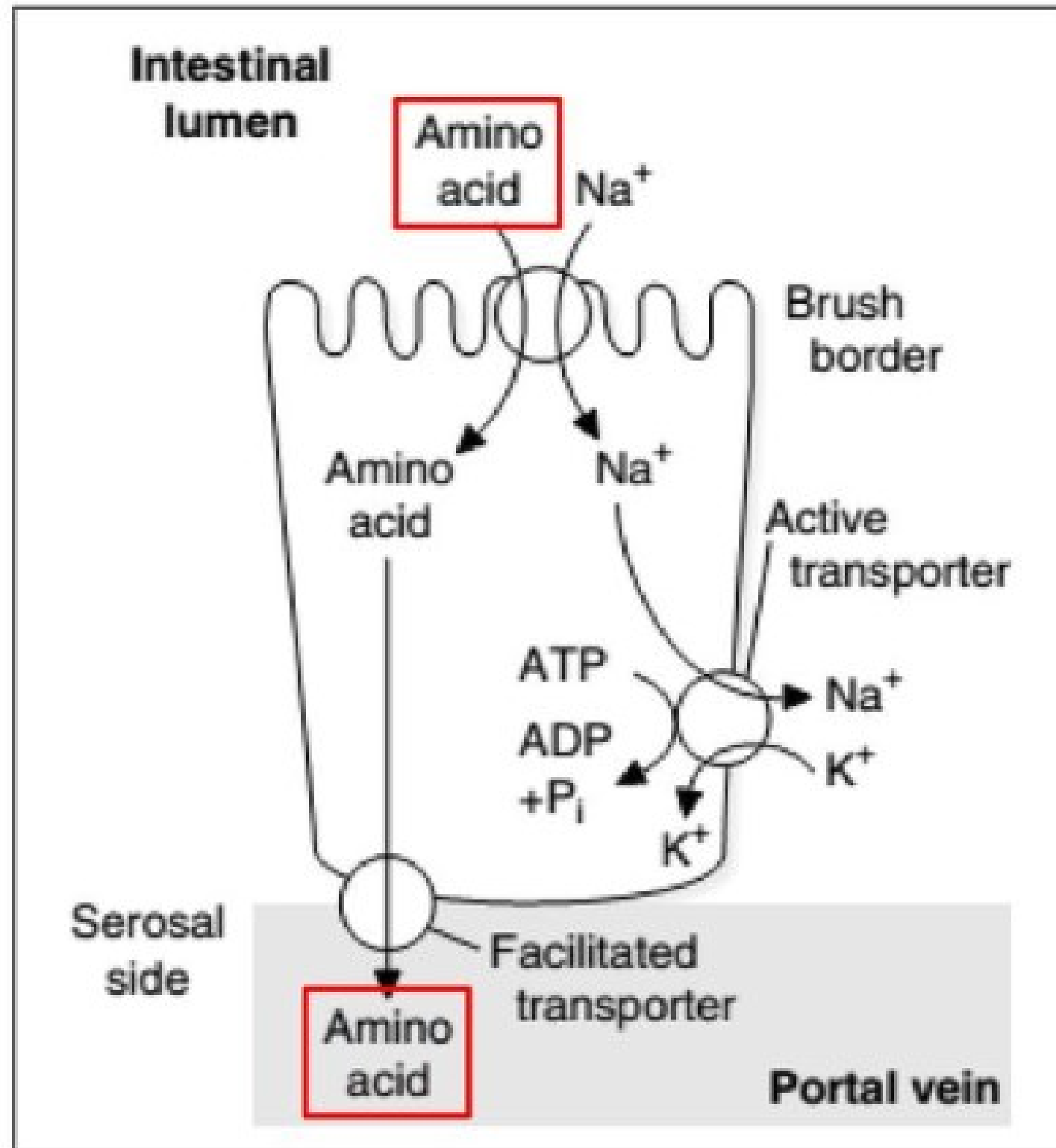
- Trypsin
- Chymotrypsin
- Elastase
- Carboxypeptidase



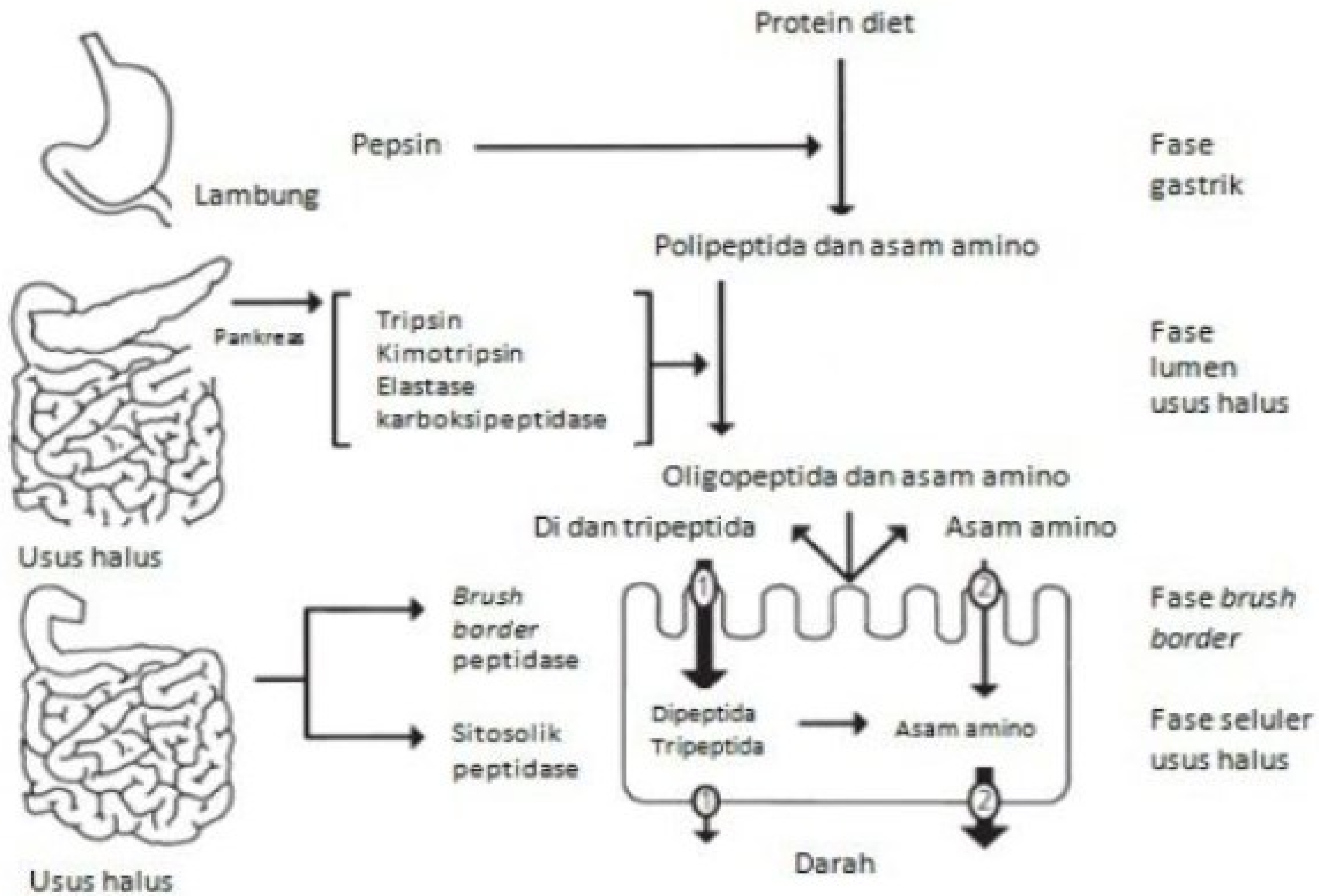
ABSORPTION

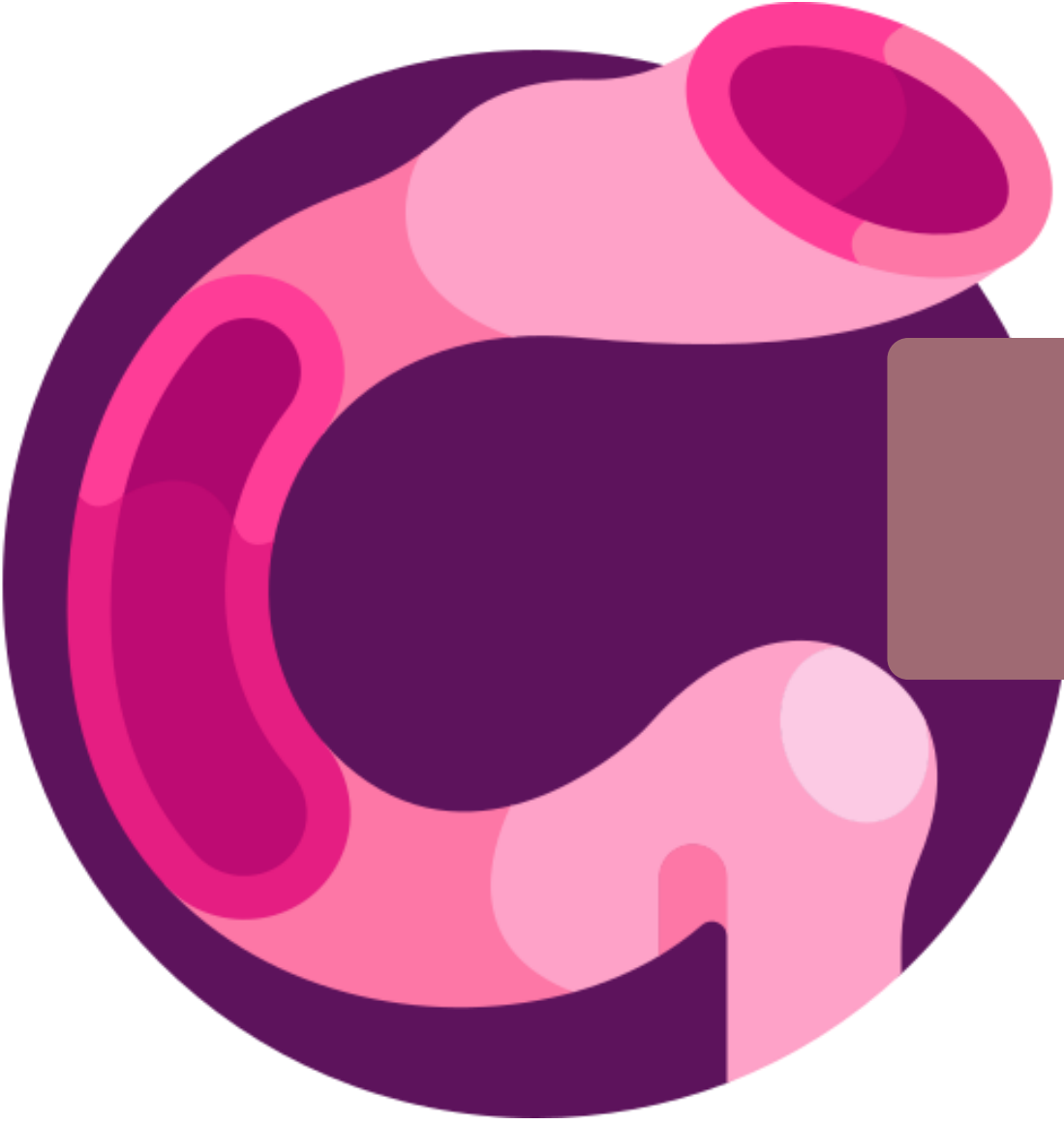
- To get into the blood for transport to tissues, amino acids must cross two intestinal membranes
1. Absorption across the Intestinal Brush Border Membrane
 2. Absorption across the Intestinal Basolateral Membrane



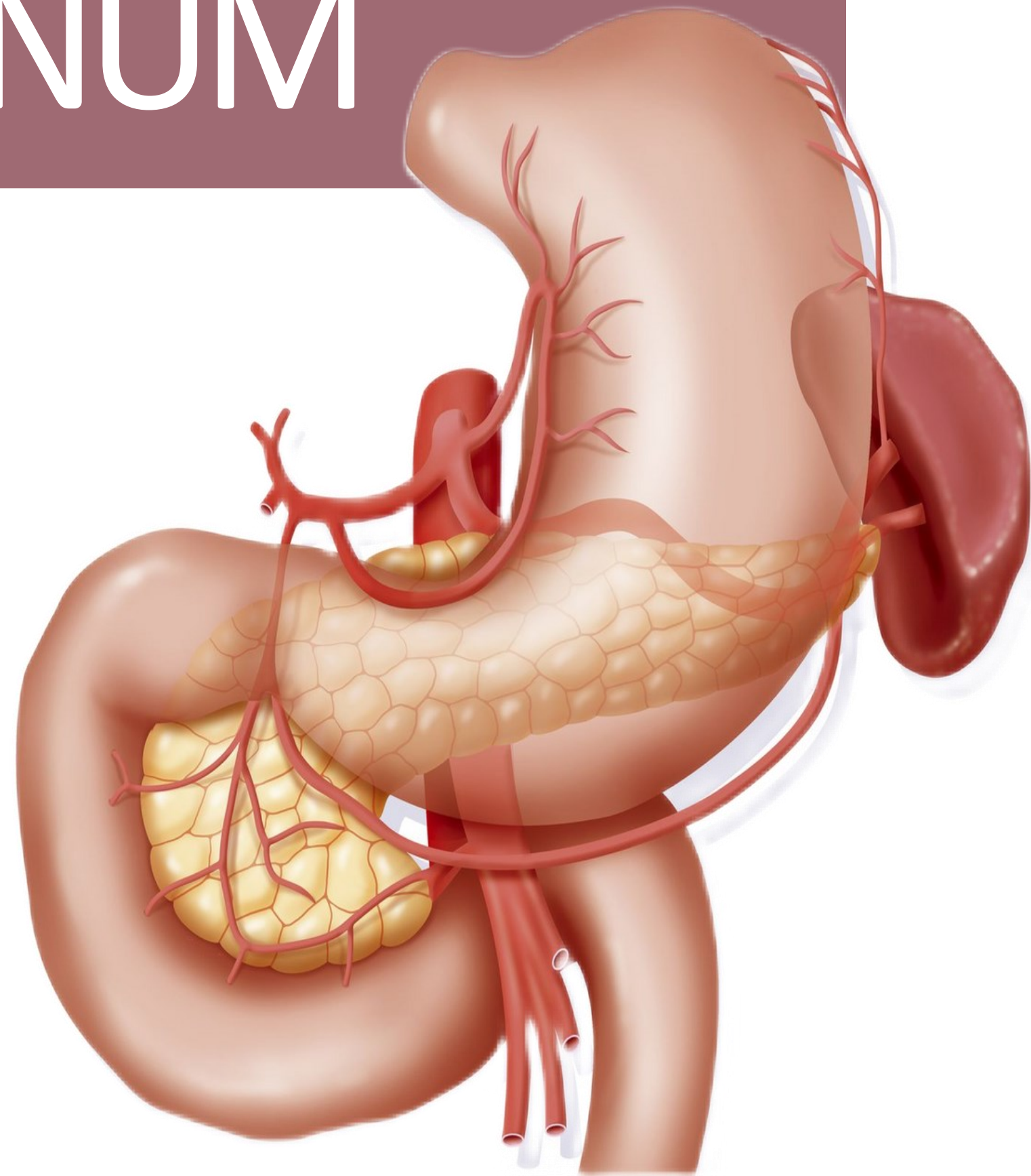


- The absorption of amino acids into enterocytes requires carriers (integral membrane proteins)
- For amino acids to enter the blood and be used by other body tissues → must be transported across the basolateral (serosal) membrane of the enterocyte → into interstitial fluid, → enter the blood through capillaries of the villi for transport → portal vein leading to the liver.



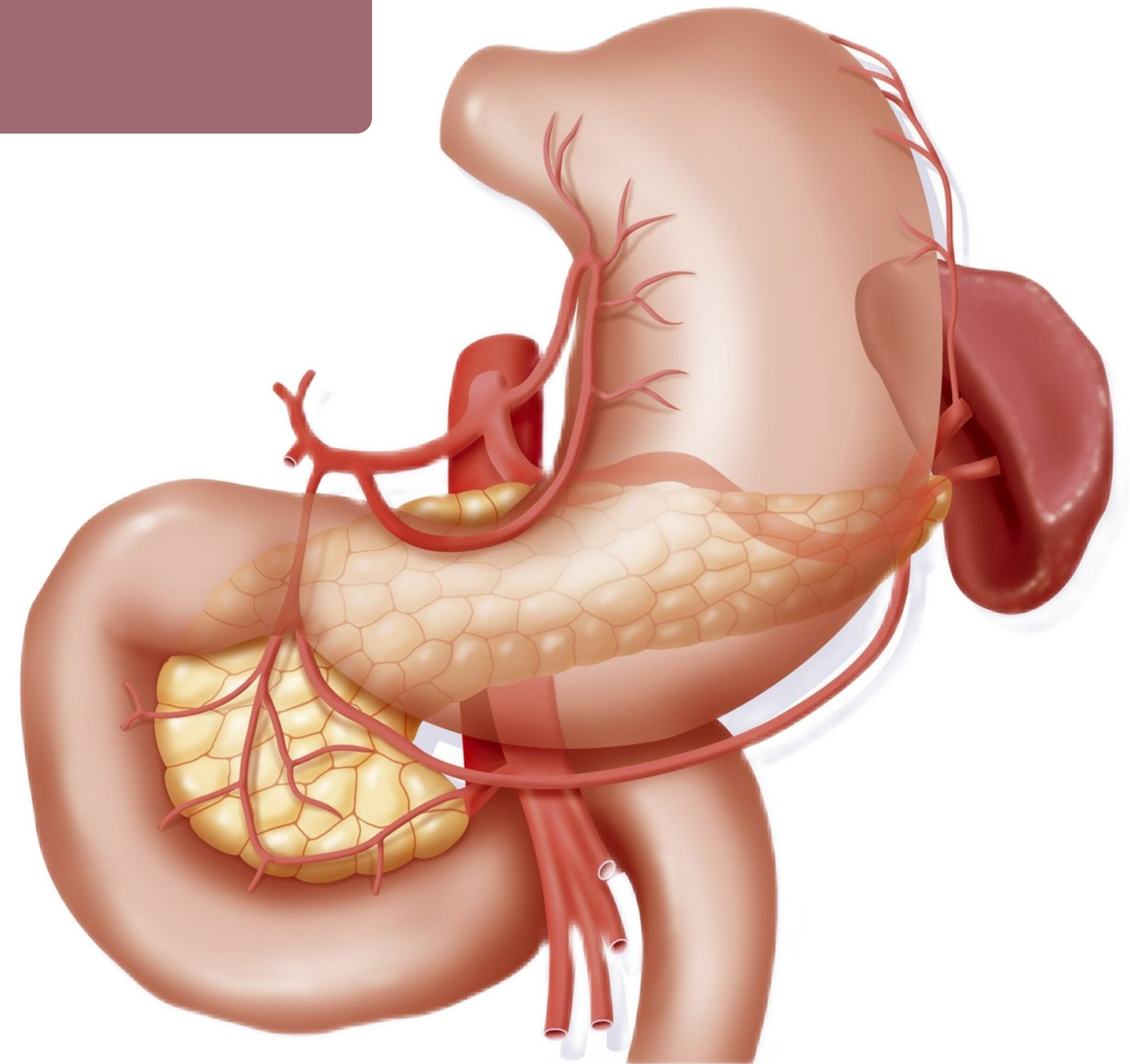


DUODENUM

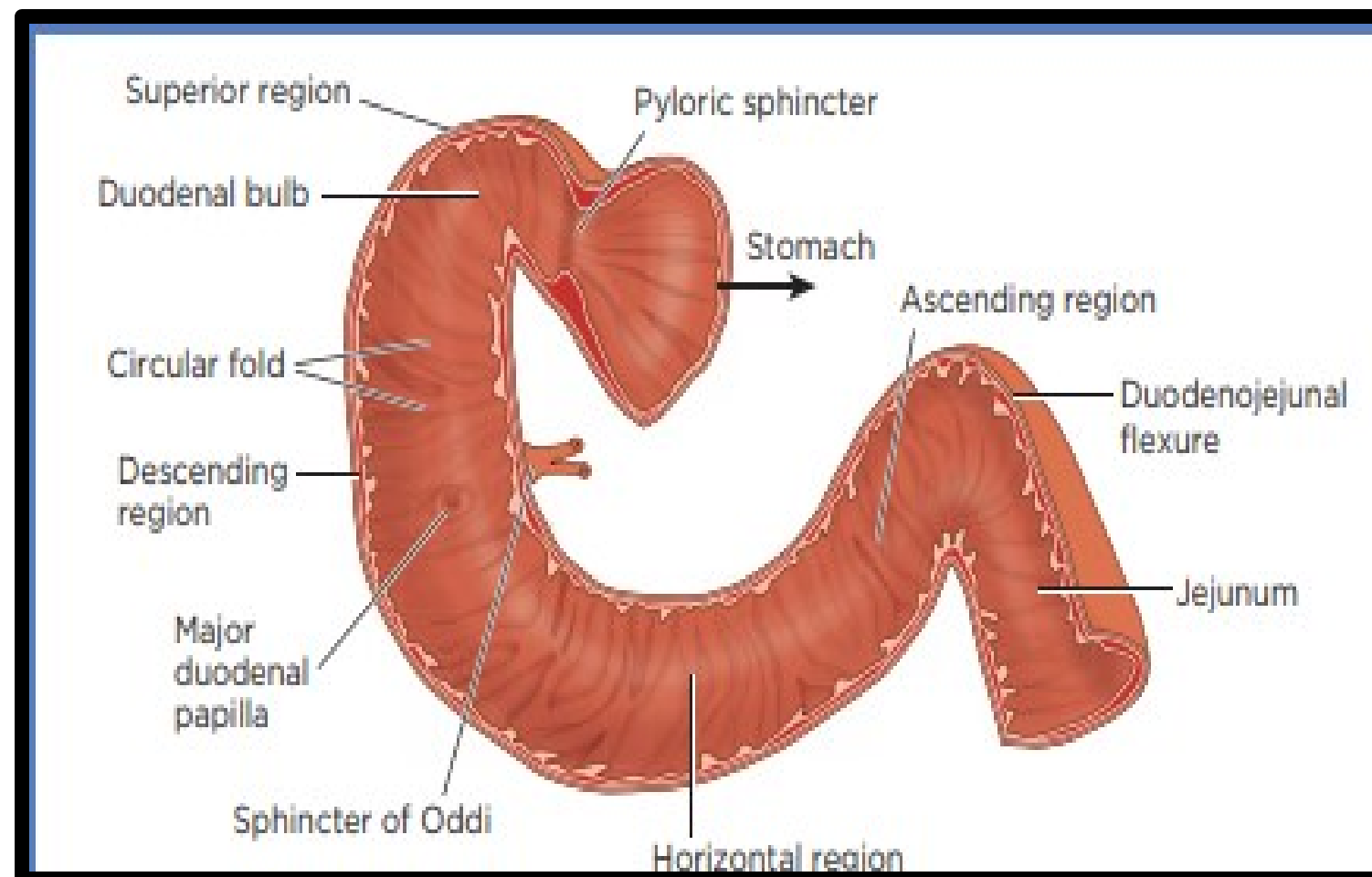


Definition

The duodenum is the initial C-shaped segment of the small intestine and is a continuation of the pylorus. Distally, it is in continuation with the jejunum and ileum, with the proximal segment being the shortest and widest



ANATOMY & PHISIOLOGY



The duodenum is primarily a region of chemical digestion. It receives secretions from the liver and pancreas, and its mucosa contains large numbers of mucusproducing (goblet) cells and Brunner's glands, which secrete a watery fluid that is rich in **mucus and bicarbonate ions**. The mucosal lining of the duodenum also contains tall columnar epithelial cells and extends into circular folds and fingerlike projections (villi), which increase the surface area for **nutrient absorption**.

Function of Duodenum

The function of the duodenum is a continuation of the digestion process that initially began in the stomach. It receives chyme generated by the stomach through a controlled valve between the stomach and the duodenum called the pylorus

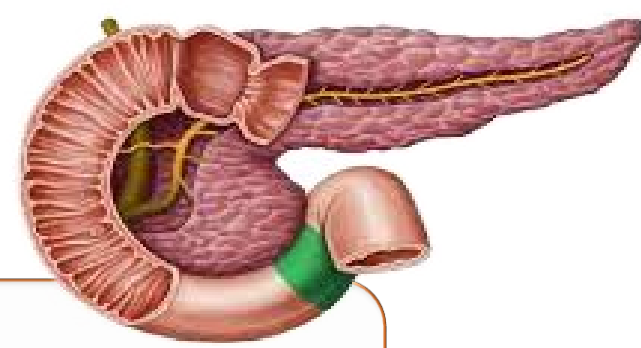
Hormon production in the duodenum

Secretin

Cholecystokinin

Gastric Inhibitory Polypeptide (GIP)

Vasoactive Intestinal Peptide (VIP)



Secretin

- secreted by the enteroendocrine S-cells of the duodenum and jejunum
- stimulates the production and release of bicarbonate ions by the pancreas

Cholecystokinin

- It is released by the enteroendocrine I-cells of the duodenum and jejunum
- Helps prevent the duodenum overfilling
- Relaxes the sphincter of Oddi, allowing bile and pancreatic juice to be released into the duodenum

GIP

- Synthesised by the enteroendocrine K-cells of the duodenum and jejunum
- GIP inhibits the release of gastric secretions in the stomach, helping to avoid damage to the small intestine

VIP

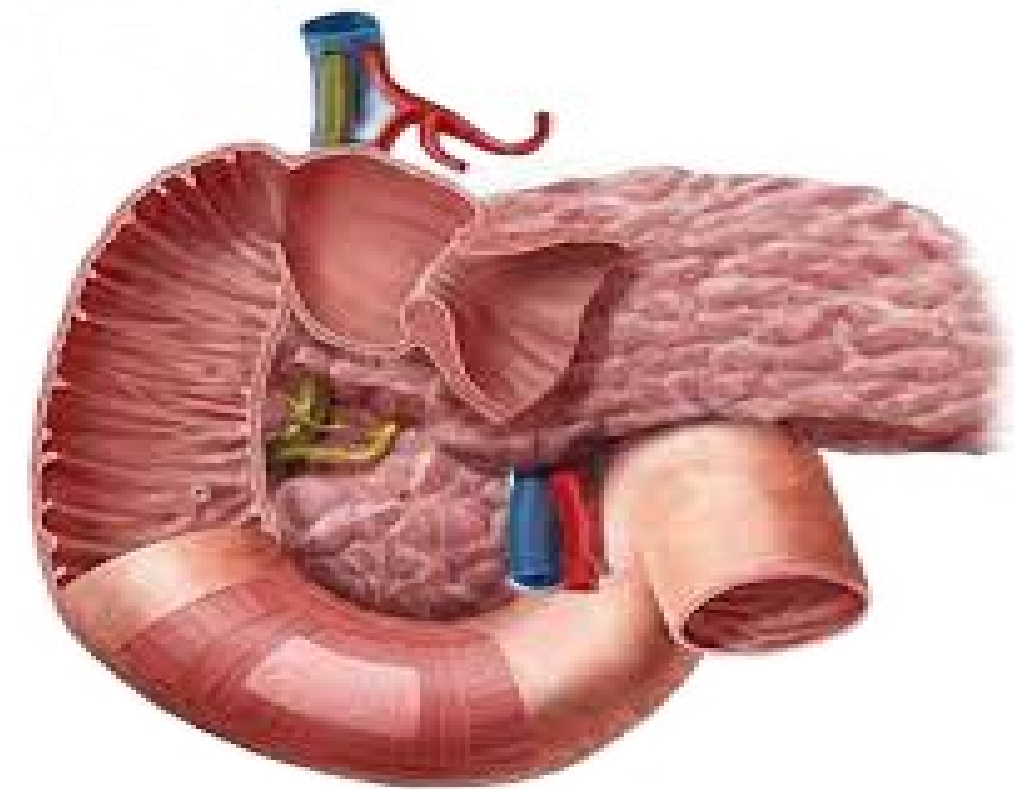
- VIP is a neuropeptide consisting of 28 amino acids and produced by nerve cells throughout the GI tract
- it is thought to regulate the composition of pancreatic juice and bile, as well as blood flow in gut tissue

Chemical digestion in the Duodenum

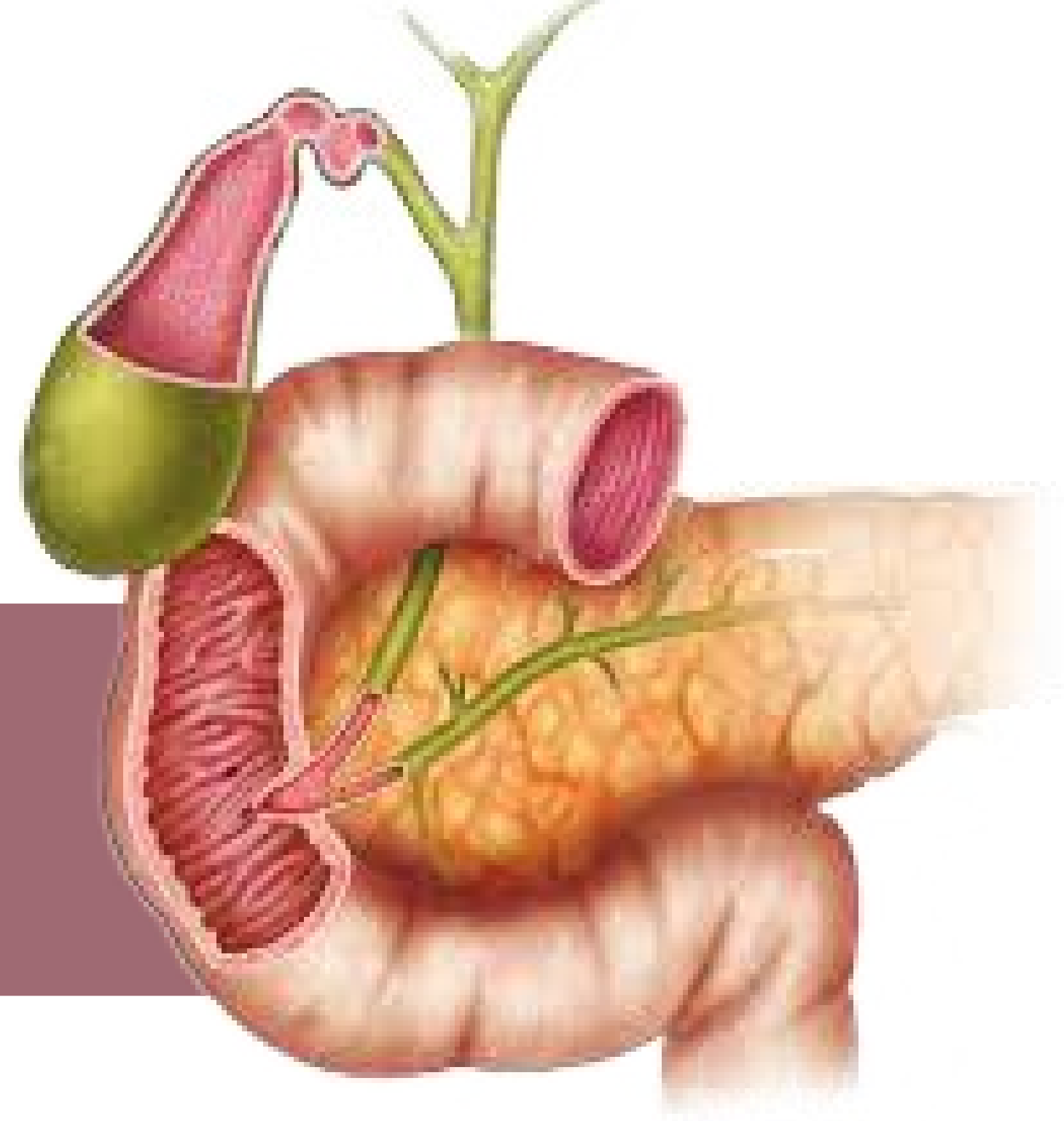
Liver, which produces
bile

Pancreas, which produces
pancreatic juice

are mixed and discharged
into the duodenum every
time the sphincter of Oddi
opens under the influence
of CCK.



Selected Disorder of Duodenum



DUODENAL ULCERATION

Peptic ulcer' is an umbrella term used to describe ulcers in the stomach or duodenum. Gastric and duodenal ulcers used to be attributed to stress and poor diet – particularly the consumption of greasy and spicy foods – but it is now accepted that infection with *Helicobacter pylori* is the major precipitating factor. The use of non-steroidal anti-inflammatory drugs significantly increases the risk of peptic ulcers (Kaur et al, 2012)

SIGN AND SYMPTOM OF DUODENAL ULCERATION

- Symptoms of gastric and duodenal ulcers are very similar. Patients often experience **a burning epigastric pain, often described as 'gnawing', and can point to its location.**
- Some with duodenal ulcers experience **pain either on an empty stomach or occurring two to three hours after food consumption, which tallies with gastric emptying times.** However, the only way to locate a peptic ulcer with absolute certainty is to visualise it, usually via endoscopy

Peptic ulcer

**Factors contributing to risk peptic ulcer
digestion of mucosa can be attributed to**

- Helicobacter pylori (bacterial pathogen)
- Anti-inflammatory drugs (eg aspirin)
- Stress ?
- Lifestyle (diet?, smoking?)

Clinical features

Epigastric pain

Peptic ulcer

Incidence

10% of adults in Western world & ↑ 'ing

Helicobacter pylori

infects 30 to 80% of world's adult pop.

higher prevalence:

- low socio-economic groups
- institutionalised individuals
- family members of infected subjects

Peptic ulcer

Dietary management

? Probiotics

compounds produced by lactic acid bacteria can inhibit the growth of H. pylori in vitro

? Milk

lactoferrin (glycoprotein)
known to possess activity against variety of bacteria

MNT for Peptic Ulcer Disease

- Identify foods that directly irritate the gastric mucosa or are not generally tolerated
- Avoid eating at least 2 hours before bedtime

Peptic Ulcer Disease

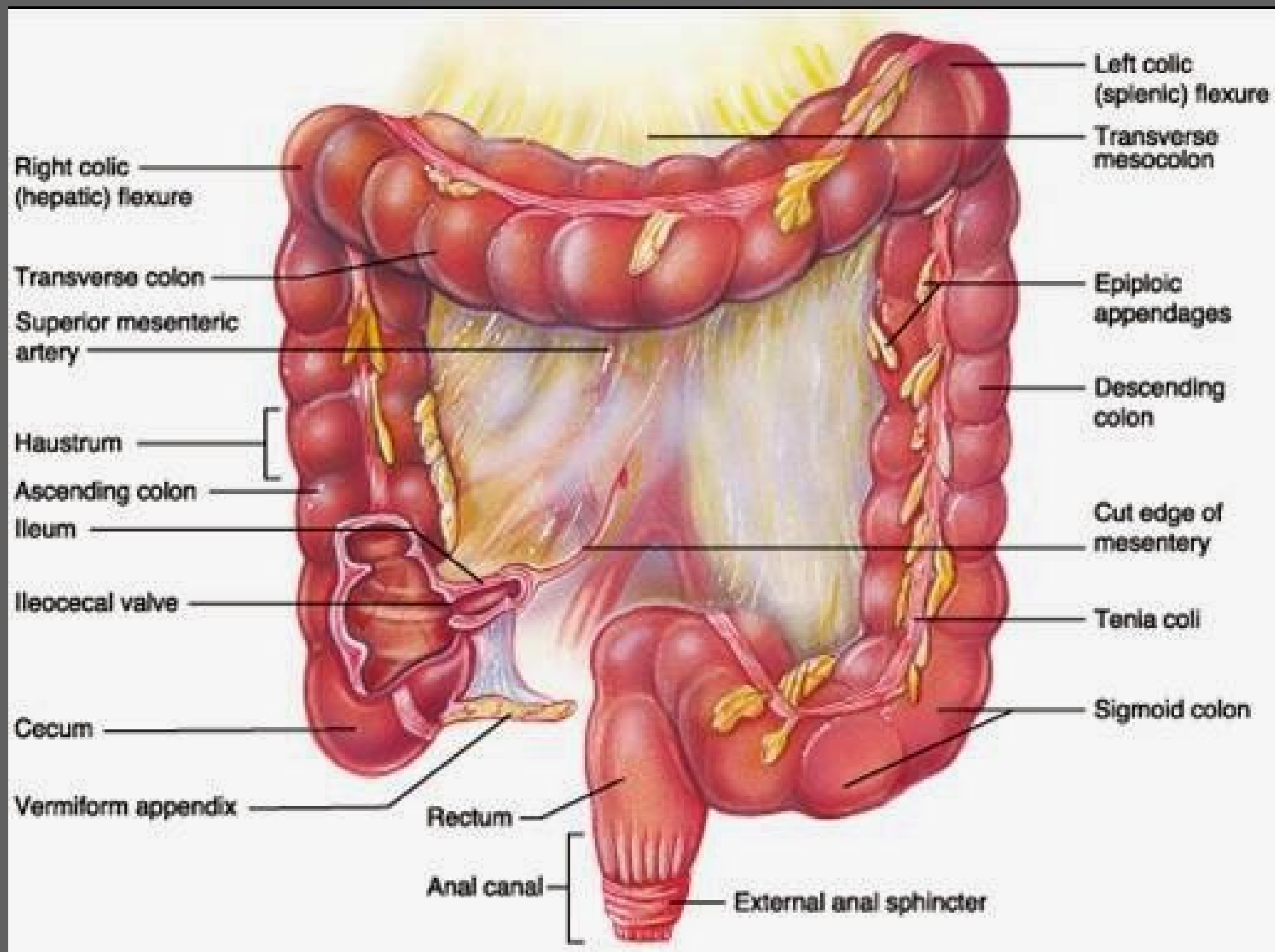
Treatment with Diet

- Meal frequency is controversial: small, frequent meals may increase comfort but may also increase acid output
- There is little evidence to support eliminating specific foods unless they cause repeated discomfort
- Overall good nutritional status helps ↓ *H. pylori*

THINK, PAIR AND SHARE

- woman, 35 years old, come to clinic because of pain in the epigastrium
- What is medical nutrition therapy for this patient?

ANATOMY AND FUNCTION



Size & length

The large intestine is approximately 6 feet in length, while the small intestine is much longer, at approximately 21 feet. The last 6 inches or so of the large intestine are called the rectum and the anal canal

3 Primary Function

- absorbing water and electrolytes
- producing and absorbing vitamins
- forming and propelling feces toward the rectum for elimination

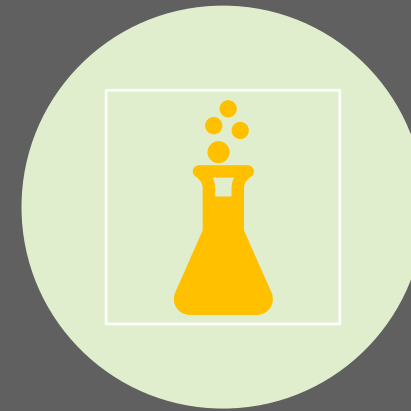
ABSORBING WATER AND ELECTROLYTES

- The proximal colonic mucosal cells typically absorb sodium, chloride, and water. About 90–95% of the water and sodium entering the colon each day is absorbed.
- Absorption of water occurs by osmosis. Water diffuses in response to an osmotic gradient established by the absorption of electrolytes.

IN THE LARGE INTESTINE, THERE IS A NET ABSORPTION OF SODIUM IONS AND CHLORIDE IONS ARE ACTIVELY ABSORBED



Sodium



Chloride and bicarbonate



Water



Potassium

PRODUCING AND ABSORBING VITAMINS

The colon plays an important role in absorbing vitamins needed for good health → through passive diffusion in the colon.

Vitamin K absorption is affected by gastrointestinal conditions

Low stomach acid → vitamin B₁₂ is not easily released from animal sources and is lost with feces.

In contrast, colonic bacteria synthesize vitamin K and biotin, which are actively absorbed and serve as the body's main source of supply.

Effects of gastric acid on microflora;

Gastric and small intestine → mostly destroyed and few colonized

Large intestine → mostly colonized

The colon houses trillions of bacteria protecting
our intestines

Microflora in the colon → fermentation process produces

Vitamin K

Biotin (vitamin B7)

thiamin,

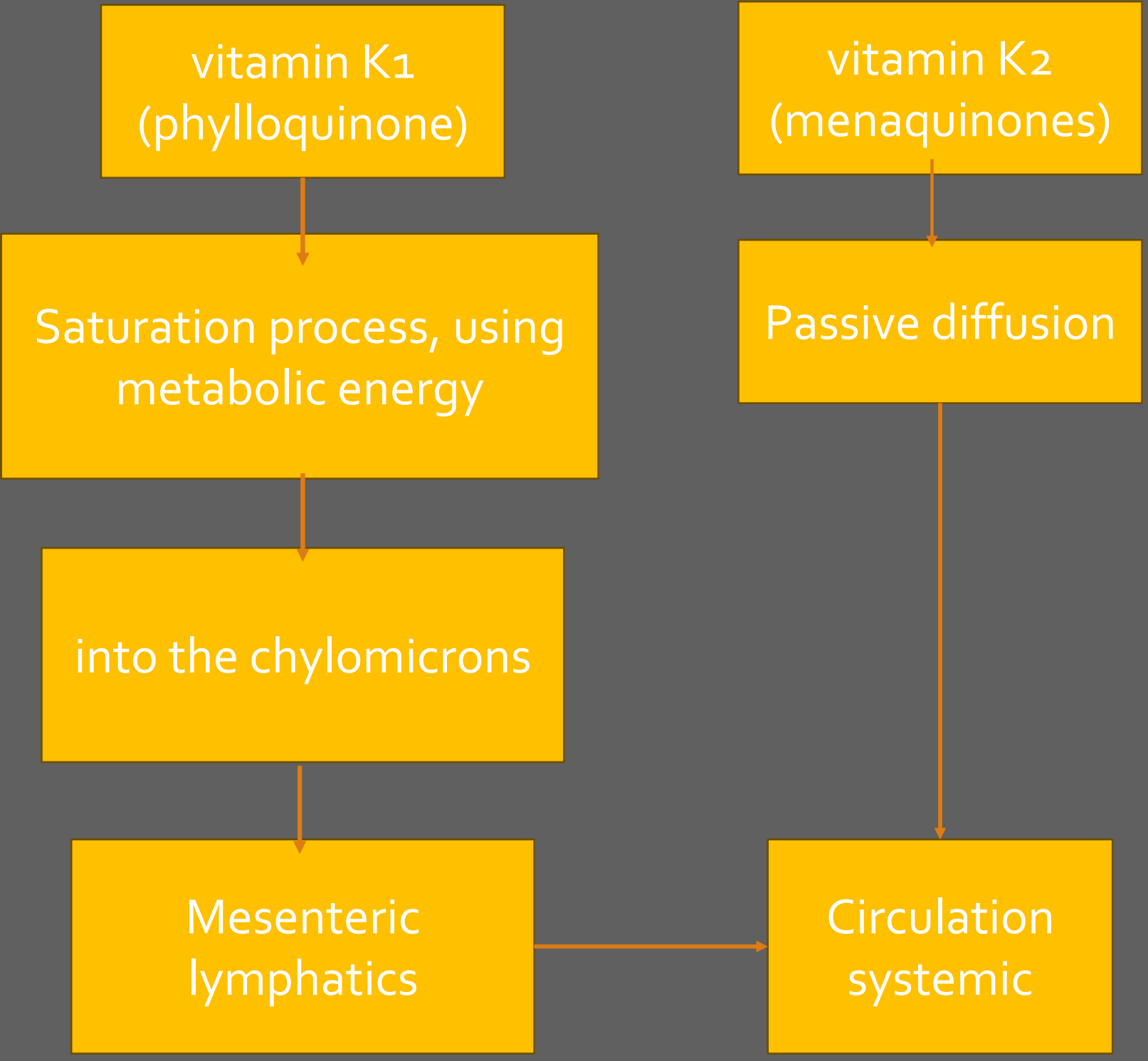
riboflavin,

vitamin B₁₂

absorbed into blood.

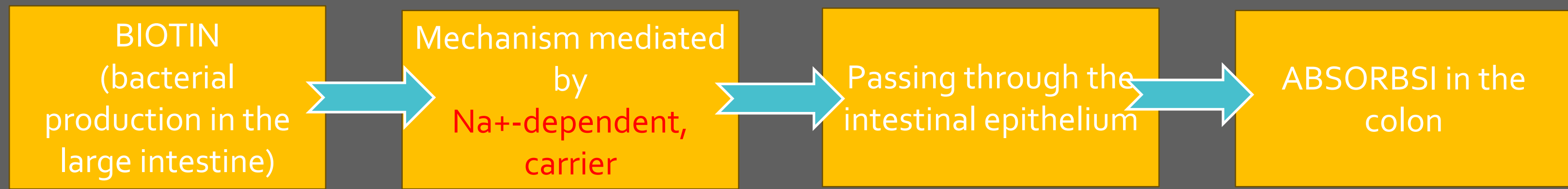
vitamin K synthesis is
essential for normal
blood clotting

Vitamin K stored in the liver
↓
Exhausted
↓
vitamin K intake from food is restricted



} Produced by intestinal and colonic bacteria
↓
important role in nutrition
vitamin K
↓
breast-fed infants

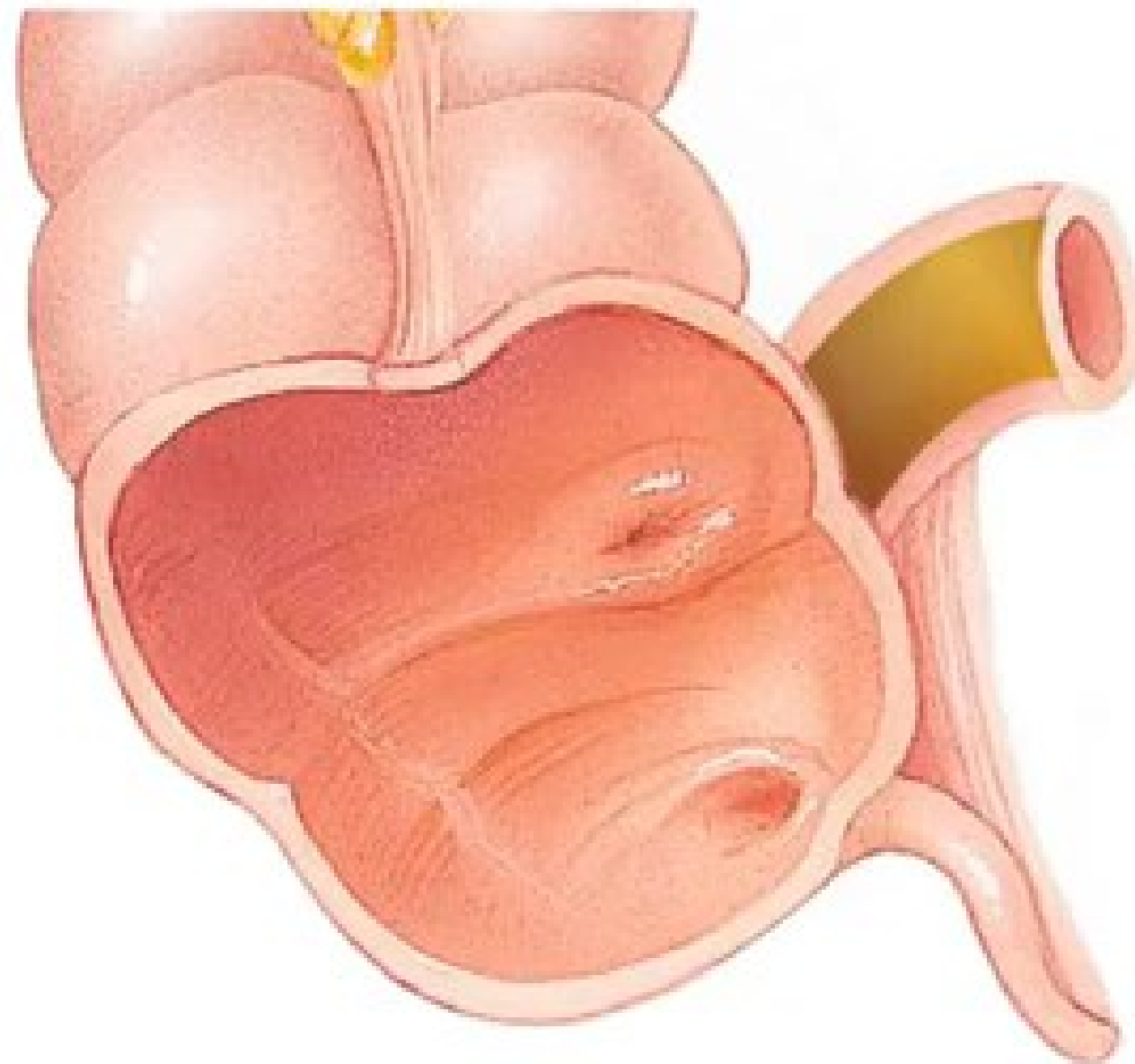
Absorbsi
5% to 15%



Homeostasis
Biotin

FORMING AND PROPELLING FECES

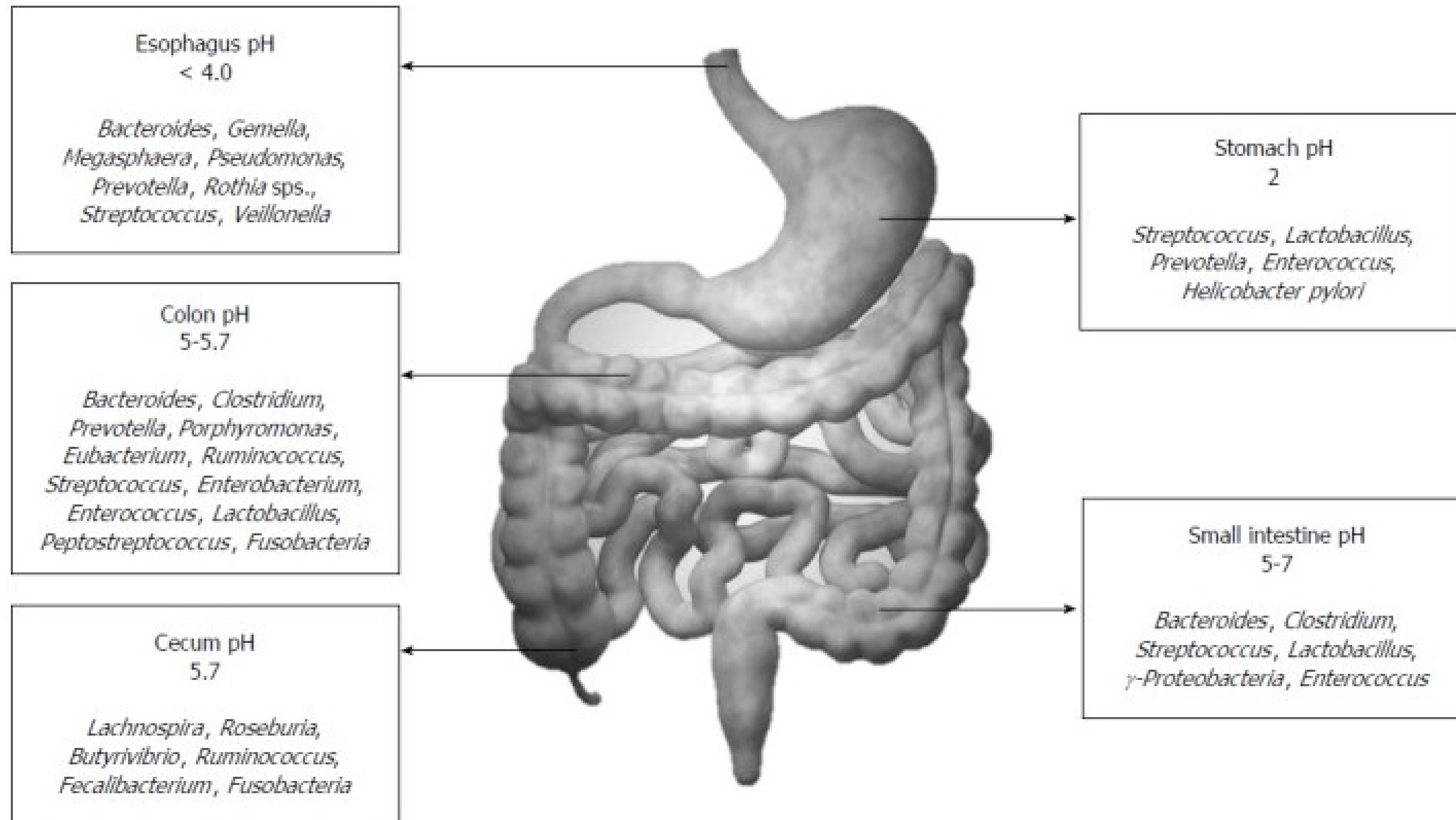
Haustral churning



FORMING AND PROPELLING FECES

- The main motility in the colon → contraction of haustra
- Mass movement occurs due to strain on the stomach & duodenum → the gastrocolic reflex and duodenocolic reflex.
- Movement pushes feces into the rectum → reflex rectal contraction and sphincter ani relaxation.
- The sphincter muscles of the rectum regulate the opening and closing of the anus consisting of smooth and striated muscles

GUT MICROBIOTA



FUNCTIONAL ASPECTS OF THE NORMAL GUT MICROBIOTA

- Nutrient metabolism
- Stimulate the immune system
- Immunomodulation
- Provide energy and serve as substrates for use within cells

MEDICAL NUTRITIONAL THERAPY OF COLITIS AND IRRITABLE BOWEL SYNDROME

- If diarrhea occurred should be give liquid diet and parenteral nutrition
- If diarrhea is not occurred can give blenderized diet and soft diet

CONSTIPATION

- Defined as hard stools, straining with defecation, infrequent bowel movements
- Normal frequency ranges from one stool q 3 days to 3 times a day
- Occurs in 5% to more than 25% of the population, depending on how defined

Constipation

What is it?

infrequent & difficult passage of hard faeces

What is the cause?

poor diet

medication

neurological disease

disorders of intestinal motility

Constipation

Common in:

disease

pregnancy

elderly

children

CAUSES OF CONSTIPATION - SYSTEMIC

- Side effect of medication, esp narcotics
- Metabolic Endocrine abnormalities, such as hypothyroidism, uremia and hypercalcemia
- Lack of exercise
- Ignoring the urge to defecate
- Vascular disease of the large bowel
- Systemic neuromuscular disease leading to deficiency of voluntary muscles
- Poor diet, low in fiber
- Pregnancy

CAUSES OF CONSTIPATION - GASTROINTESTINAL

- Diseases of the upper gastrointestinal tract
 - Celiac Disease – Gastric cancer
 - Duodenal ulcer – Cystic fibrosis
- Diseases of the large bowel resulting in:
 - Failure of propulsion along the colon (colonic inertia)
 - Failure of passage through anorectal structures (outlet obstruction)
- Irritable bowel syndrome
- Anal fissures or hemorrhoids
- Laxative abuse

TREATMENT OF CONSTIPATION

- Encourage physical activity as possible
- Bowel training: encourage patient to respond to urge to defecate
- Change drug regimen if possible if it is contributory
- Use laxatives and stool softeners judiciously
- Use stool bulking agents such as psyllium (metamucil) and pectin

MNT FOR CONSTIPATION

- Depends on cause
- Use high fiber or high residue diet as appropriate
- If caused by medication, may be refractory to diet treatment



Treatment of constipation

Eliminate non-dietary causes

↑ dietary fibre by ↑ intake & variety of plant foods

↑ fluid intake

Dietary fibre/resistant starch

Insoluble fibre

resists microdegradation

Soluble fibre

microdegradation

Resistant starch

microdegradation

FIBER, ROUGHAGE, AND RESIDUE

- Fiber or roughage
 - From plant foods
 - Not digestible by human enzymes
- Residue
 - Fecal contents, including bacteria and the net remains after ingestion of food, secretions into the GI tract, and absorption

High Fibre Diets

Not just the amount!

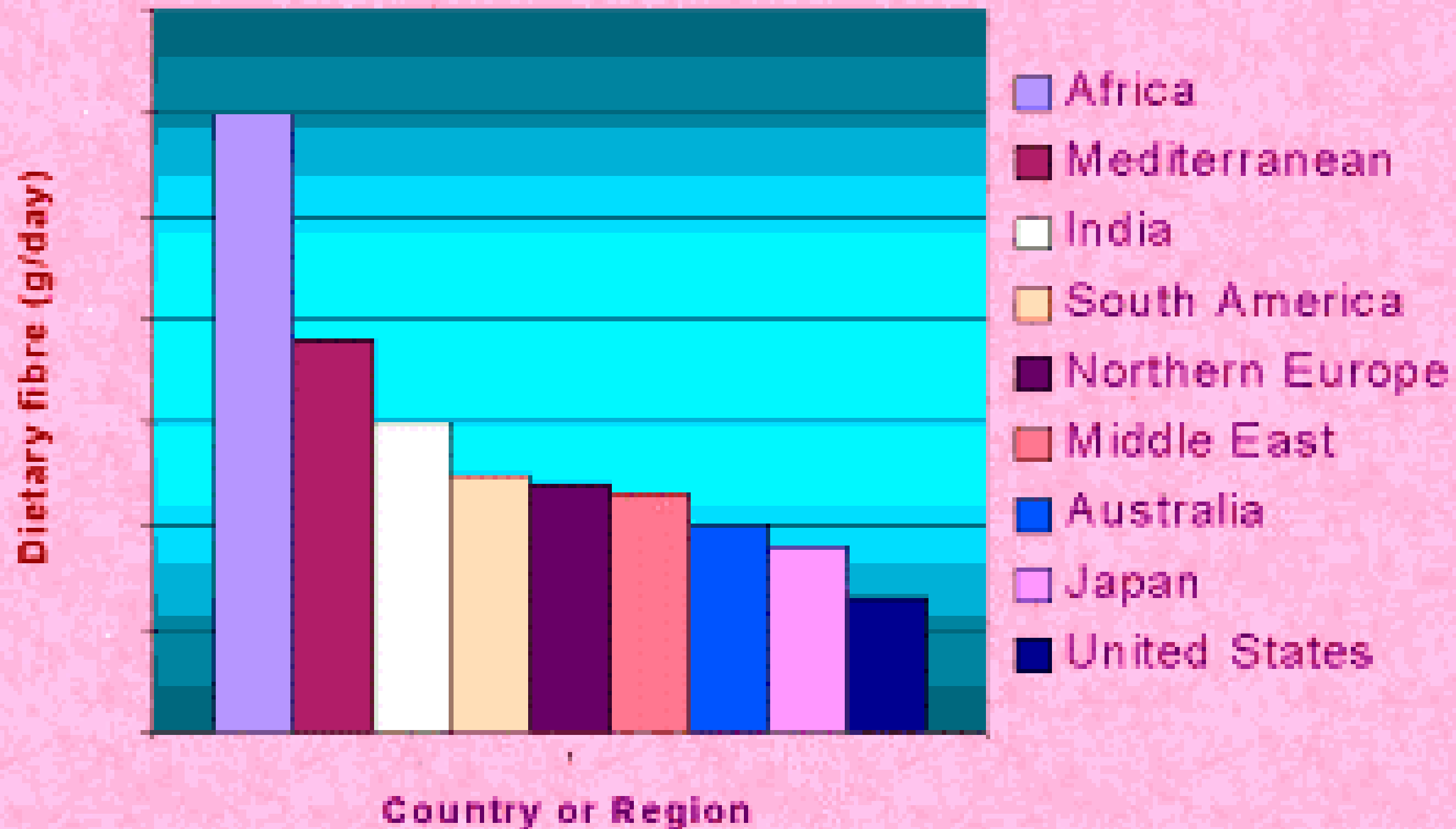
It's the variety of different types that
is important.

HIGH-FIBER DIETS

- Most Americans = 10 – 15 g/day
- Recommended = 25 g/day
- More than 50g/day = no added benefit, may cause problems



Dietary fibre intakes by country or region



Spiller CRC Press 1993

HIGH-FIBER DIET

- Increase consumption of whole-grain breads, cereals, flours, other whole-grain products
- Increase consumption of vegetables, especially legumes, and fruits, edible skins, seeds, hulls
- Consume high-fiber cereals, granolas, legumes to increase fiber to 25 g/day
- Increase consumption of water to at least 2 qts (eight 8 oz cups)

HIGH-FIBER DIETS: CAUTIONS

- Gastric obstruction, fecal impaction may occur when insufficient fluid consumed
- With GI strictures, motility problems, increase fiber slowly (~1mo.)
- Unpleasant side effects
 - Increased flatulence
 - Borborygmus
 - Cramps, diarrhea

PHYSIOLOGIC EFFECT OF DIETARY FIBER

- stimulating chewing, saliva flow, gastric juice secretion
- fills the stomach and provides a sense of satiety
- Increase fecal bulk
- Normalizes intestinal transit time
- Become a substrat for colonic fermentation
- Delay gastric emptyng
- Slows the rate of digestion and absorption
- Lower serum cholesterol

DIARRHEA

- Characterized by frequent evacuation of liquid stools
- Accompanied by loss of fluid and electrolytes, especially sodium and potassium
- Occurs when there is excessively rapid transit of intestinal contents through the small intestine, decreased absorption of fluids, increased secretion of fluids into the GI tract

DIARRHEA ETIOLOGY

- Inflammatory disease
- Infections with fungal, bacterial, or viral agents
- Medications (antibiotics, elixirs)
- Overconsumption of sugars
- Insufficient or damaged mucosal absorptive surface
- Malnutrition

DIARRHEA TREATMENT FOR ADULTS

- Identify and treat the underlying problem
- Manage fluid and electrolyte replacement using oral glucose electrolyte solutions (see WHO guidelines)
- Initiate minimum-residue diet
- Avoid large amounts of sugars and sugar alcohols
- Prebiotics in modest amounts including pectin, oligosaccharides, inulin, oats, banana flakes
- Probiotics, cultured foods and supplements that are sources of beneficial gut flora

LOW- OR MINIMUM RESIDUE DIET

- Foods completely digested, well absorbed
- Foods that do not increase GI secretions
- Used in:
 - Maldigestion
 - Malabsorption
 - Diarrhea
 - Temporarily after some surgeries, e.g. hemorrhoidectomy

FOODS TO LIMIT IN A LOW- OR MINIMUM RESIDUE DIET

- Lactose (in lactose malabsorbers)
- Fiber >20 g/day
- Resistant starches
 - Raffinose, stachyose in legumes
- Sorbitol, mannitol, xylitol >10g/day
- Caffeine
- Alcohol, esp. wine, beer

What is Irritable bowel syndrome (IBS)?



IBS

Clinical Features

- recurrent abdominal pain
- sense of bloating
- altered bowel habit
 - diarrhoea
 - constipation

Management

- high fibre diet
- exclusion diet if suspect food intolerance
- ? peppermint oil (menthol)

INFLAMMATORY BOWEL DISEASE

- Crohn's Disease and Ulcerative Colitis
- Autoimmune diseases of unknown origin
- Genetic component and environmental factors
- Onset usually between 15 to 30 years of age

Gut motility

Food derived factors which may affect gut function

- ***peppermint oil***
relaxes GI smooth muscle by ↓ calcium influx
- ***casein***
casomorphins → mu receptors ↓ gut motility
- ***caffeine***
↑ gut motility

what does caffe latte do to gut motility?

IBS: NUTRITIONAL CARE

- ID individual food intolerances
 - Keep food record, include symptoms, time they occur in relation to meals
- Avoid offending foods, substances
 - Milk, milk products (lactose) only in presence of lactose deficiency
 - Fatty foods
 - Gas-forming foods, beverages
 - Caffeine, alcohol
 - Foods w/ ↑ fructose or sorbitol

IBS: NUTRITIONAL CARE

- Eat small frequent meals at relaxed pace, regular times
- Gradually add dietary fiber to diet
 - 20 – 30 g
 - Fiber supplements may help (psyllium)
- Fluids – 2 – 3 qts w/ fiber supp.
- Regular physical activity to reduce stress

**High fibre diet recommended for:
constipation, diverticular disease &
irritable bowel syndrome**



Whitney et al., *Understanding Normal & Clin Nutr* WEST 1994

Nutrient profile

	<u>Apple</u>	<u>Juice</u>
Energy (kJ)	320	350
Energy (cals)	80	85
Fibre (g)	2-3	-

Dietary fibre content of foods



Whitney et al., *Understanding Normal & Clin Nutr* WEST 1994

Dietary fibre content of foods



Nutrient profile

	<u>Burger</u>	<u>Sand</u>	<u>Banana</u>
Energy (kJ)	2100	1220	420
Energy (cals)	500	290	100
Fat (g)	26	16	-
Fibre (g)	3	6	3
Vitamin C (mg)	4	17	15
Na+ (mg)	1480	650	1
K+ (mg)	410	450	370
Water (g)	93	200	80

Inflammatory bowel disease

Ulcerative colitis

chronic inflammation of colon-rectum

Crohn's disease

chronic inflammation of any part of GIT
(from mouth to anus)

Ulcerative colitis

Theory of Ulcerative colitis

carrageenan induces UC in guinea pigs

carrageenan seaweed extract

used widely: ice-cream, confectionery,
biscuits

Marcus R, Watt J. Lancet 1969:489-90

Substrates for maintaining the integrity of intestinal mucosa

Glutamine

- small intestine**

beneficial in metabolically stressed patients

? needed in healthy people

Butyrate

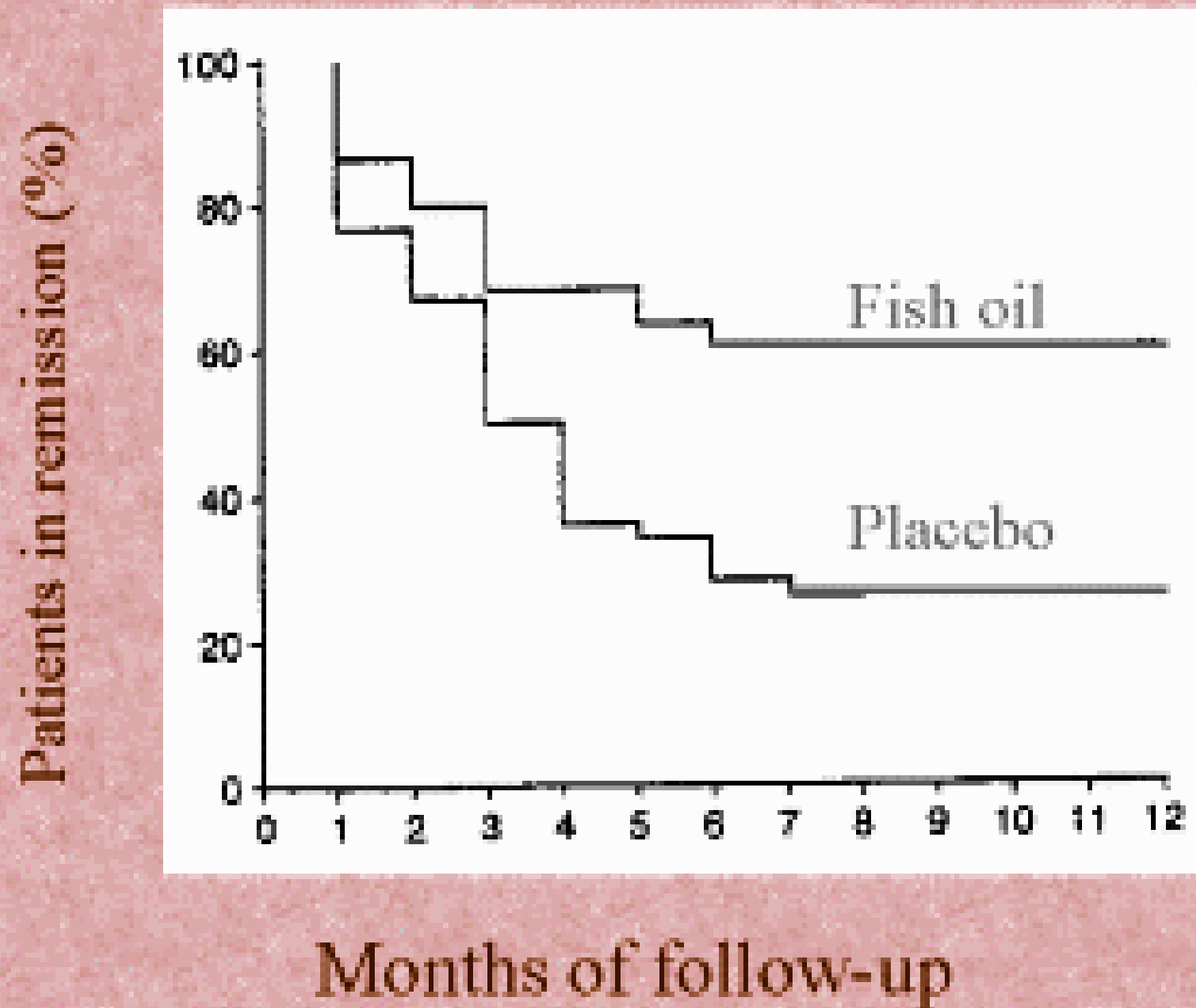
- large bowel**

colonocytes utilize butyrate (SCFA)

i) substrate & ii) cell differentiation

? increased requirements in UC

Crohn's disease



Other nutritional considerations in the management of CIBD

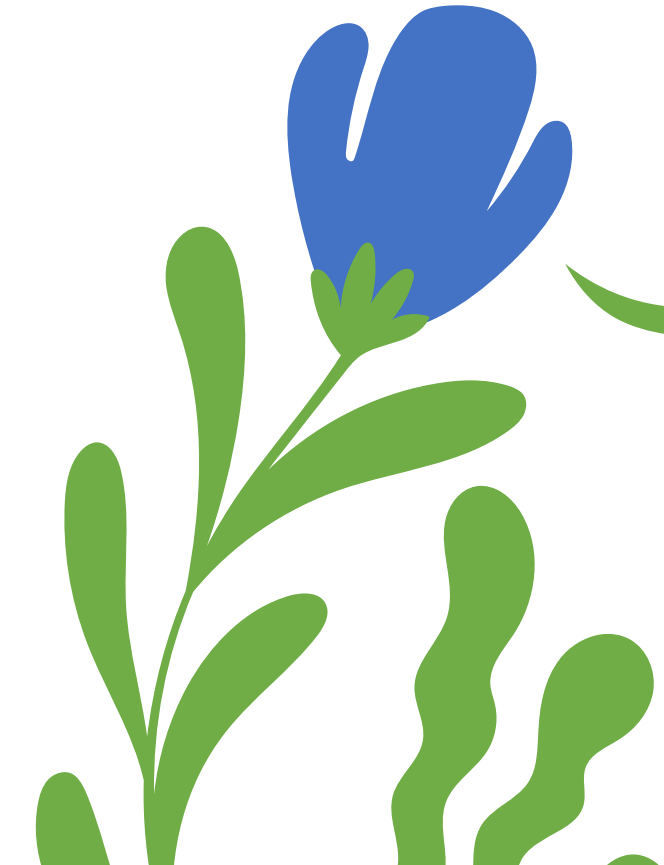
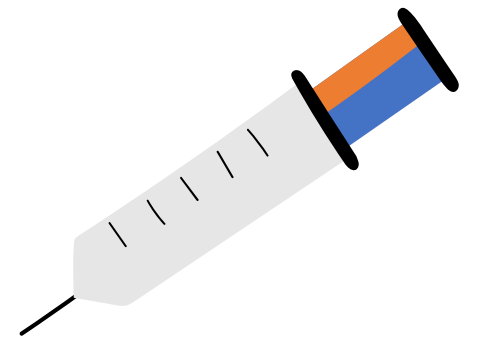
- **high protein and energy diet**
- **vitamin and mineral supplements**
- **low fibre during acute attacks**
- **low fat if steatorrhea occurs**

THINK, PAIR AND SHARE

- Man, 50 years old, come to clinic because of diarrhea and diagnosed IBS
- What is medical nutrition therapy for this patient?

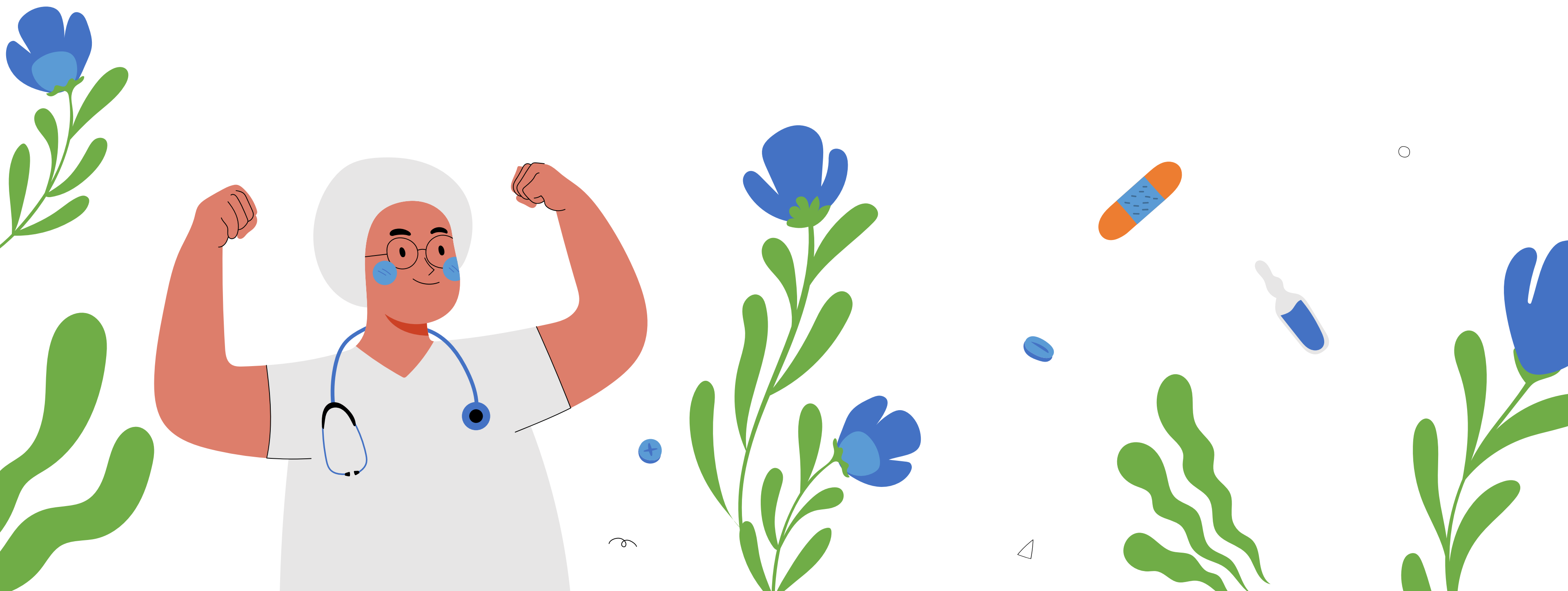


PANCREAS



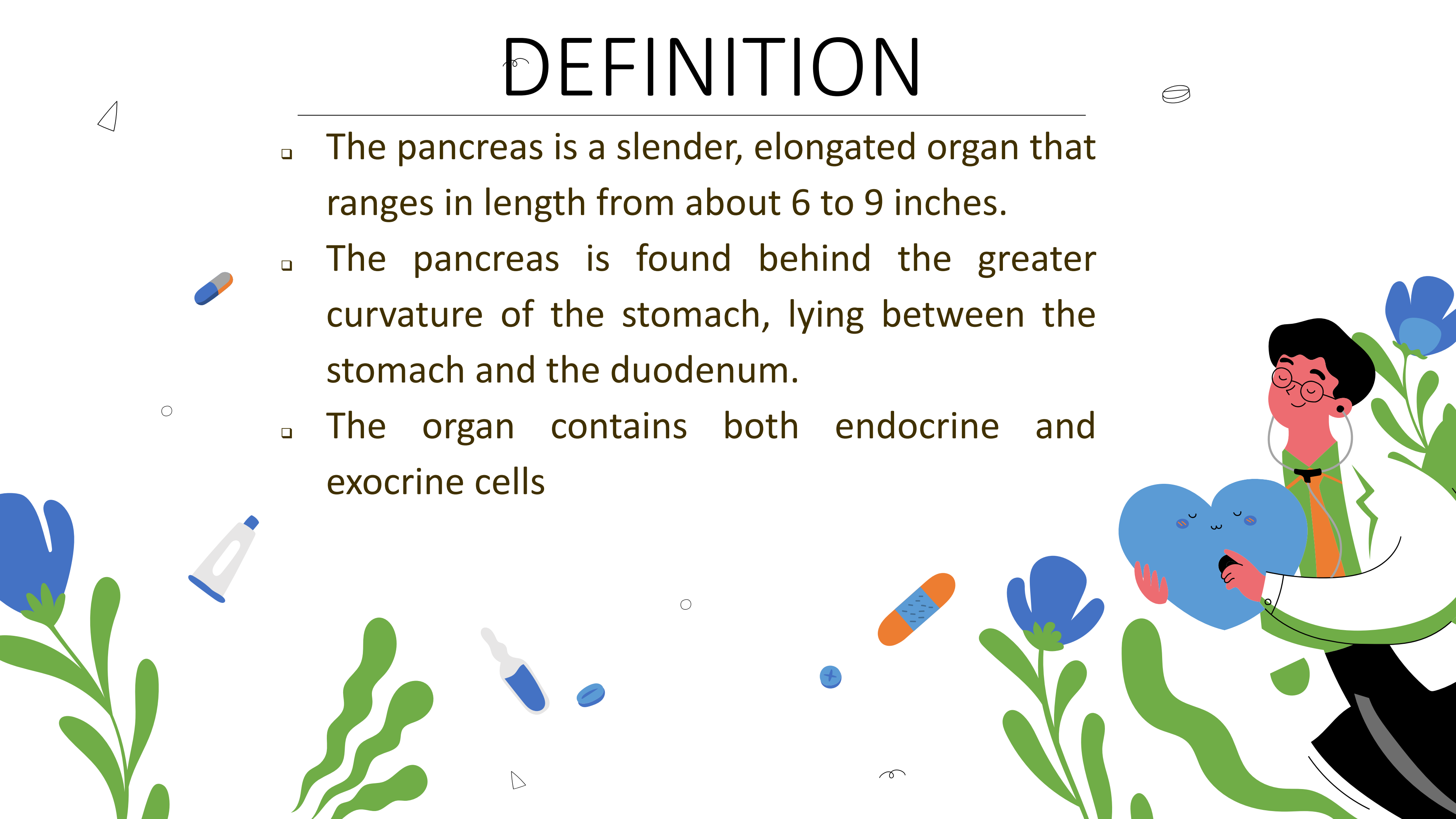
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Defining PANCREAS

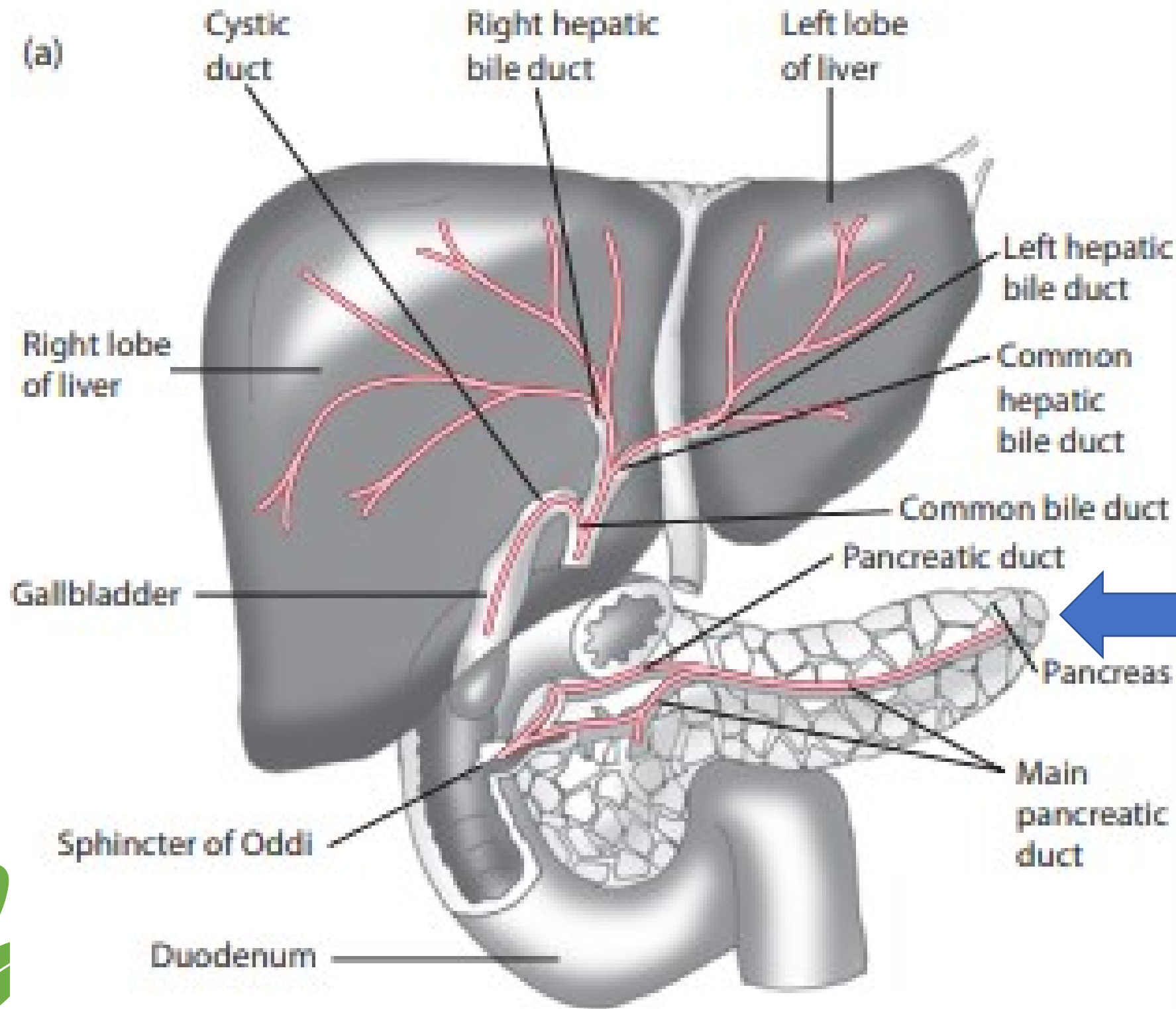


DEFINITION

- ❑ The pancreas is a slender, elongated organ that ranges in length from about 6 to 9 inches.
- ❑ The pancreas is found behind the greater curvature of the stomach, lying between the stomach and the duodenum.
- ❑ The organ contains both endocrine and exocrine cells



ANATOMY

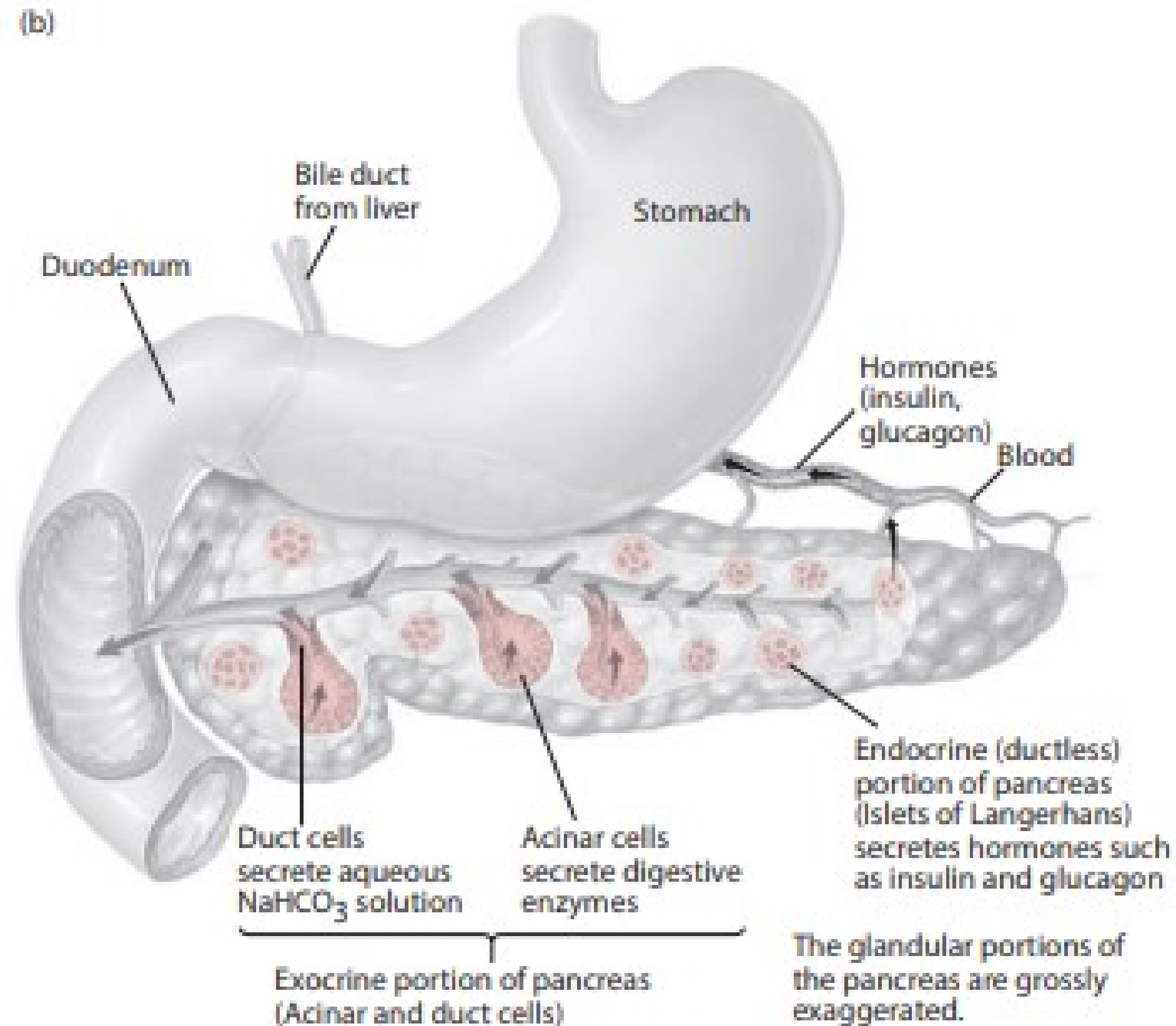


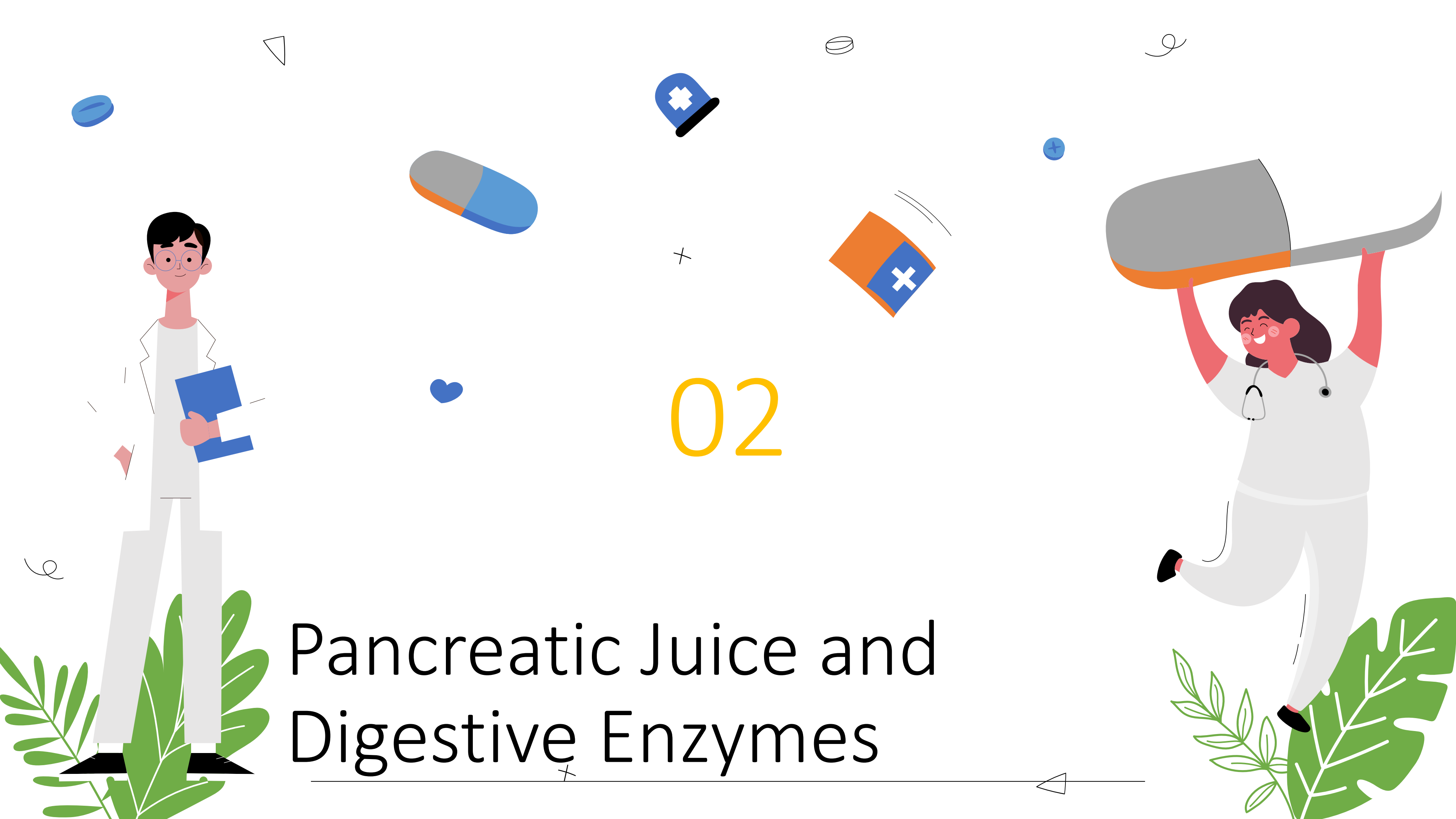
□ The endocrine cells are found among the 1–2 million cells that make up the islets of Langerhans, located primarily in the tail region of the pancreas.

- + - The A or α cells secrete glucagon
- The B or β cells secrete insulin
- The D or δ cells secrete somatostatin.

□ The exocrine cells of the pancreas that are more involved in the digestive processes with the production of pancreatic juice and enzymes

Schematic Representation of The Exocrine and Endocrine Portions of the Pancreas





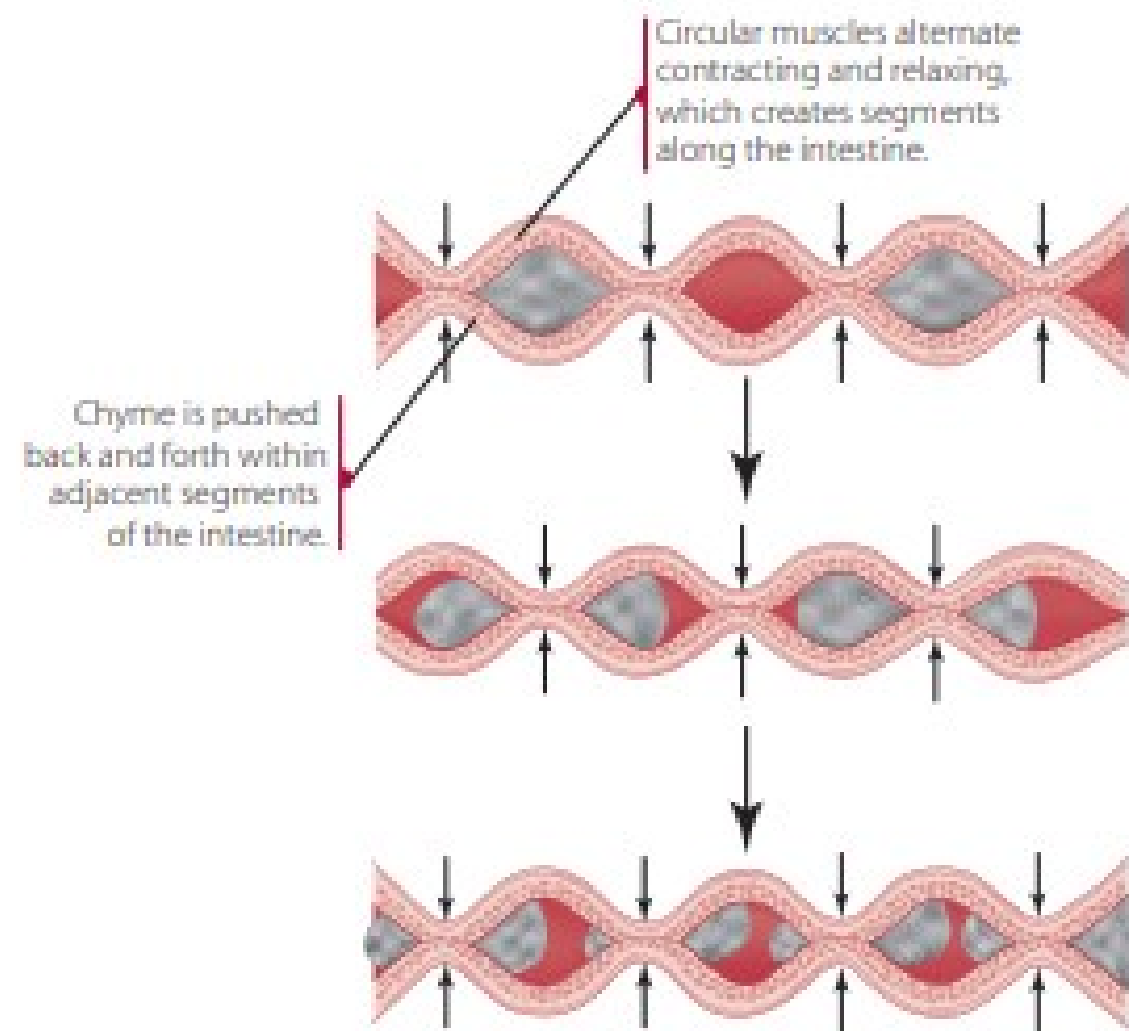
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Pancreatic Juice and Digestive Enzymes

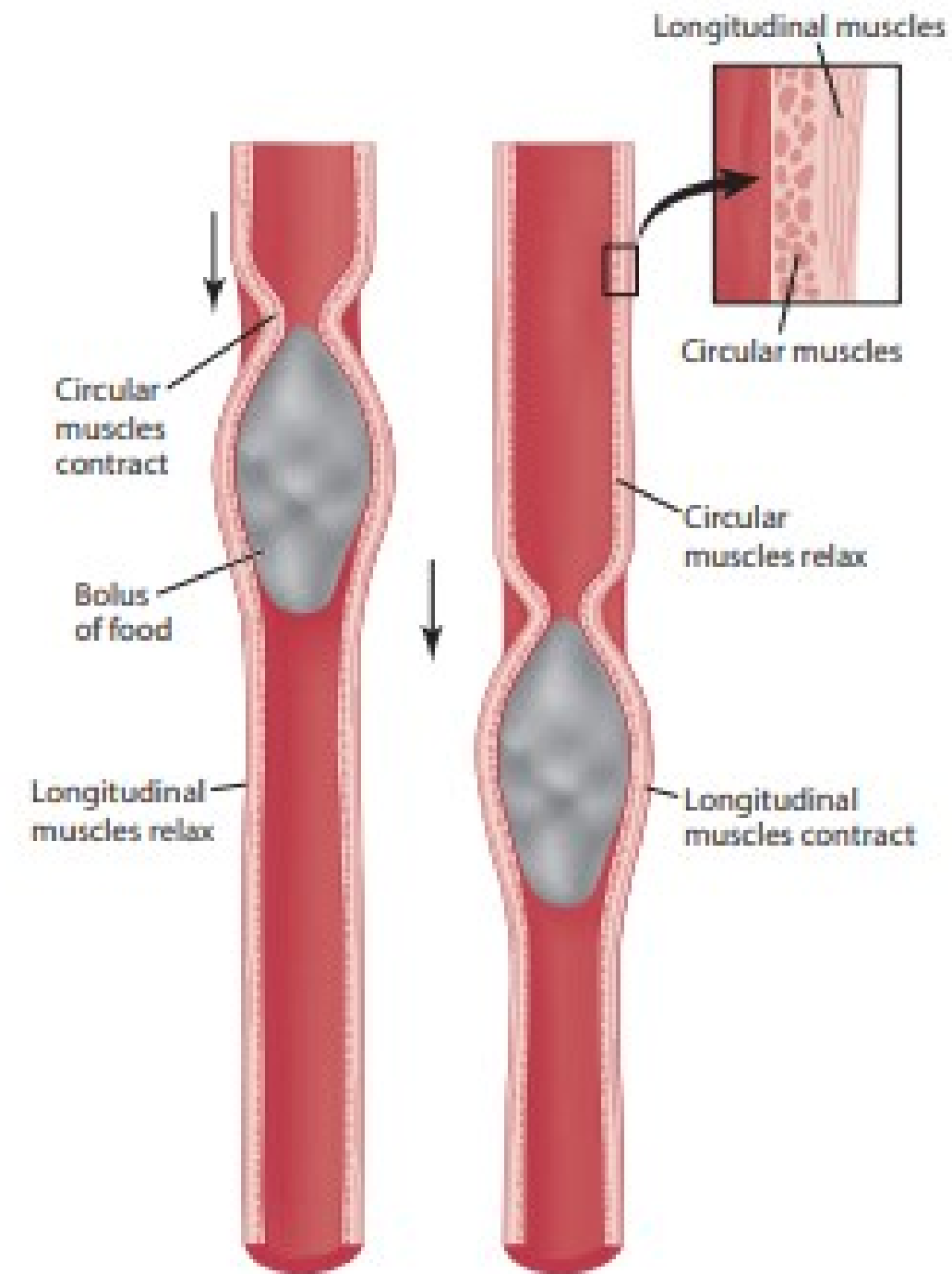
- 
- 
- 
- ✓ The pancreas releases up to about 2 L of its secretions daily into the duodenum.
 - ✓ The juice contains mostly water, electrolytes (the cations sodium, potassium, and calcium and the anion chloride), and bicarbonate (as NaHCO_3).
 - ✓ The bicarbonate neutralizes the acidic chyme released from the stomach into the duodenum and creates a more alkaline pH needed for enzyme activity within the intestinal lumen.
- 



Movement of Chyme in The Gastrointestinal Tract



Segmentation. Segmentation mixes food in the GI tract by moving the food mass back and forth. The circular muscles contract and relax, which creates a "chopping" motion.



Peristalsis. Peristalsis consists of a series of wave-like rhythmic contractions and relaxation involving the muscles of the gastrointestinal tract. This action propels food forward through the GI tract.

Enzyme or Zymogen/Enzyme	Site of Secretion	Preferred Substrate(s)	Primary Site of Action
Salivary α -amylase	Mouth	α (1-4) bonds in starch, dextrins	Mouth, stomach
Lingual lipase	Mouth	Triacylglycerol	Mouth, stomach
Pepsinogen/pepsin	Stomach	Carboxyl end of phe, tyr, trp, met, leu, glu, asp*	Stomach
Gastric lipase	Stomach	Triacylglycerol (mostly medium chain)	Stomach
Trypsinogen/trypsin	Pancreas	Carboxyl end of lys, arg*	Small intestine
Chymotrypsinogen/chymotrypsin	Pancreas	Carboxyl end of phe, tyr, trp, met, asn, his*	Small intestine
Procarboxypeptidase/carboxypeptidase A	Pancreas	C-terminal neutral amino acids	Small intestine
Carboxypeptidase B	Pancreas	C-terminal basic amino acids	Small intestine
Proelastase/elastase	Pancreas	Fibrous connective tissue proteins—elastin	Small intestine
Collagenase	Pancreas	Collagen	Small intestine
Ribonuclease	Pancreas	Ribonucleic acids	Small intestine
Deoxyribonuclease	Pancreas	Deoxyribonucleic acids	Small intestine
Pancreatic α -amylase	Pancreas	α (1-4) bonds, in starch, maltotriose	Small intestine
Pancreatic lipase and colipase	Pancreas	Triacylglycerol	Small intestine
Phospholipase	Pancreas	Lecithin and other phospholipids	Small intestine
Cholesterol esterase	Pancreas	Cholesterol esters	Small intestine
Retinyl ester hydrolase	Pancreas	Retinyl esters	Small intestine
Amino peptidases	Small intestine	N-terminal amino acids	Small intestine
Dipeptidases	Small intestine	Dipeptides	Small intestine
Nucleotidase	Small intestine	Nucleotides	Small intestine
Nucleosidase	Small intestine	Nucleosides	Small intestine
Alkaline phosphatase	Small intestine	Organic phosphates	Small intestine
Monoglyceride lipase	Small intestine	Monoglycerides	Small intestine
Alpha dextrinase or isomaltase	Small intestine	α (1-6) bonds in dextrins, oligosaccharides	Small intestine
Glucoamylase, glucosidase, and sucrase	Small intestine	α (1-4) bonds in maltose, maltotriose	Small intestine
Trehalase	Small intestine	Trehalose	Small intestine
Disaccharidases	Small intestine		Small intestine
Sucrase		Sucrose	
Maltase		Maltose	
Lactase		Lactose	

* Amino acid abbreviations: phe, phenylalanine; tyr, tyrosine; trp, tryptophan; met, methionine; leu, leucine; glu, glutamic acid; asp, aspartic acid; lys, lysine; arg, arginine; asn, asparagine; and his, histidine.

Table : Digestive Enzymes & Their Action

○

△

03

Regulation of Pancreatic Secretions



The primary stimuli for the release of pancreatic juice are the hormones **secretin** and **cholecystokinin**



Secretin :

produced by S cells in the proximal small intestine, is secreted into the blood primarily in response to the presence of unneutralized acidic chyme in the duodenum

Cholecystokinin :

secreted by I cells of the proximal small intestine and enteric nerves in response to the presence of fat and partially digested proteins in the duodenum

04

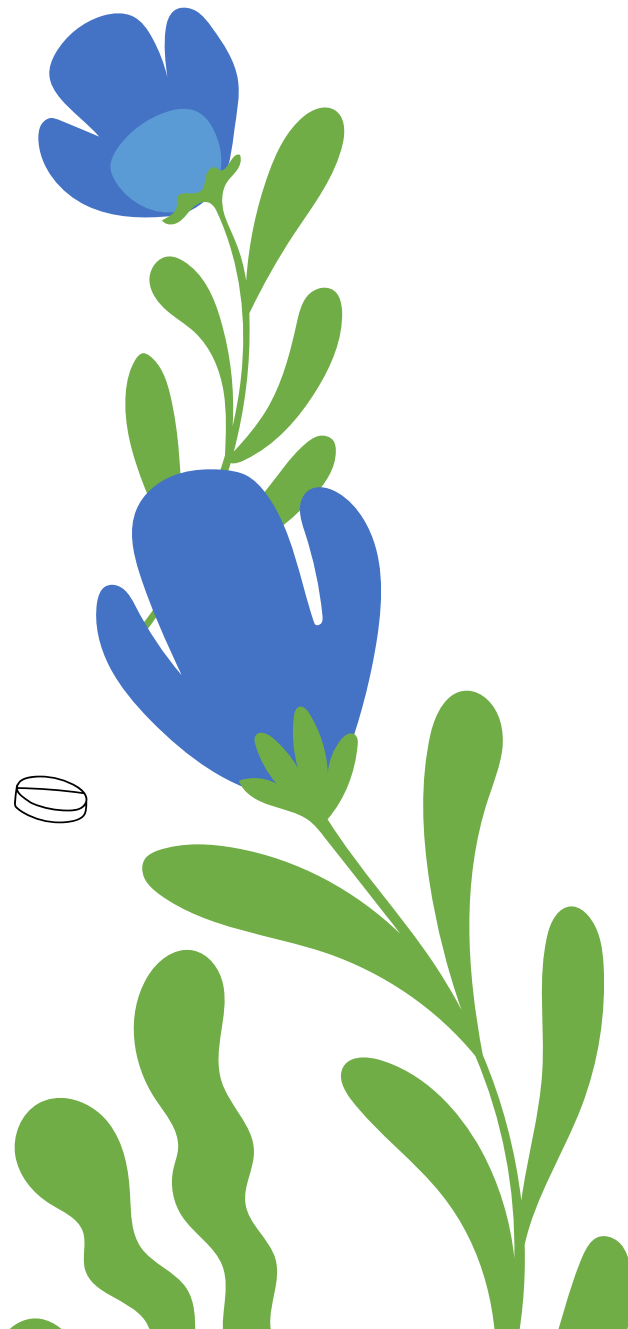
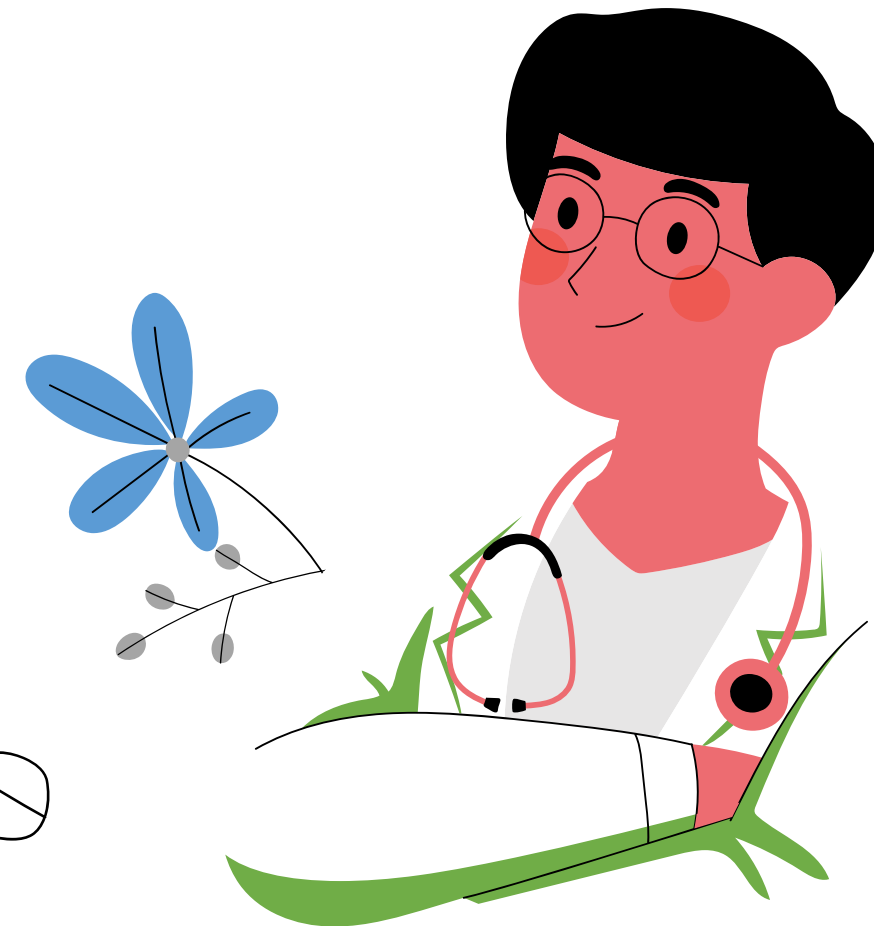
Selected Disorders of the Pancreas





PANCREATITIS

- ✓ It occurs when zymogens become activated within the pancreas and digest pancreatic tissue and sometimes associated tissues including blood vessels and fat.
- ✓ Can occur with excessive alcohol consumption, hypertriglyceridemia (serum triglycerides in excess of usually about 1,000 mg/dL), blockage of the pancreatic duct (e.g., from gallstones), viral infections, and pancreatic injury, among others.



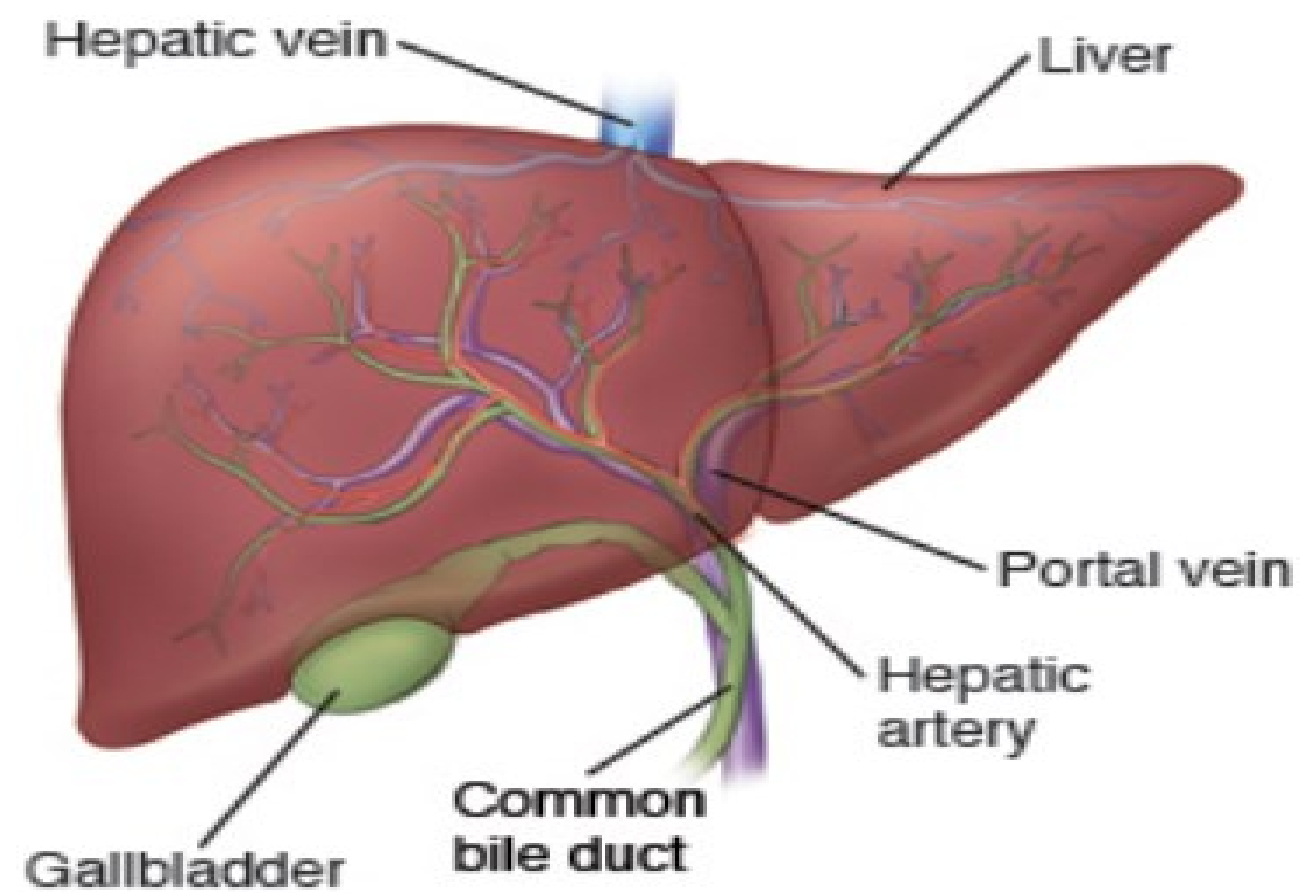
Nutrition care in Pancreatitis disease

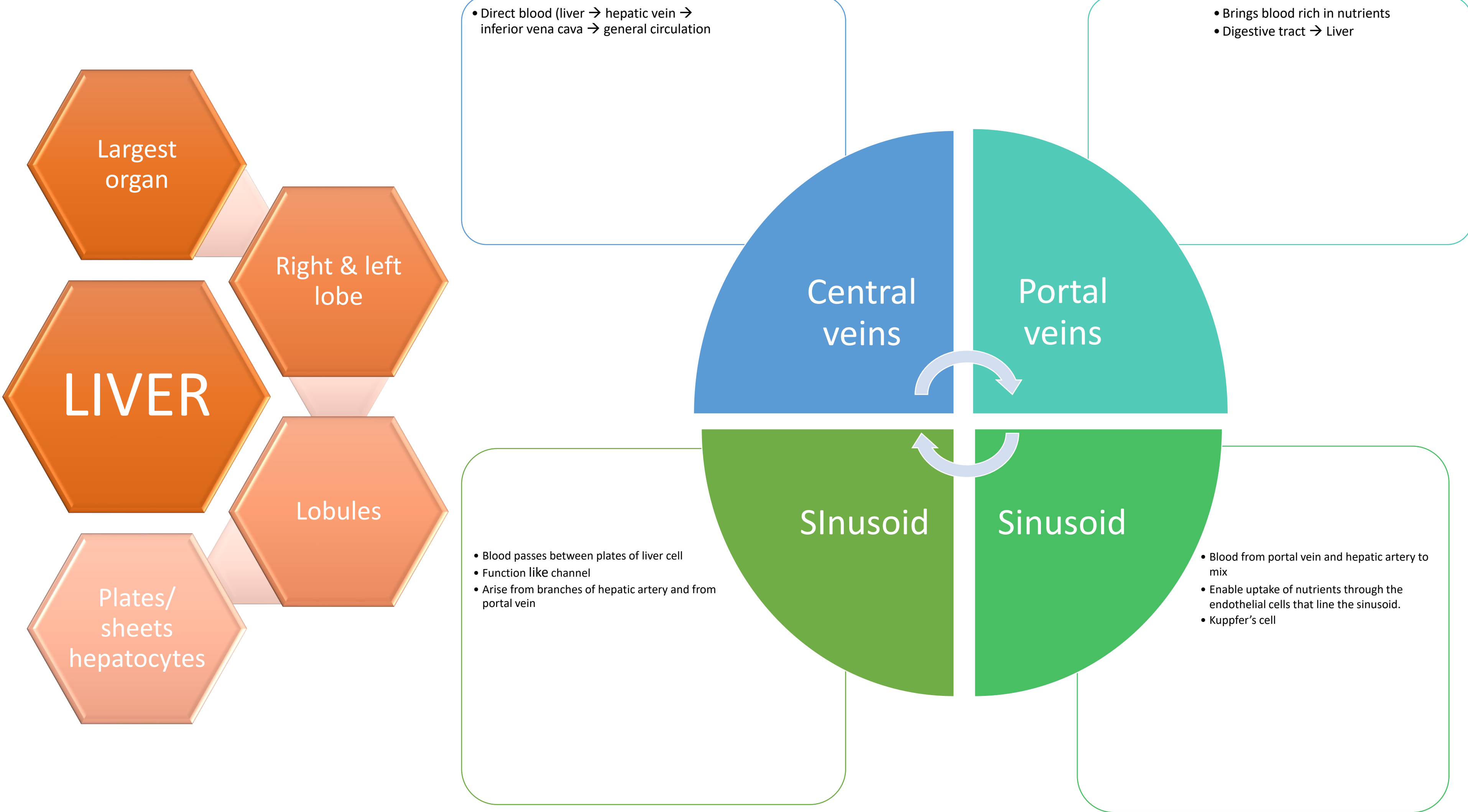
- Acute and severe attacks— oral feeding is withheld and hydration is maintained intravenous
- After 24-48 hours – clear liquid diet as tolerance
- Formula diet consisting of amino acid, glucose, and small amount of fat will not stimulate pancreatic secretions.
- Prolonged severe pancreatitis—TPN
- Fat emulsion can be used as long as –acute pancreatittis is not the basis for hypertriglyceridemia

Signs & Symptoms

- **Abdominal pain**, usually in the upper left quadrant and that worsens with food intake, is a major symptom.
- Most notably, the enzymes pancreatic **amylase** and **lipase**, which are normally released into the duodenum, leak out of the damaged pancreas and become elevated in the blood.
- The nutritional management of individuals with especially acute, severe pancreatitis is quite involved.
- Because the damaged pancreas cannot produce enzymes in sufficient quantities, the patient often requires the provision of nutrients that are already partially hydrolyzed (rather than intact) or supplements of pancreatic enzymes, especially lipase, to replace those not being released by the malfunctioning pancreas.
- Because bicarbonate secretion from the pancreas is frequently diminished with pancreatitis, and because this bicarbonate is needed to neutralize acid from the stomach and increase the pH of intestinal juices for enzyme function, **medications such as antacids** are sometimes provided.

LIVER





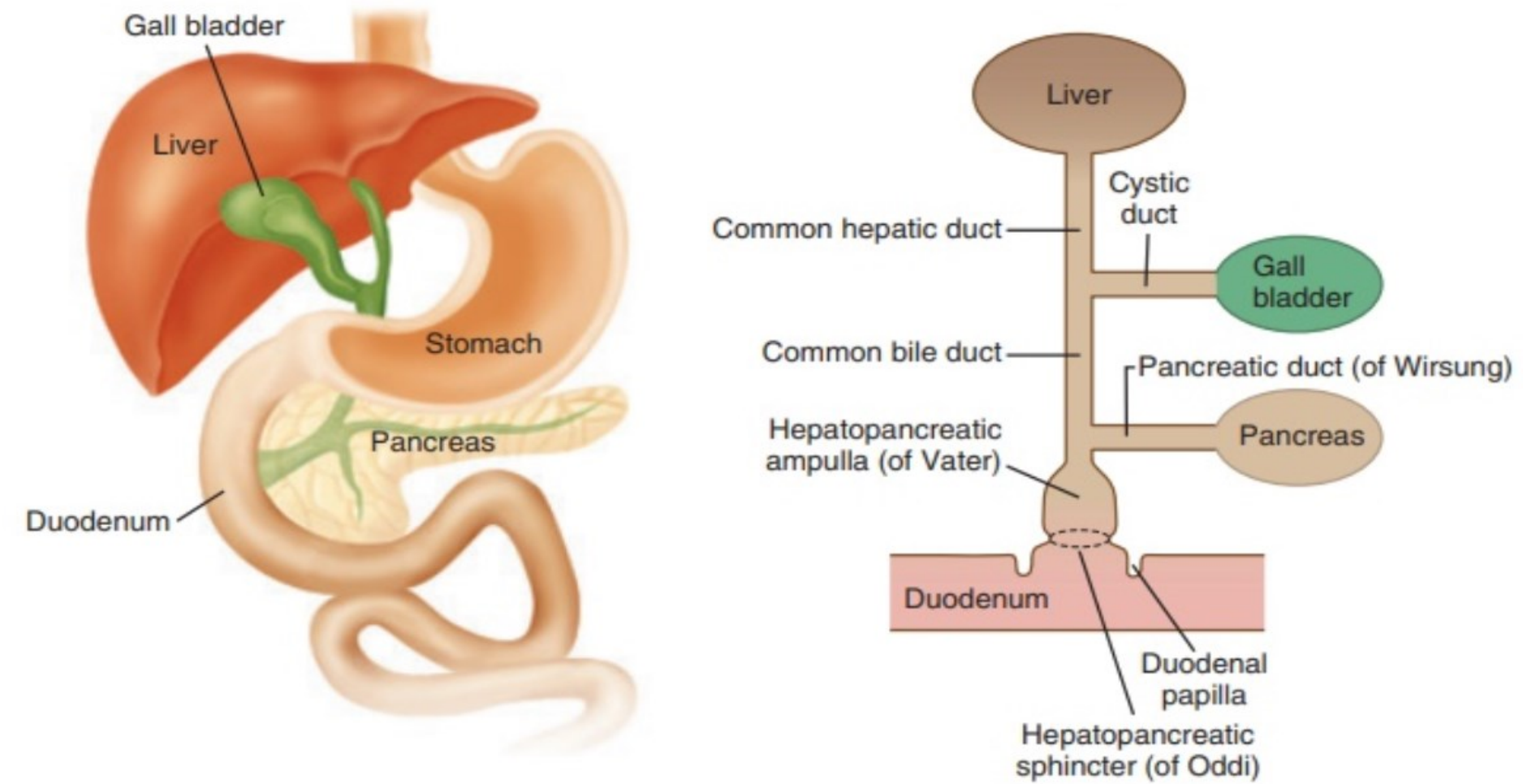
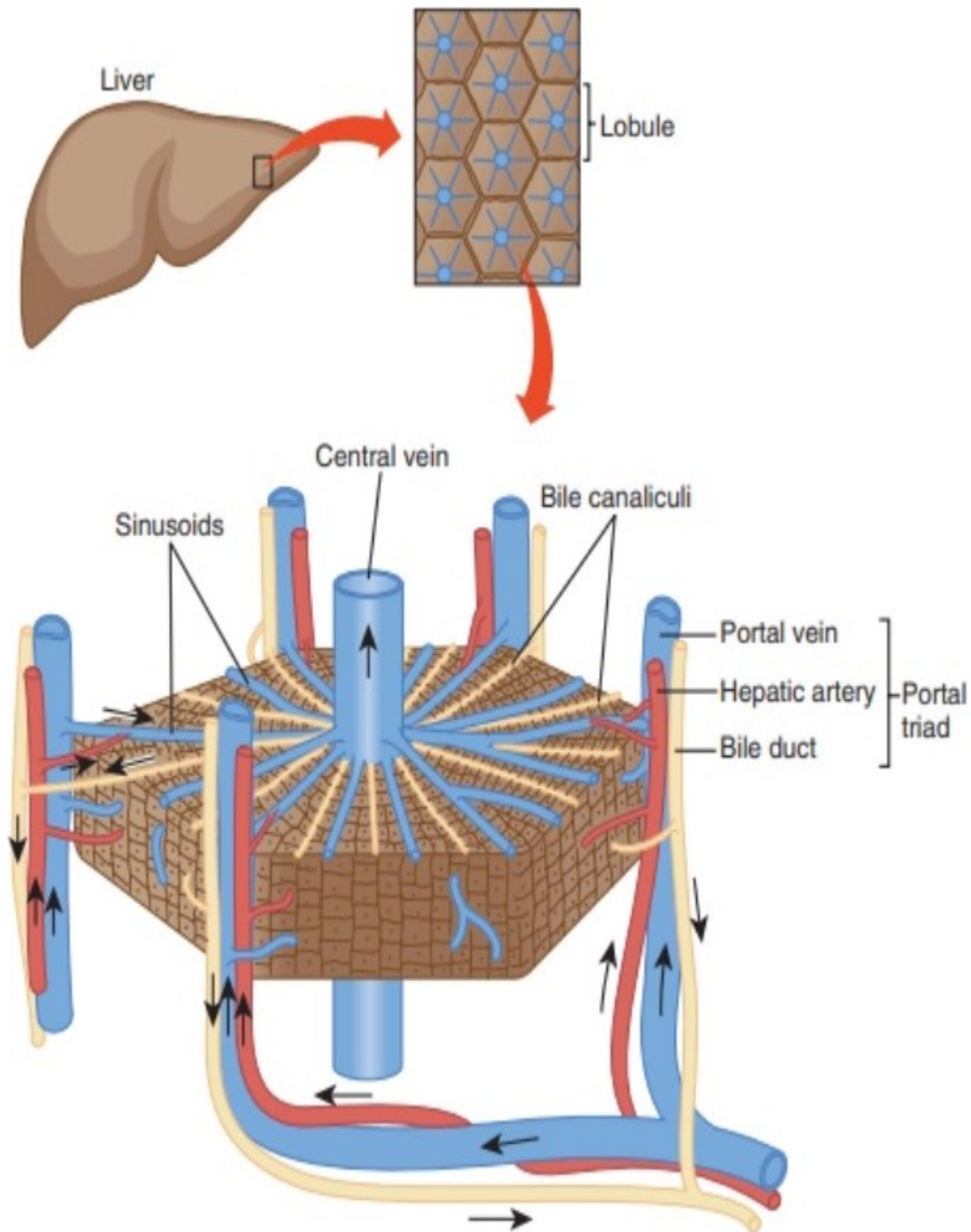


FIGURE Gross anatomy of the human liver and gallbladder.

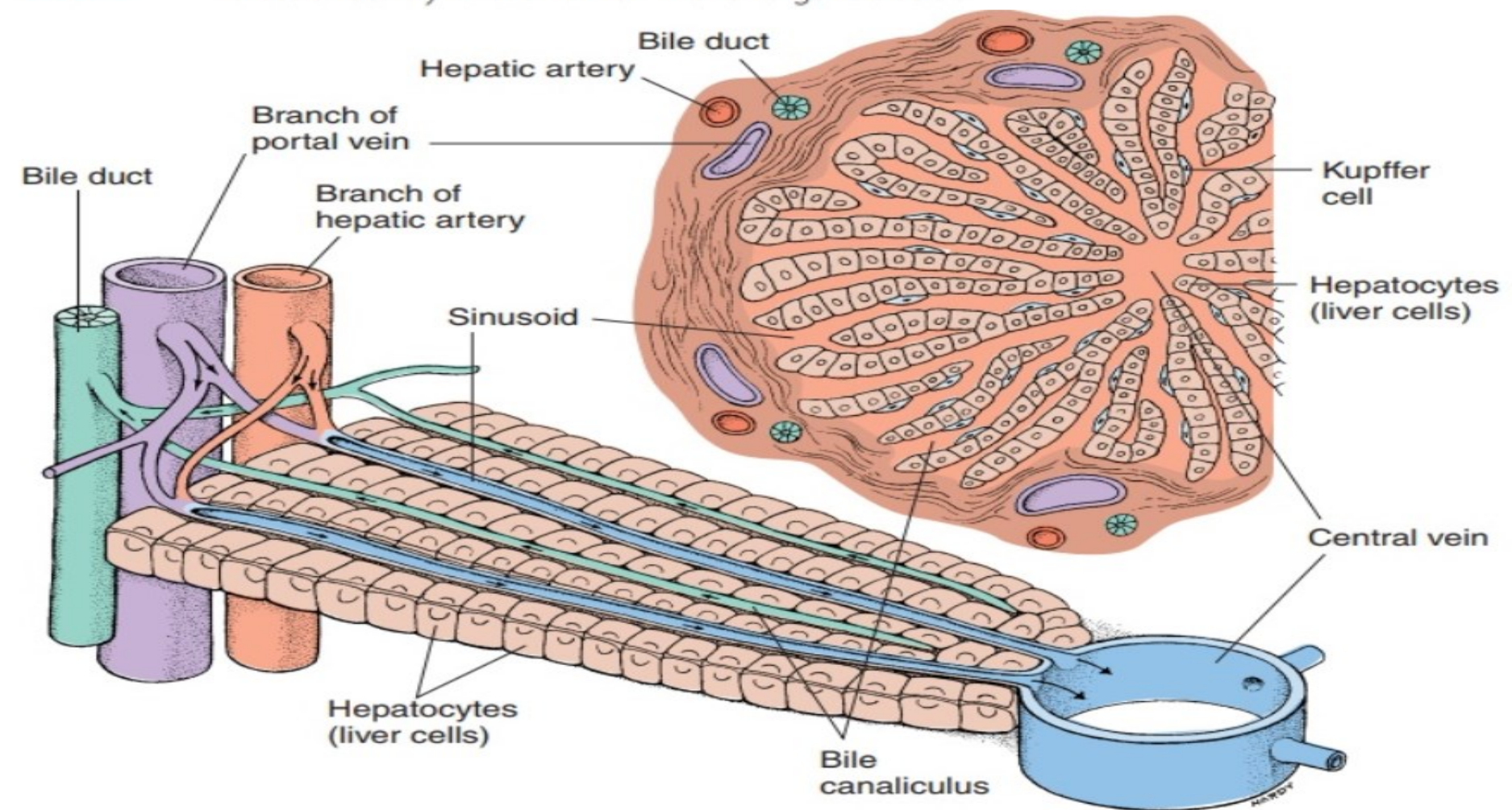


FIGURE Microscopic Structure of the Liver

LIVER

Bile production

- Aids in absorption & digestion of lipids (secretion of bile salts and acids)

Fat soluble vitamin storage and metabolism

- Via intestinal absorption (form : chylomicrons or VLDL)

Drug metabolism

- Metabolism and or detoxification of xenobiotics

Bilirubin metabolism

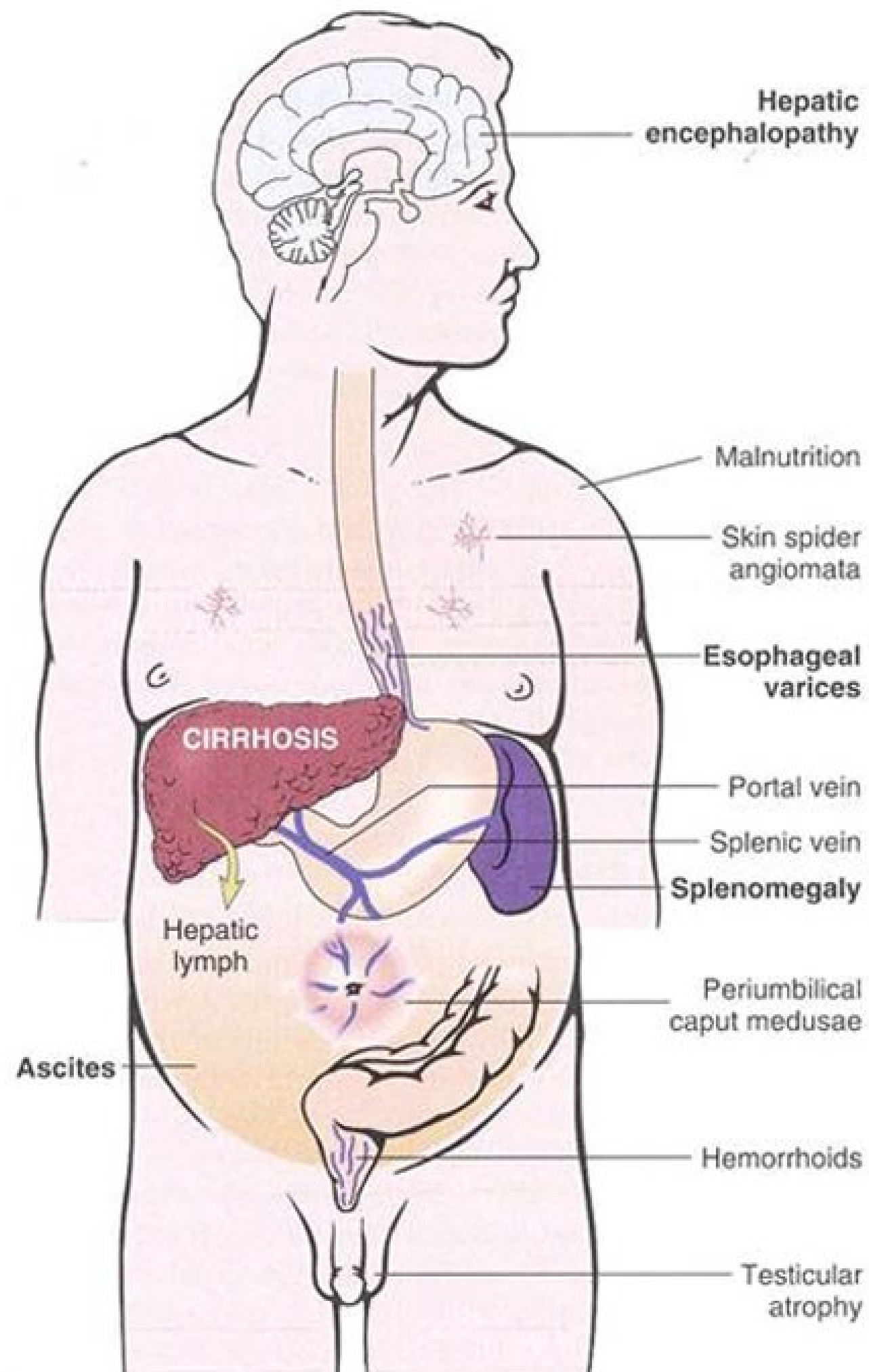
- Breakdown of heme

Other function

- Thyroid hormone function (site of deiodination of T4 to T3)
- Synthesis of plasma protein in body (albumin, binding globulin, protein C, protein S)
- Clotting factors (intrinsic and extrinsic factors – exc factor VIII)

Malnutrition In Liver Disease

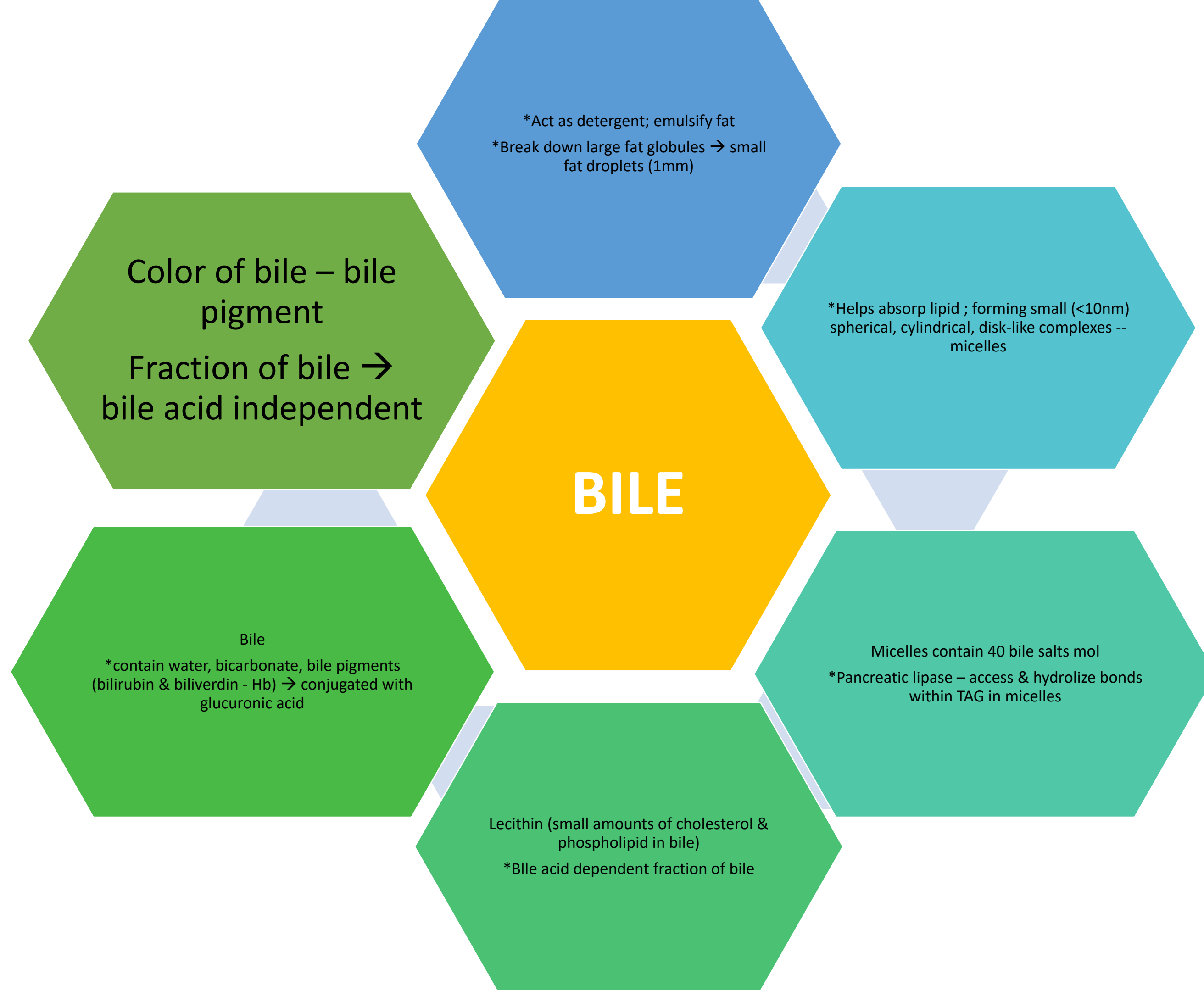
- Malnutrition is an early and typical aspect of hepatic cirrhosis
 - Contributes to poor prognosis and complications
- Degree of malnutrition related to severity of liver dysfunction and disease etiology (higher in alcoholics)
 - Mortality doubled in cirrhotic patients with malnutrition (35% vs 16%)
 - Complications more frequent than in well-nourished (44% vs 24%)
 - Usually more of a clinical problem than hepatic encephalopathy itself



Cirrhosis is common end result of many chronic liver disorders

- Severe damage to structure & function of normal cells
- Inhibits normal blood flow
- Decrease in # functional hepatocytes
- Results in portal hypertension & ascites
- Portal systemic shunting
 - Blood bypasses the liver via shunt, thus bypassing detoxification
 - Toxins remain in circulating blood
 - Neurtoxic substances can precipitate hepatic encephalopathy





LIVER

Fat soluble vitamin

- Vitamin A – stored in Ito cells
- Vitamin D3 – undergo 25-hydroxylation by hepatic CYP-450 system
- Liver – vitamin E – α and γ tocopherol
- Vitamin K – liver enzyme

Drug metabolism

- Metabolism & detoxification of xenobiotics
- Biotransformation
- (phase I - oxidation, reduction, hydrolysis
- Phase II - conjugation)

Bilirubin metabolism

- Breakdown of heme
- Hemolysis – liver, spleen, bone marrow
- Heme $\rightarrow$ biliverdin $\rightarrow$ unconjugated bilirubin $\rightarrow$ conjugation UGT system $\rightarrow$ conjugated bilirubin (hydrophylic) $\rightarrow$ bile $\rightarrow$ feces ; some converted to urobilinogen or unconjugated bilirubin by gut bacteria (resorption – enterohepatic circulation).

LIVER

Hematology

- Heme breakdown into unconjugated bilirubin and conjugates it
- Clotting factor and proteins synthesis

Interact with endocrine & gastrointestinal system

Aiding in digestion and metabolism

- Stores and metabolizes fat-soluble vitamins

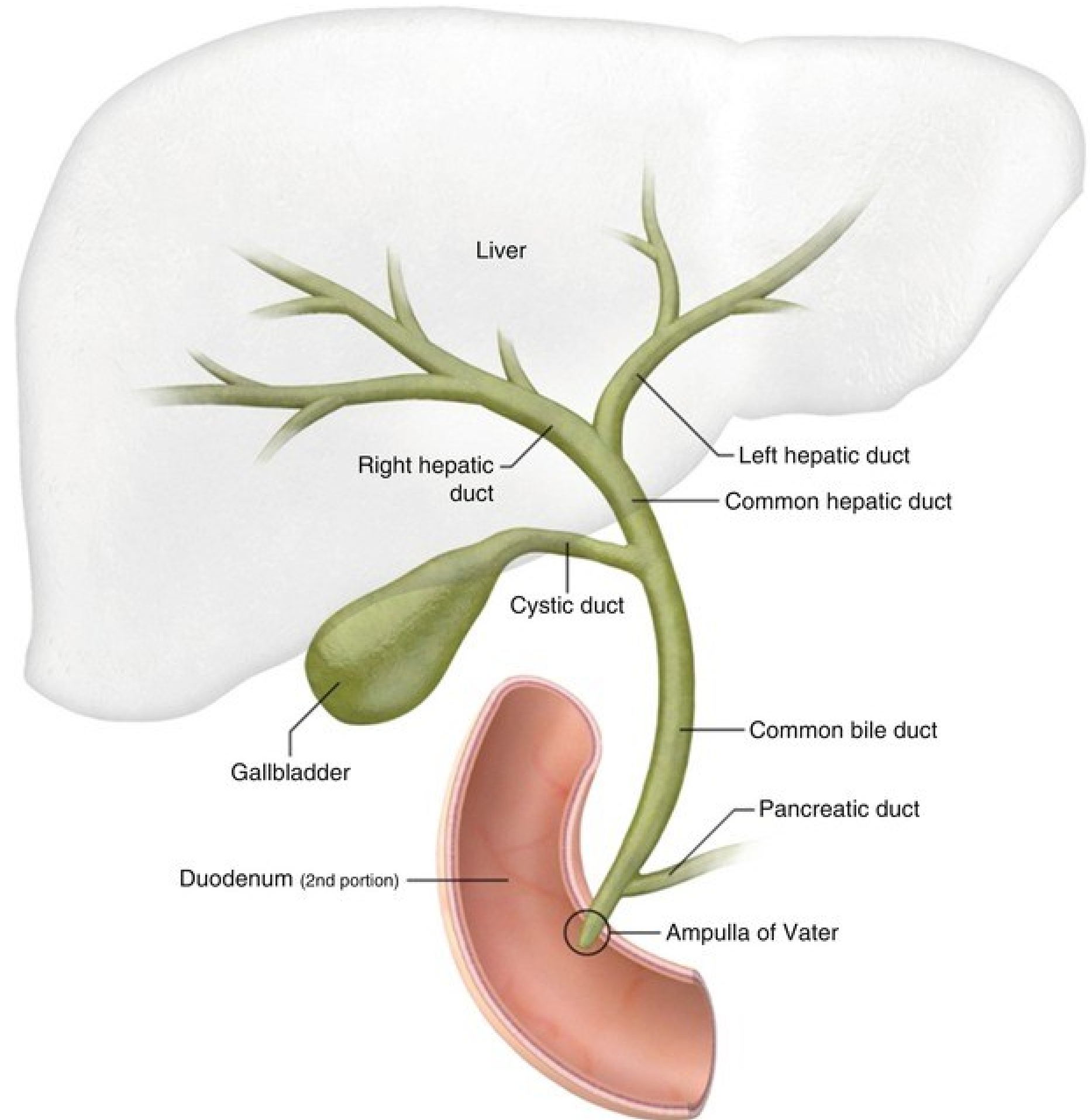
Reproduction and Development

- Sex hormone metabolism
- Produces carrier protein

Immunologic System

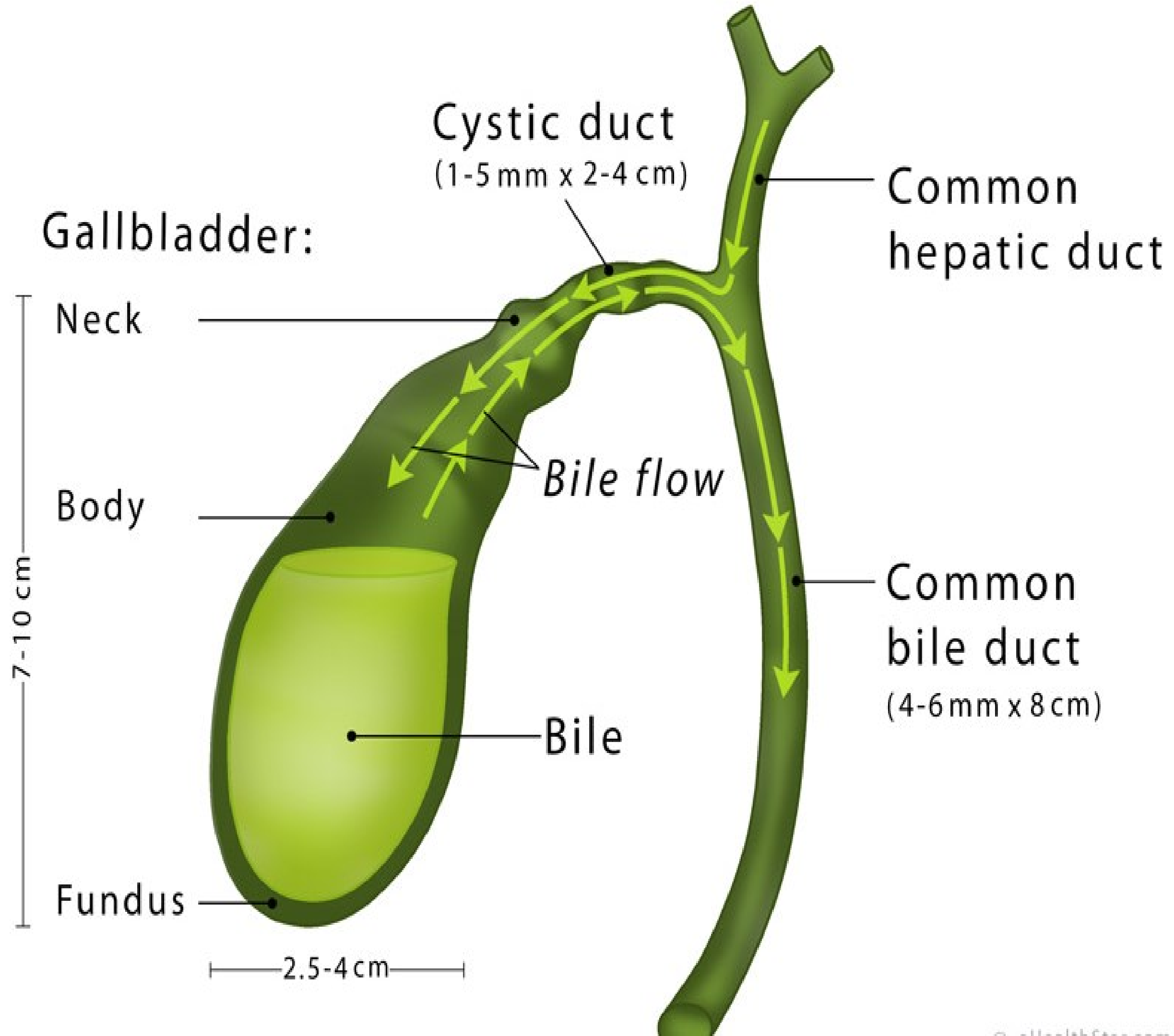
- Kupffer cells
- Pit cells

GALLBLADDER



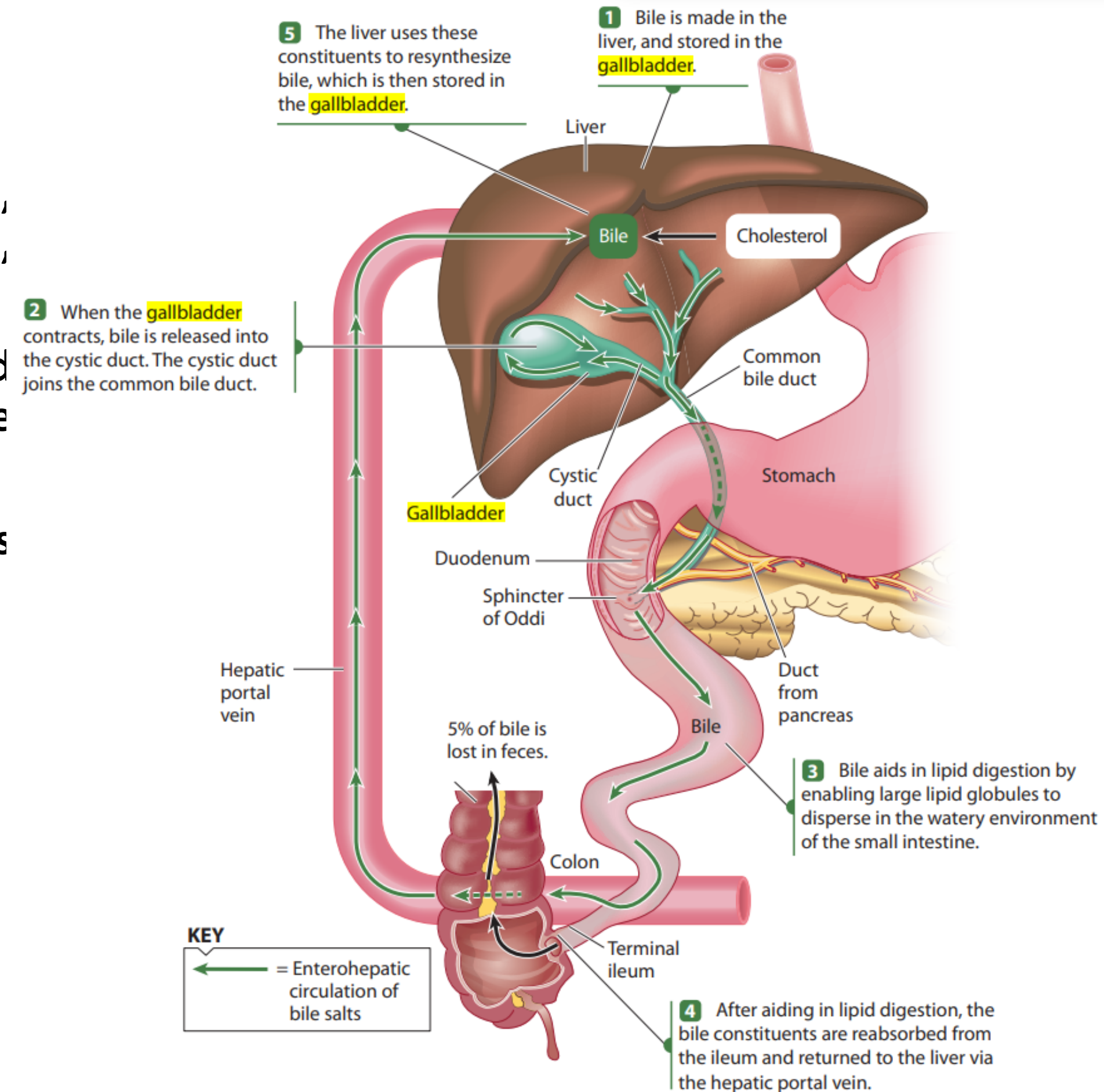
GALLBLADDER

- ❑ A small organ with a capacity of approximately 40–50 ml
- ❑ Located on the surface of the liver
- ❑ The human body has a total bile pool of about 2.5–5.0 g
- ❑ While ingesting a meal, the presence of fats and proteins in the intestines stimulate the release of cholecystokinin

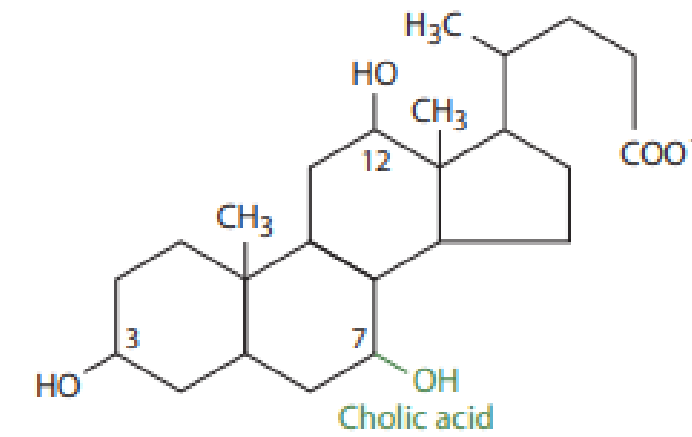


The Recirculation and Excretion of Bile

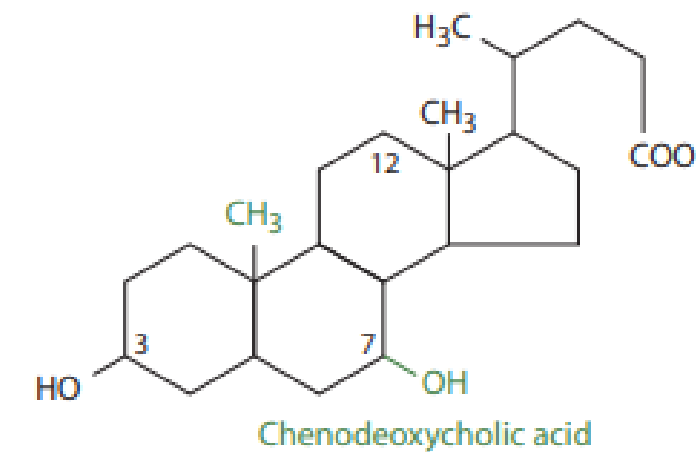
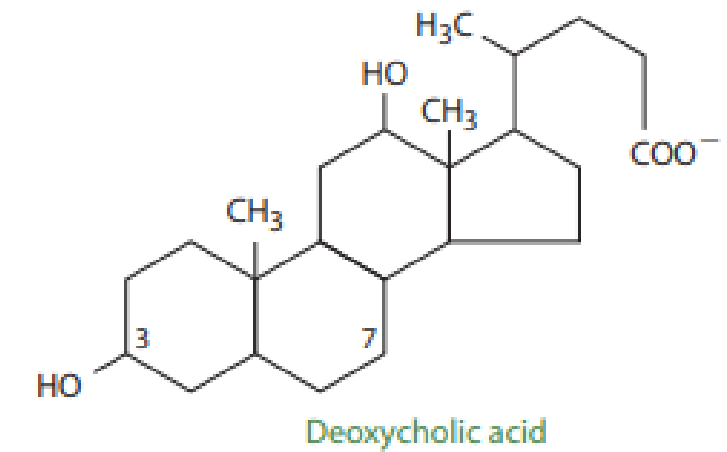
- ❑ Absorbed bile enters the portal vein and is transported, attached to the plasma protein albumin in the blood, back to the liver
- ❑ The reabsorbed bile is reconstituted to amino acids, and secreted along with the newly synthesized bile into the duodenum during digestion
- ❑ New bile acids are typically synthesized in amounts about equal to those lost in the feces.



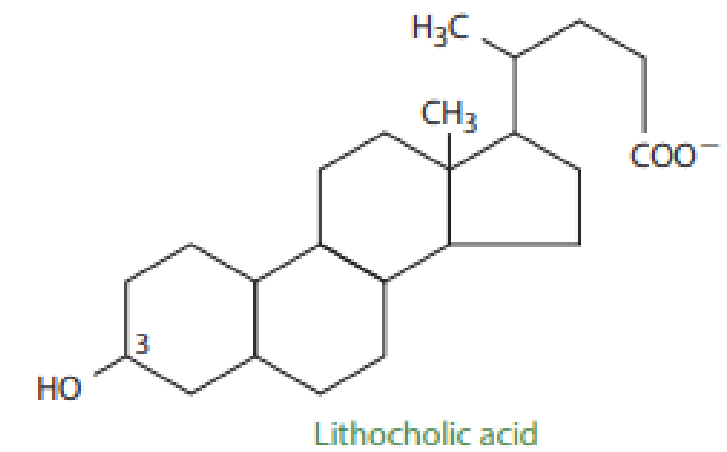
- ❑ Some of the bile acids that are not reabsorbed in the small intestine may be deconjugated by bacteria to form secondary bile acid
- ❑ Bacterial enzymatic activity on bile salts may regulate in part both cholesterol metabolism and energy balance in the host



Intestinal
bacteria →



Intestinal
bacteria →



Hormone in the Gallbladder

CHOLECYSTOCININE

secreted into the blood by I cells of the proximal small intestine in response to the presence of fat-laden chyme in the duodenum, stimulates the gallbladder to contract and the sphincter of Oddi to relax

SECRETIN

stimulates the secretion of bicarbonate and water by cholangiocytes in the bile duct, protecting it from bile acid by controlling the pH and promoting the flow in the duct

GASTRIN

can stimulate the gallbladder to empty its store of bile and the pancreas to secrete enzymes

Disorders of the Gallbladder

- **Cholelithiasis**
- **Hypercholesterolemia**

Acute – related to obstruction:

- ◇ stop oral
- ◇ low fat diet (<50gr)

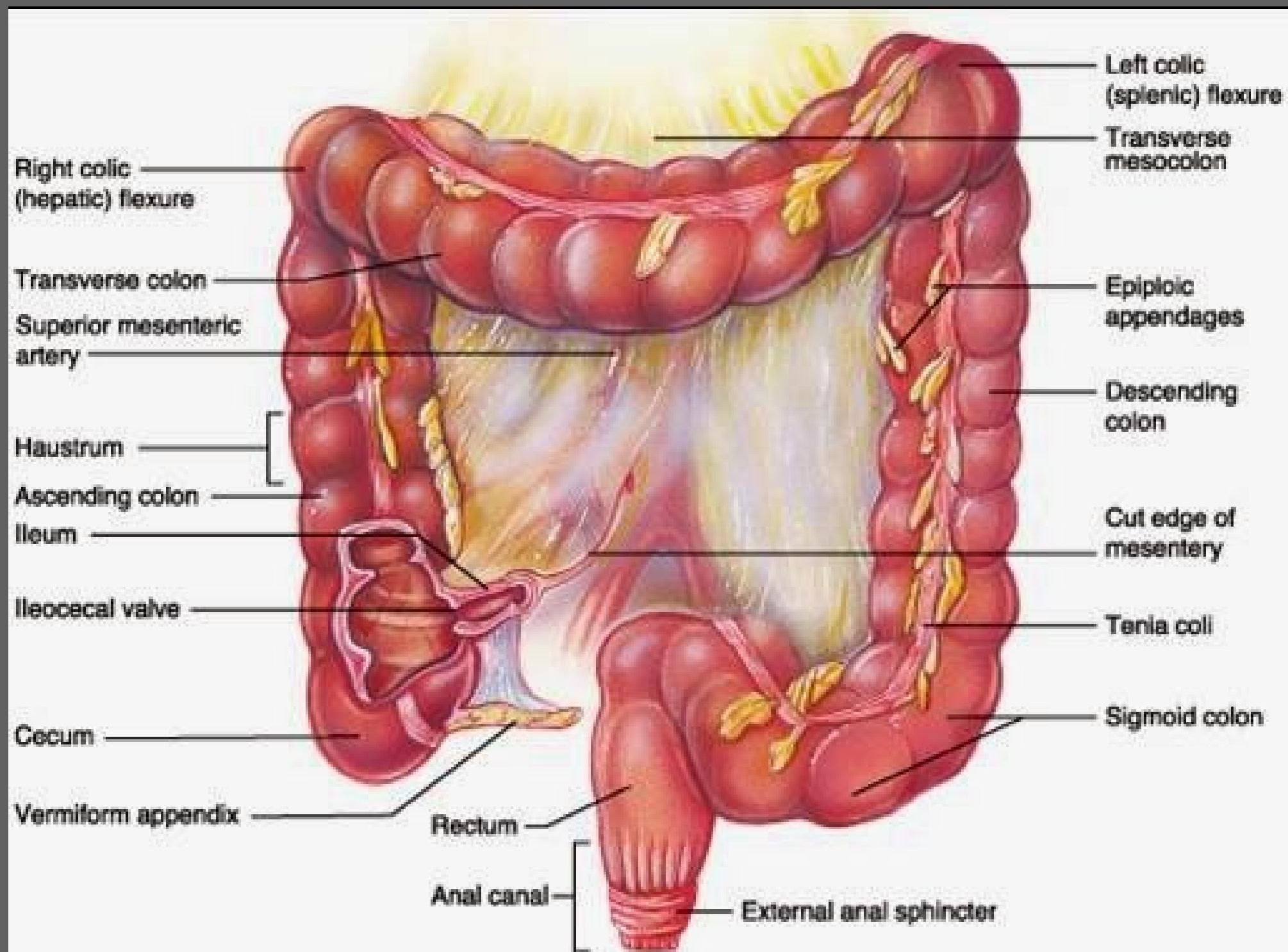
Chronic-- cholecystitis

- ◇ Low fat
- ◇ Decreased Body Weight
- ◇ Limitation high food content gas
- ◇ Supl. Vitamin ADEK

Nutrition care in Gallbladder disease

- **Adequate Food**
- **Low fat → decreased contraction**
- **Moderate intake of Energie, protein, carbohidrate**
- **High intake of Fluid**
- **Small portion**
- **not irritated**

ANATOMY AND FUNCTION



Size & length

The large intestine is approximately 6 feet in length, while the small intestine is much longer, at approximately 21 feet. The last 6 inches or so of the large intestine are called the rectum and the anal canal

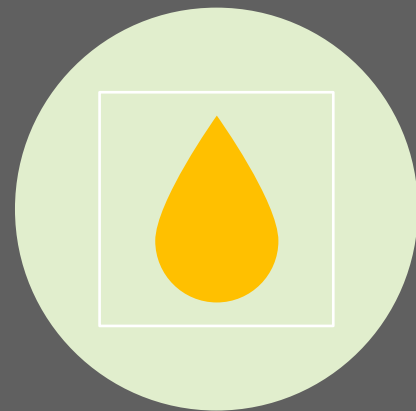
3 Primary Function

- absorbing water and electrolytes
- producing and absorbing vitamins
- forming and propelling feces toward the rectum for elimination

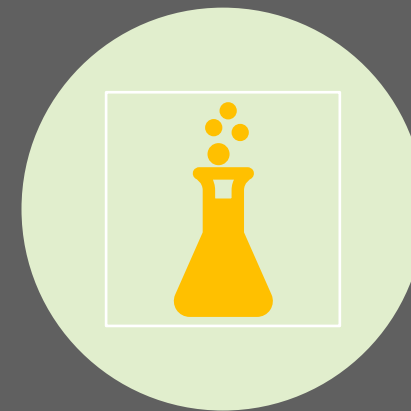
ABSORBING WATER AND ELECTROLYTES

- The proximal colonic mucosal cells typically absorb sodium, chloride, and water. About 90–95% of the water and sodium entering the colon each day is absorbed.
- Absorption of water occurs by osmosis. Water diffuses in response to an osmotic gradient established by the absorption of electrolytes.

IN THE LARGE INTESTINE, THERE IS A NET ABSORPTION OF SODIUM IONS AND CHLORIDE IONS ARE ACTIVELY ABSORBED



Sodium



Chloride and bicarbonate



Water



Potassium

PRODUCING AND ABSORBING VITAMINS

The colon plays an important role in absorbing vitamins needed for good health → through passive diffusion in the colon.

Vitamin K absorption is affected by gastrointestinal conditions

Low stomach acid → vitamin B₁₂ is not easily released from animal sources and is lost with feces.

In contrast, colonic bacteria synthesize vitamin K and biotin, which are actively absorbed and serve as the body's main source of supply.

Effects of gastric acid on microflora;

Gastric and small intestine → mostly destroyed and few colonized

Large intestine → mostly colonized

The colon houses trillions of bacteria protecting
our intestines

Microflora in the colon → fermentation process produces

Vitamin K

Biotin (vitamin B7)

thiamin,

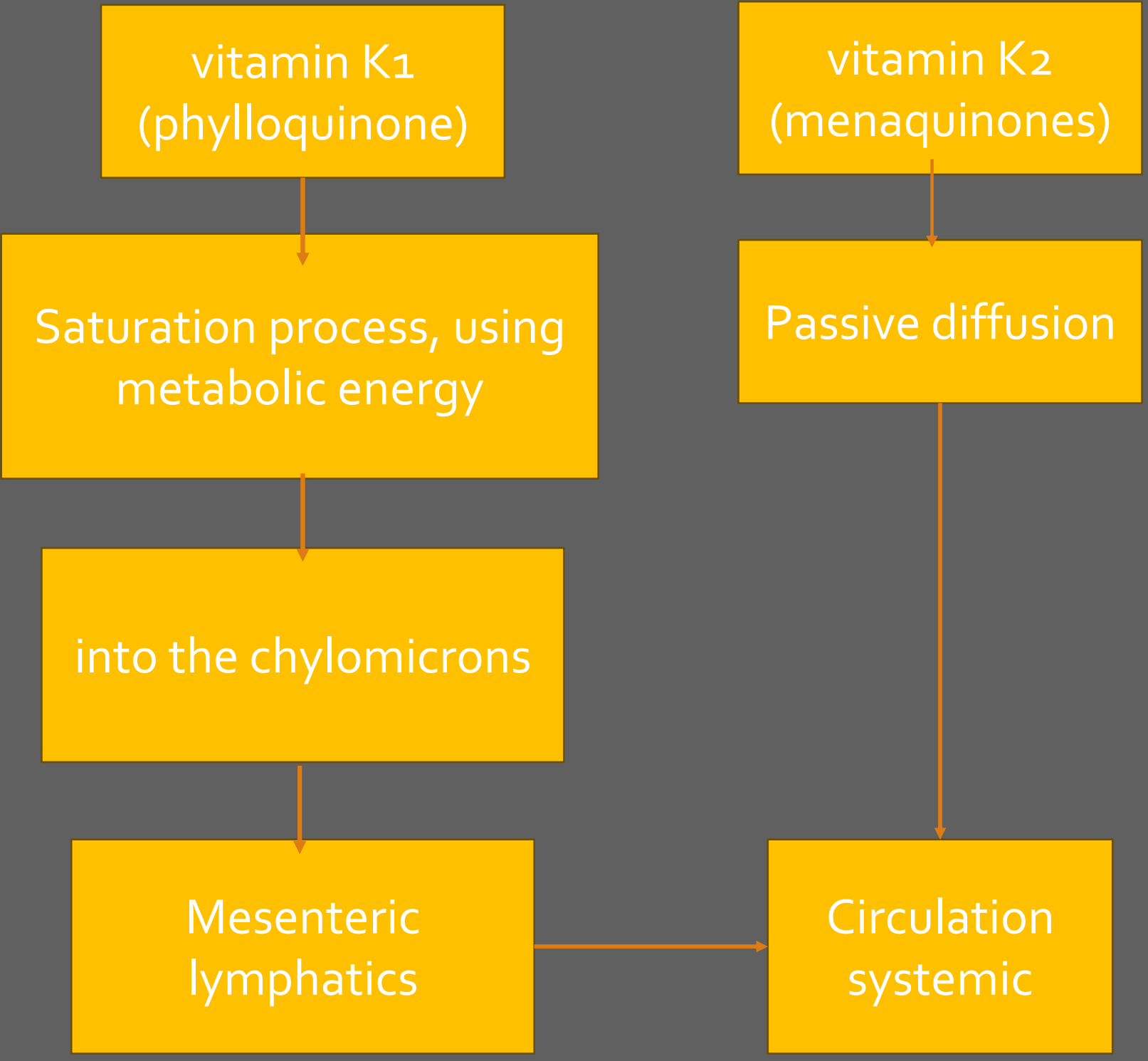
riboflavin,

vitamin B₁₂

absorbed into blood.

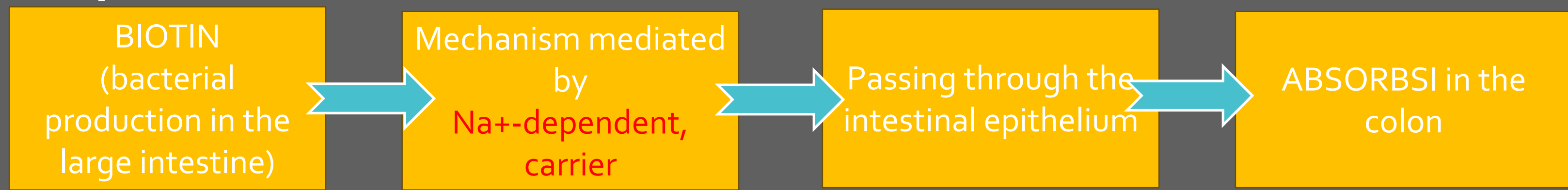
vitamin K synthesis is
essential for normal
blood clotting

Vitamin K stored in the liver
↓
Exhausted
↓
vitamin K intake from food is restricted



} Produced by intestinal and colonic bacteria
↓
important role in nutrition
vitamin K
↓
breast-fed infants

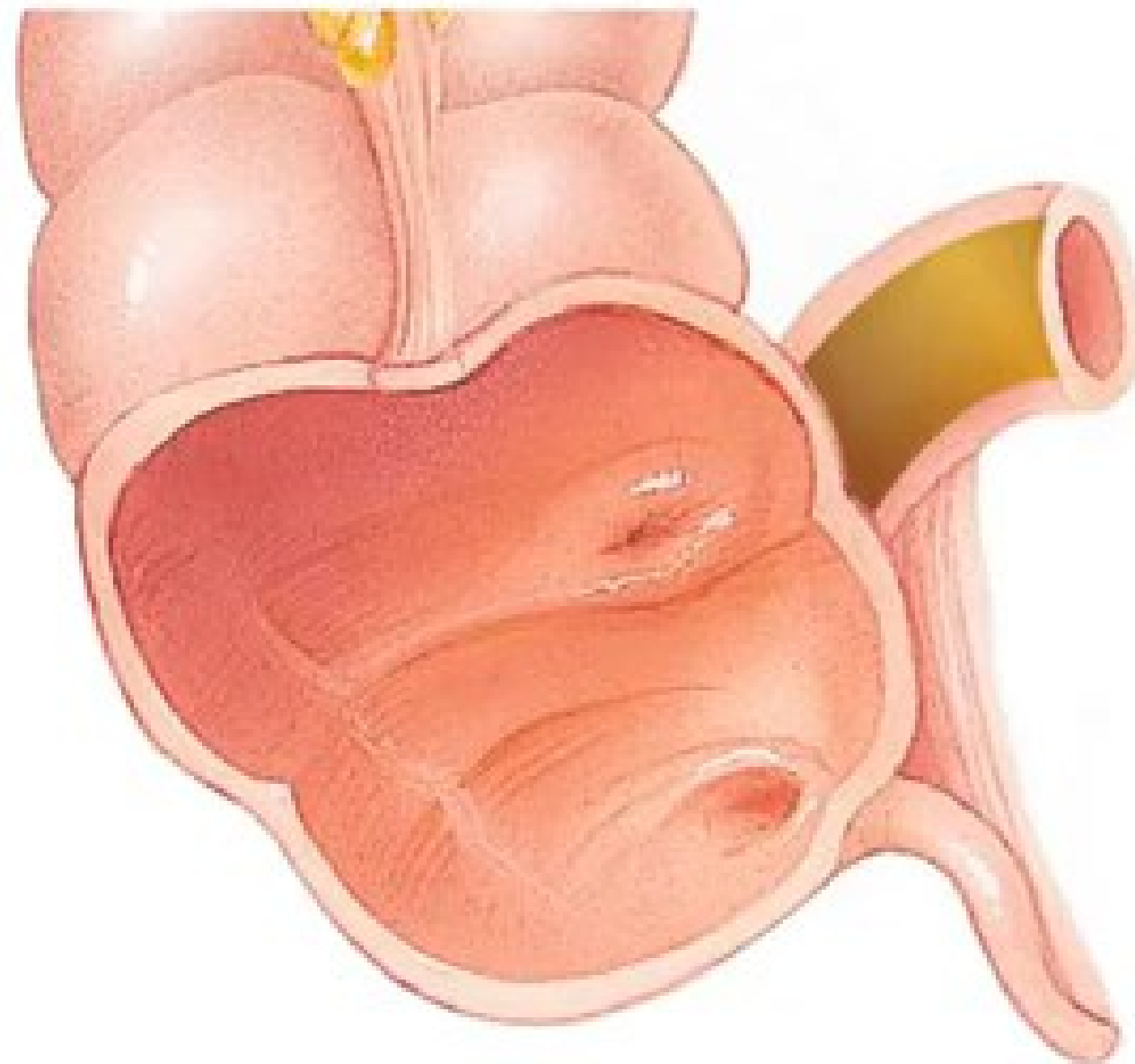
Absorbsi
5% to 15%



Homeostasis
Biotin

FORMING AND PROPELLING FECES

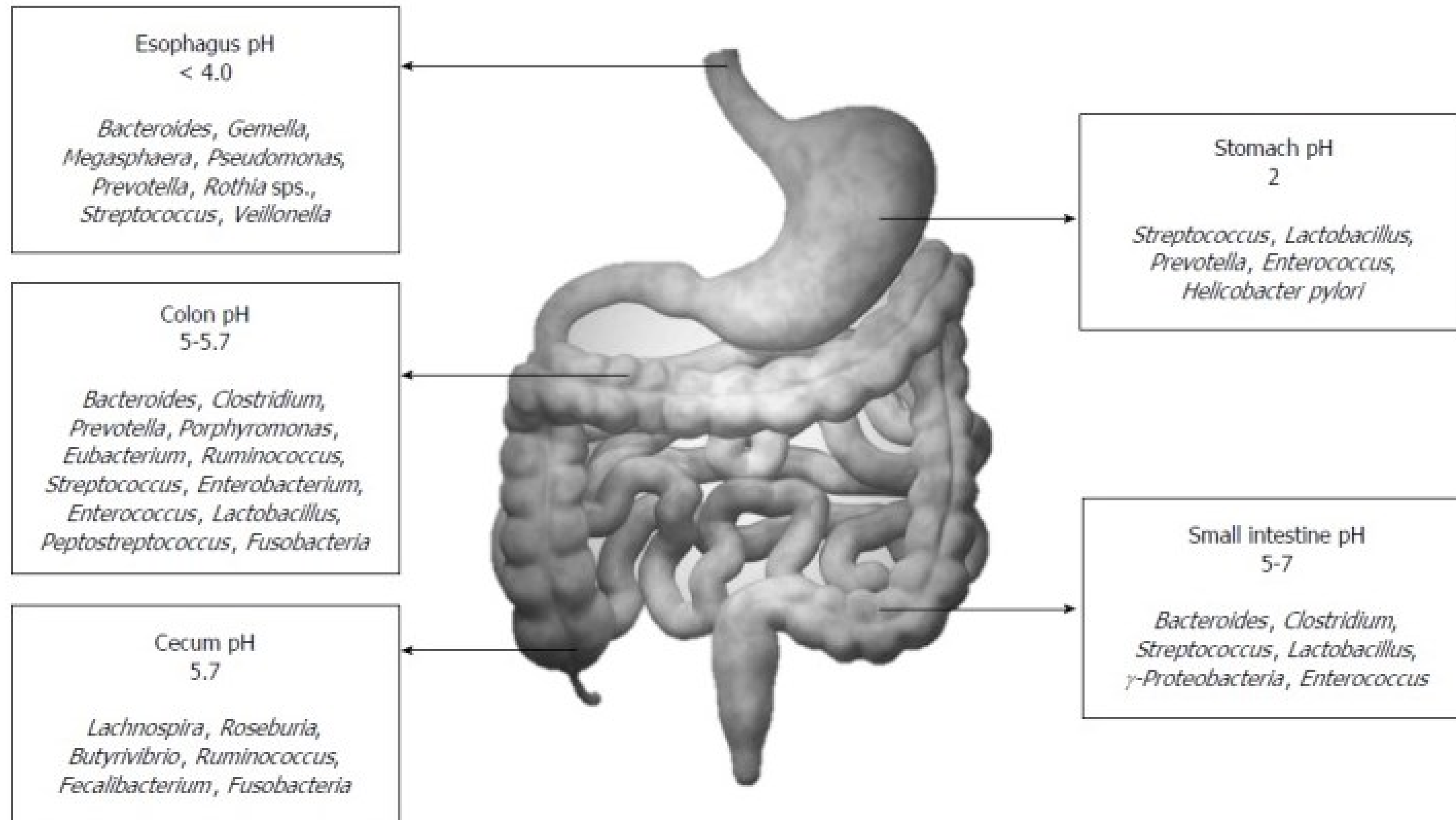
Haustral churning



FORMING AND PROPELLING FECES

- The main motility in the colon → contraction of haustra
- Mass movement occurs due to strain on the stomach & duodenum → the gastrocolic reflex and duodenocolic reflex.
- Movement pushes feces into the rectum → reflex rectal contraction and sphincter ani relaxation.
- The sphincter muscles of the rectum regulate the opening and closing of the anus consisting of smooth and striated muscles

GUT MICROBIOTA



FUNCTIONAL ASPECTS OF THE NORMAL GUT MICROBIOTA

- Nutrient metabolism
- Stimulate the immune system
- Immunomodulation
- Provide energy and serve as substrates for use within cells

Modifikasi Diet, Klasifikasi Makanan dan Minuman Berdasarkan Kekentalan dan Tekstur

Derajat	Deskripsi	Terminologi Umum
0	Encer	Mengalir seperti air
1	Sedikit kental	Lebih kental dari air; membutuhkan sedikit lebih banyak usaha
2	Agak kental	Mengalir dari sendok; lebih lambat dari minuman encer
3	Kekentalan moderat	Bisa diminum dari cangkir atau dimakan dengan sendok
4	Puree	Tidak bisa diminum dari cangkir; dimakan dengan sendok; tidak mengunyah
5	Cincang	Dimakan dengan garpu atau sendok; lembut dan lembab
6	Lembut dan diblender	Dimakan dengan garpu atau sumpit; tidak perlu pisau; perlu mengunyah
7	Reguler	Normal, makanan harian



Diet disfagia I



Diet disfagia III

Diet disfagia



Diet disfagia II



Diet disfagia III, II, I